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DEPARTMENT OF ARTIFICAL INTELLIGENCE AND DATA SCIENCE



**BUSINESS
ANALYTICS**

20AD501 – BUSINESS ANALYTICS

HANDOUT NOTES

UNIT – 1

INTRODUCTION TO BUSINESS ANALYTICS

Analytics and Data Science – Analytics Life Cycle – Types of Analytics – Business Problem Definition – Data Collection – Data Preparation – Hypothesis Generation – Modeling – Validation and Evaluation – Interpretation – Deployment and Iteration.

1. INTRODUCTION

What is Business Analysis?

- ❖ Business Analysis is the set of tasks, knowledge, and techniques required to identify business needs and determine solutions to enterprise business problems.
- ❖ Business analytics refers to new skills, naive practices, marketing, new technologies, and business tools algorithms for continuous iterative exploration.

Who is a Business Analyst?

A business analyst is someone who analyzes an organization or business domain (real or hypothetical) and documents its business, processes, or systems, assessing the business model or its integration with technology. However, organizational titles vary such as analyst, business analyst, business systems analyst or maybe systems analyst.

Why a Business Analyst?

Organizations employ business analysis for the following reasons –

- To understand the structure and the dynamics of the organization in which a system is to be deployed.
- To understand current problems in the target organization and identify improvement potentials.

Types of Analytics:

Business analytics techniques can be segmented in the following four ways:

1. Descriptive Analytics: This technique describes the past or present situation of the organization's activities.
2. Diagnostic Analytics: This technique discovers factors or reasons for past or current performance.
3. Predictive Analytics: This technique predicts figures and results using a combination of business analytics tools.

4. Prescriptive Analytics: This technique recommends specific solutions for businesses to drive their growth forward.

Business Analytics Applications:

Business Analytics is now systematically integrated across several applications in the field of

- ✓ supply chain management,
- ✓ customer relationship management,
- ✓ financial management,
- ✓ human resources,
- ✓ manufacturing, and
- ✓ even build smart strategies for sports too.

Artificial Intelligence in Business Analytics:

- ❖ We know that artificial intelligence is a field that builds intelligent machines that are capable of solving problems and performing tasks that would require human intelligence.
- ❖ For artificial intelligence systems to learn and function as intended, they need data. That's why AI in business analytics is a perfect use case, as business analytics is all about data and that's exactly what AI needs.
- ❖ Implementing AI solutions in business intelligence means processing this massive amount of data in a matter of seconds. In this way, managers can easily analyze data and export valuable information that can be used to improve decision-making and strategy building.

1.1 Analytics and Data Science

Data Science: Data Science is a field that deals with extracting meaningful information and insights by applying various algorithms, processes, scientific methods from structured and unstructured data. This field is related to big data and one of the most demanded skills currently.

Data Analytics: Data Analytics is used to get conclusions by processing the raw data. It is helpful in various businesses as it helps the company to make decisions based upon the conclusions from the data. Basically, data analytics helps to convert a Large number of figures in the form of data into Plain English i.e., conclusions which are further helpful in making the decisions.

Below is a table of differences between Data Science and Data Analytics:

Feature	Data Science	Data Analytics
Coding Language	Python is the most commonly used language for data science along with the use of other languages such as C++, Java, Perl, etc.	The Knowledge of Python and R Language is essential for Data Analytics.
Programming Skills	In- depth knowledge of programming is required for data science.	Basic Programming skills is necessary for data analytics.
Use of Machine Learning	Data Science makes use of machine learning algorithms to get insights.	Data Analytics doesn't makes use of machine learning.
Other Skills	Data Science makes use of Data mining activities for getting meaningful insights.	Hadoop Based analysis is used for getting conclusions from raw data.
Scope	The scope of data science is large.	The Scope of data analysis is micro i.e., small.
Goals	Data science deals with explorations and new innovations.	Data Analysis makes use of existing resources.
Data Type	Data Science mostly deals with unstructured data.	Data Analytics deals with structured data.
Statistical Skills	The statistical skills are necessary in the field of Data Science..	The statistical skills are of minimal or no use in data analytics.

Table 1.1: differences between Data Science and Data Analytics

1.2 Analytics Life Cycle

Life Cycle of Data Analytics

The Data analytics lifecycle was designed to address Big Data problems and data science projects. The process is repeated to show the real projects. To address the specific demands for conducting analysis on Big Data, the step-by-step methodology is required to plan the various tasks associated with the acquisition, processing, analysis, and recycling of data.

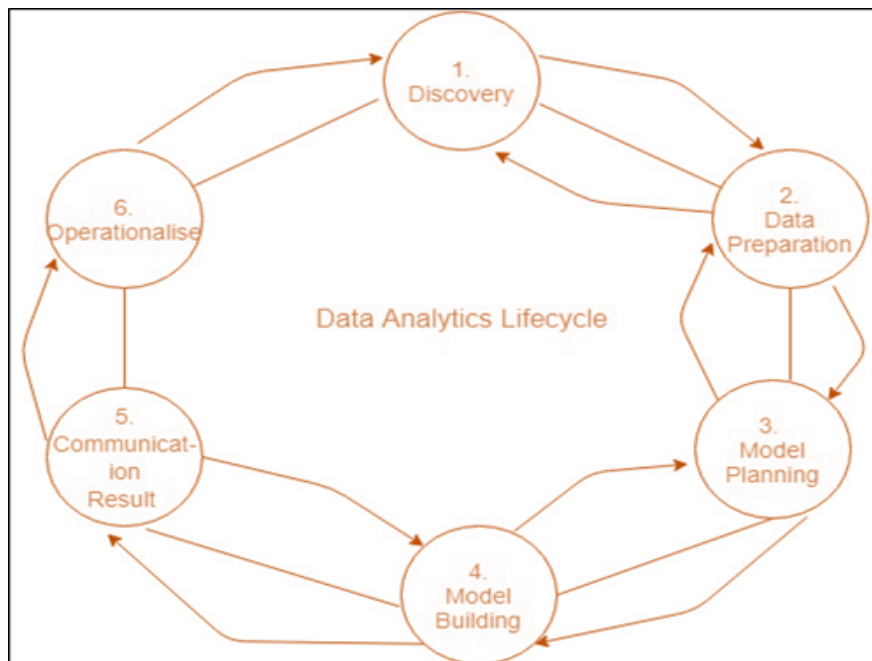


Figure 1.1: Data Analytics Lifecycle

Phase 1: Discovery -

- The data science team is trained and researches the issue.
- Create context and gain understanding.
- Learn about the data sources that are needed and accessible to the project.
- The team comes up with an initial hypothesis, which can be later confirmed with evidence.

Phase 2: Data Preparation -

- Methods to investigate the possibilities of pre-processing, analysing, and preparing data before analysis and modelling.
- It is required to have an analytic sandbox. The team performs, loads, and transforms to bring information to the data sandbox.
- Data preparation tasks can be repeated and not in a predetermined sequence.

- Some of the tools used commonly for this process include - Hadoop, Alpine Miner, Open Refine, etc.

Phase 3: Model Planning -

- The team studies data to discover the connections between variables. Later, it selects the most significant variables as well as the most effective models.
- In this phase, the data science teams create data sets that can be used for training for testing, production, and training goals.
- The team builds and implements models based on the work completed in the modelling planning phase.
- Some of the tools used commonly for this stage are MATLAB and STASTICA.

Phase 4: Model Building -

- The team creates datasets for training, testing as well as production use.
- The team is also evaluating whether its current tools are sufficient to run the models or if they require an even more robust environment to run models.
- Tools that are free or open-source or free tools R and PL/R, Octave, WEKA.
- Commercial tools - MATLAB, STASTICA.

Phase 5: Communication Results -

- Following the execution of the model, team members will need to evaluate the outcomes of the model to establish criteria for the success or failure of the model.
- The team is considering how best to present findings and outcomes to the various members of the team and other stakeholders while taking into consideration cautionary tales and assumptions.
- The team should determine the most important findings, quantify their value to the business and create a narrative to present findings and summarize them to all stakeholders.

Phase 6: Operationalize -

- The team distributes the benefits of the project to a wider audience. It sets up a pilot project that will deploy the work in a controlled manner prior to expanding the project to the entire enterprise of users.

- This technique allows the team to gain insight into the performance and constraints related to the model within a production setting at a small scale and then make necessary adjustments before full deployment.
- The team produces the last reports, presentations, and codes.
- Open source or free tools such as WEKA, SQL, MADlib, and Octave.

1.3 Types of Analytics

For different stages of business analytics huge amount of data is processed at various steps. Depending on the stage of the workflow and the requirement of data analysis.

There are four main kinds of analytics :

- ✓ descriptive,
- ✓ diagnostic,
- ✓ predictive and
- ✓ prescriptive.

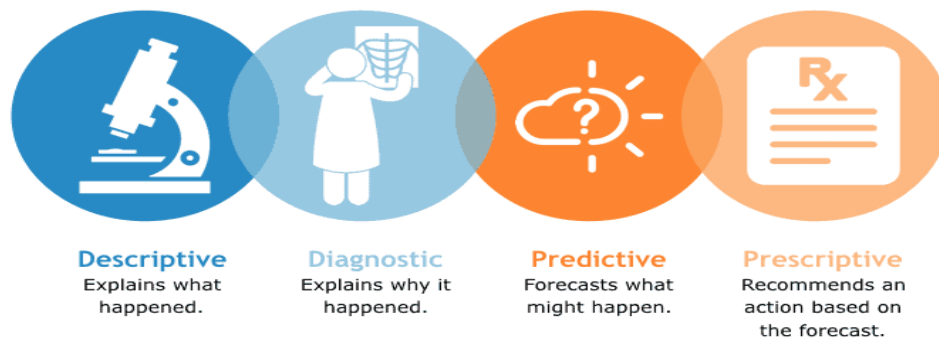


Figure 1.2: Kinds of Analytics

Four types of analytics:

- 1) **Descriptive Analytics**: Describing or summarising the existing data using existing business intelligence tools to better understand what is going on or what has happened.
- 2) **Diagnostic Analytics**: Focus on past performance to determine what happened and why. The result of the analysis is often an analytic dashboard.
- 3) **Predictive Analytics**: Emphasizes on predicting the possible outcome using statistical models and machine learning techniques.
- 4) **Prescriptive Analytics**: It is a type of predictive analytics that is used to recommend one or more course of action on analyzing the data. Let's understand these in a bit more depth.

1.3.1 Descriptive Analytics:

This can be termed as the simplest form of analytics.

- Descriptive analytics answers the question of “*what happened*”.
- Let us bring an example from ScienceSoft’s practice: having analyzed monthly revenue and income per product group, and the total quantity of metal parts produced per month, a manufacturer was able to answer a series of ‘what happened’ questions and decide on focus product categories.
- The two main techniques involved are
 - ✓ data aggregation and
 - ✓ data mining.
- The tools used in this phase are MS Excel, MATLAB, SPSS, STATA, etc.

1.3.2 Diagnostic analytics

- At this stage, historical data can be measured against other data to answer the question of “*why something happened*” and used to determine why something happened in the past.

1.3.3 Predictive analytics

- Predictive analytics tells “*what is likely to happen*” and used to predict future outcomes. It uses the findings of descriptive and diagnostic analytics to detect clusters and exceptions, and to predict future trends, which makes it a valuable tool for forecasting.
- The most popular tools for predictive analytics include Python, R, RapidMiner, etc.

1.3.4 Prescriptive analytics

- The purpose of prescriptive analytics is to literally prescribe “*what action to take*” to eliminate a future problem or take full advantage of a promising trend.

1.4 Business Problem

Running a business is an extremely tedious task. Almost always, there’s a problem or two that needs your immediate attention, and addressing these problems head-on to find their solutions is a part of the process. Luckily, there’s a tool that can help you with that: **a business problem statement**.

You may know how necessary it is to write a business plan for your company. What you may not know of, however, are the tools and techniques that can help you solve your business problems.

A business problem statement can help analyze the problem and come up with new and creative ways to solve it.

Essentially, it enables you to approach the problem in a more systematic, measurable, and objective way.

1.4.1 What Is a Business Problem Statement?

It defines the problem that a company is facing. Also, it involves an intricate analysis of the problem, details relevant to the situation, and a solution that can solve the problem. This is a simple yet effective way to present a problem and its solution concisely.

A business problem statement is a compact communication tool that helps you convey what you want to change.

1.4.2 How to Write a Business Problem Statement?

Before writing a business problem statement, it is crucial to conduct a complete analysis of the problem and everything related. You should have the knowledge to describe your problem and also suggest a solution to it.



Figure 1.3: Key aspects of business problem

1. Define the problem

Defining the problem is the primary aspect of a business problem statement. Summarize your problem in simple and layman terms. It is highly recommended to avoid industrial lingo and buzzwords. Write a 3-5 sentences long summary, avoid writing more than it.

2. Provide the problem analysis

Adding statistics and results from surveys, industry trends, customer demographics, staffing reports, etc., helps the reader understand the problem distinctly. These references should describe your problem and its effects on various attributes of your business.

Don't mention numbers in your problem statement if they are irrelevant to the situation.

3. Propose a solution

Your business problem statement should conclude with a solution to the problem that was previously described. The solution should describe how the current state can be improved.

Solutions must not exceed than two sentences.

4. Consider the audience

When you start writing your business problem statement, or any formal document, it is important to be aware of the reader. Write your problem statement keeping in mind the reader's knowledge about the situation, requirements, and expectations.

If the readers don't know the context of the problem, make sure to introduce it to them first before proceeding with the proposed solutions.

1.4.3 How to Develop a Business Problem Statement:



A popular method that is used while writing a problem statement is the **5W2H** (What, Why, Where, Who, When, How, How much) method. These are the questions that need to be asked and answered while writing a business problem statement.

1.4.4 Business Problem Statement Framework:



A problem statement consists of four main components. They are:

- The problem: The problem statement begins with mentioning and explaining the current state.
- Who it affects: Mention the people who are affected by the problem.
- How it impacts: Explain the impacts of the problem.
- The solution: Your problem statement ends with a proposed solution.

Business Problem Statement Templates

Below is a common template used for writing a problem statement. It includes the four key components mentioned in the above framework.

Template : The problem of _____, affects _____, the impact of which is _____, so an apt solution for this is _____.
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1.4.5 Business Problem Statement Examples

Here are a few problem statement examples to help you understand how to write your business problem statement:

Example 1: A problem statement by a software company

The problem of a manual auditing process affects the finance department and the staff as the process creates a lot of hassle because of the searching and verifying physical documents.

The impact of this is a lengthy auditing process where many mishaps and misplacements of documents happen. A successful solution would be to create an online database with search filters that would make it easy to find and verify documents.

Example 2: A problem statement by a manufacturing company

The problem of an inefficient manual assembly process affects the productivity of the company, and the workers have to manually install some parts, which consumes more time.

This impacts the production goals and incremental loss for this year. An efficient solution would be to install conveyor belts to optimize the manufacturing process.

1.4.6 The business analyst's way to solve the problem:

Here's the 8 step Business Analyst way to solve the problem:



Figure 1.4: 8 step Business Analyst way to solve the problem:

1. **Identify the problem** — ‘What’s going on?’

This basic question will help the BA get an idea about problems being faced in production. As a BA, you should always keep an eye on production issues.

2. **Gather data** — ‘What do we know about the problem?’

This helps to gather the data with regards to the issue. The data should be collected for a period of 3 to 6 months on the reported problem, as well as linked issues.

3. **Analyze data** — ‘What are the root causes?’

Identifying the root cause is the next activity in analyzing the data. Proper root cause analysis will help in subsequent phases of problem-solving.

4. **Identify solutions** — ‘What could be the solution?’

This is the most important step in the problem-solving phase. Based on root causes, the BA needs to work with the architect team and look out for possible solutions available in-house or

look for solution in-market. If need be, do the comparison between in-house solutions and the products available in the market.

5. Define solution — ‘What’s the best solution?’

Once available solutions are put into perspective, the next step is to define the best solution to implement. At this stage, POC (Proof of concept) of the solution can help to finalize the solution.

6. Plan implementation — ‘What’s the best approach?’

This is predominantly a project management activity where the BA needs to work alongside the Project Manager to chalk up the plan and define the approach.

7. Test solution — ‘Is the problem solved successfully?’

Test review of the solution implemented in production should be done at this stage. This should be a continuous process to test the solution until you get the confidence of a successful resolution of the problem.

8. Improve solution — ‘Is there any further scope of Improvement?’

This is a retrospective to verify if there is any further scope for improvement.

1.5 DATA COLLECTION

Data collection is the methodological process of gathering information about a specific subject.

Data collection is a part of nearly every research project. You can do it in many different ways, and the best method to use often depends on the type of data you are collecting. And speaking of the type of data, generally, there are three types of consumer data:

1. **First-party data.** Collected directly from users by your organization.
2. **Second-party data.** Data shared by another organization about their customers.
3. **Third-party data.** Aggregated and rented or sold data by organizations.

Data can be qualitative (meaning contextual in nature) or quantitative (meaning numeric in nature). Many data collection methods apply to either type, but some are better suited to one over the other.

1.5.1 Quantitative vs Qualitative Data Collection

The **data collection methods in research** can be divided into two main categories: qualitative and quantitative.

Qualitative data is descriptive in nature and deals with non-numerical information. This data type is used to understand concepts, feelings, or points of view. It allows analysts to get at the heart of an issue and understand its behavior or motivations. Qualitative data is typically collected through interviews, focus groups, or observations.

1.5.2 Purpose of data collection

The main purpose of data collection is to use it to glean actionable insights that can help the organization to succeed in terms of better productivity, faster results, lesser time spent and greater profit ratio and finally improved return on investment.

Diving more into the specificity of data collection, here are some of the points why an organization must keep in mind the purpose of collecting accurate data:

- 1) Collect data to improvise on data strategies
- 2) Collect data to probe to the right customer
- 3) Collect data to make processes more efficient and effective
- 4) Collect data to get deep into facts
- 5) Collect data to reduce risk

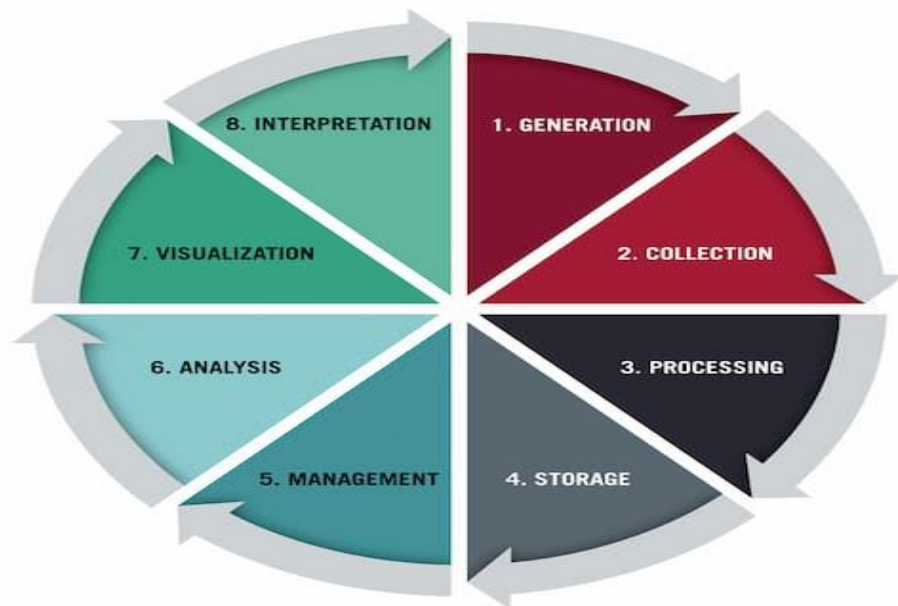


Figure 1.5: Data Life Cycle

Before collecting data, there are several factors you need to define:

- The question you aim to answer
- The data subject(s) you need to collect data from
- The collection timeframe
- The data collection method(s) best suited to your needs

1.5.3 7 DATA COLLECTION METHODS USED IN BUSINESS ANALYTICS

1. Surveys

Surveys are physical or digital questionnaires that gather both qualitative and quantitative data from subjects. One situation in which you might conduct a survey is gathering attendee feedback after an event. This can provide a sense of what attendees enjoyed, what they wish was different, and areas in which you can improve or save money during your next event for

2. Transactional Tracking

Each time your customers make a purchase, tracking that data can allow you to make decisions about targeted marketing efforts and understand your customer base better.

3. Interviews and Focus Groups

Interviews and focus groups consist of talking to subjects face-to-face about a specific topic or issue. Interviews tend to be one-on-one, and focus groups are typically made up of several people. You can use both to gather qualitative and quantitative data.

4. Observation

Observing people interacting with your website or product can be useful for data collection because of the candor it offers. If your user experience is confusing or difficult, you can witness it in real-time.

5. Online Tracking

To gather behavioral data, you can implement pixels and cookies. These are both tools that track users' online behavior across websites and provide insight into what content they're interested in and typically engage with.

6. Forms

Online forms are beneficial for gathering qualitative data about users, specifically demographic data or contact information. They're relatively inexpensive and simple to set up, and you can use them to gate content or registrations, such as webinars and email newsletters.

7. Social Media Monitoring

Monitoring your company's social media channels for follower engagement is an accessible way to track data about your audience's interests and motivations. Many social media platforms have analytics built in, but there are also third-party social platforms that give more detailed, organized insights pulled from multiple channels.

1.5.4 The Advantages and Disadvantages of Data Collection Methods

Each data collection method has its advantages and disadvantages. On the one hand, some **benefits of qualitative data** include:

- Allowing for in-depth, rich descriptions
- Providing context and background information
- Helping to generate new ideas or hypotheses
- Allowing for flexibility and changes to be made during the data collection process

1.6 DATA PREPARATION

Data preparation is the process of gathering, combining, structuring and organizing data so it can be used in business intelligence (BI), analytics and data visualization applications.

The components of data preparation include data preprocessing, profiling, cleansing, validation and transformation; it often also involves pulling together data from different internal systems and external sources.

Data preparation work is done by information technology (IT), BI and data management teams as they integrate data sets to load into a data warehouse, NoSQL database or data lake repository, and then when new analytics applications are developed with those data sets.

1.6.1 Purposes of data preparation

One of the primary purposes of data preparation is to ensure that raw data being readied for processing and analysis is accurate and consistent so the results of BI and analytics applications will be valid.

Data is commonly created with missing values, inaccuracies or other errors, and separate data sets often have different formats that need to be reconciled when they're combined. Correcting data errors, validating data quality and consolidating data sets are big parts of data preparation projects.

Data preparation also involves finding relevant data to ensure that analytics applications deliver meaningful information and actionable insights for business decision-making.

1.6.2 What are the benefits of data preparation?

Data scientists often complain that they spend most of their time gathering, cleansing and structuring data instead of analyzing it. A big benefit of an effective data preparation process is that they and other end users can focus more on data mining and data analysis -- the parts of their job that generate business value. For example, data preparation can be done more quickly, and prepared data can automatically be fed to users for recurring analytics applications.

Done properly, data preparation also helps an organization do the following:

- ensure the data used in analytics applications produces reliable results;
- identify and fix data issues that otherwise might not be detected;
- enable more informed decision-making by business executives and operational workers;
- reduce data management and analytics costs;
- avoid duplication of effort in preparing data for use in multiple applications; and
- get a higher ROI from BI and analytics initiatives.

1.6.3 Steps in the data preparation process

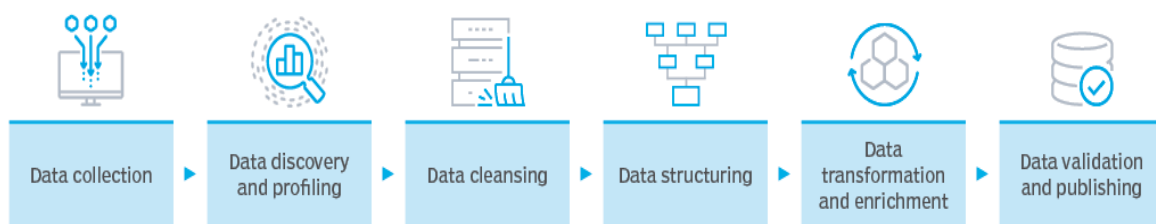


Figure 1.6: Steps in the data preparation process

Data preparation is done in a series of steps. There's some variation in the data preparation steps listed by different data professionals and software vendors, but the process typically involves the following tasks:

1. **Data collection.** Relevant data is gathered from operational systems, data warehouses, data lakes and other data sources. During this step, data scientists, members of the BI team,

other data professionals and end users who collect data should confirm that it's a good fit for the objectives of the planned analytics applications.

2. **Data discovery and profiling.** The next step is to explore the collected data to better understand what it contains and what needs to be done to prepare it for the intended uses. To help with that, data profiling identifies patterns, relationships and other attributes in the data, as well as inconsistencies, anomalies, missing values and other issues so they can be addressed.
3. **Data cleansing.** Next, the identified data errors and issues are corrected to create complete and accurate data sets. For example, as part of cleansing data sets, faulty data is removed or fixed, missing values are filled in and inconsistent entries are harmonized.
4. **Data structuring.** At this point, the data needs to be modeled and organized to meet the analytics requirements. For example, data stored in comma-separated values (CSV) files or other file formats has to be converted into tables to make it accessible to BI and analytics tools.
5. **Data transformation and enrichment.** In addition to being structured, the data typically must be transformed into a unified and usable format. For example, data transformation may involve creating new fields or columns that aggregate values from existing ones. Data enrichment further enhances and optimizes data sets as needed, through measures such as augmenting and adding data.
6. **Data validation and publishing.** In this last step, automated routines are run against the data to validate its consistency, completeness and accuracy. The prepared data is then stored in a data warehouse, a data lake or another repository and either used directly by whoever prepared it or made available for other users to access.

1.6.4 What are the challenges of data preparation?

Data preparation is inherently complicated. Data sets pulled together from different source systems are highly likely to have numerous data quality, accuracy and consistency issues to resolve. The data also must be manipulated to make it usable, and irrelevant data needs to be weeded out. As noted above, it's a time-consuming process: The 80/20 rule is often applied to analytics applications, with about 80% of the work said to be devoted to collecting and preparing data and only 20% to analyzing it.

In an article on [common data preparation challenges](#), Rick Sherman, managing partner of consulting firm Athena IT Solutions, detailed the following seven challenges along with advice on how to overcome each of them:

- **Inadequate or nonexistent data profiling.** If data isn't properly profiled, errors, anomalies and other problems might not be identified, which can result in flawed analytics.
- **Missing or incomplete data.** Data sets often have missing values and other forms of incomplete data; such issues need to be assessed as possible errors and addressed if so.
- **Invalid data values.** Misspellings, other typos and wrong numbers are examples of invalid entries that frequently occur in data and must be fixed to ensure analytics accuracy.
- **Name and address standardization.** Names and addresses may be inconsistent in data from different systems, with variations that can affect views of customers and other entities.
- **Inconsistent data across enterprise systems.** Other inconsistencies in data sets drawn from multiple source systems, such as different terminology and unique identifiers, are also a pervasive issue in data preparation efforts.
- **Data enrichment.** Deciding how to enrich a data set -- for example, what to add to it -- is a complex task that requires a strong understanding of business needs and analytics goals.
- **Maintaining and expanding data prep processes.** Data preparation work often becomes a recurring process that needs to be sustained and enhanced on an ongoing basis

1.6.5 Data preparation tools

BI, analytics and data management vendors that offer data preparation tools or capabilities include the following:

- ✓ Altair
- ✓ Boomi
- ✓ Datameer
- ✓ DataRobot

- ✓ IBM
- ✓ Informatica
- ✓ Microsoft
- ✓ Precisely
- ✓ SAP
- ✓ SAS
- ✓ Tableau
- ✓ Talend
- ✓ Tamr
- ✓ Tibco Software

1.7 HYPOTHESIS GENERATION

Hypothesis are statement about the given problem. Hypothesis testing is a statistical method that is used in making a statistical decision using experimental data. Hypothesis testing is basically an assumption that we make about a population parameter. It evaluates two mutually exclusive statements about a population to determine which statement is best supported by the sample data.

There are different kinds of hypotheses. They are:

- **Simple Hypothesis:** Simple hypothesis, also known as a basic hypothesis, proposes that an independent variable is accountable for the corresponding dependent variable. In simpler words, the occurrence of independent variable results in the existence of the dependent variable. Generally, simple hypotheses are thought of as true and they create a causal relationship between the two variables. One example of a simple hypothesis is smoking cigarettes daily leads to cancer.
- **Complex Hypothesis:** This type of hypothesis is also termed a modal. It holds for the relationship between two variables that are independent and result in a dependent variable. This means that the amalgamation of independent variables results in the dependent variables. An example of this kind of hypothesis can be “adults who don’t drink and smoke are less likely to have liver-related problems.
- **Null Hypothesis:** A null hypothesis is created when a researcher thinks that there is no connection between the variables that are being observed. An example of this kind of hypothesis can be “A student’s performance is not impacted if they drink tea or coffee before classes.
- **Alternative Hypothesis:** If a researcher wants to disapprove of a null hypothesis, then the researcher has to develop an opposite assumption—known as an alternative hypothesis. For example, beginning your day with tea instead of coffee can keep you more alert.
- **Logical Hypothesis:** A proposed explanation supported by scant data is called a logical hypothesis. Generally, you wish to test your hypotheses or postulations by converting a logical

hypothesis into an empirical hypothesis. For example, waking early helps one to have a productive day.

- **Empirical Hypothesis:** This type of hypothesis is based on real evidence, evidence that is verifiable by observation as opposed to something that is correct in theory or by some kind of reckoning or logic. This kind of hypothesis depends on various variables that can result in specific outcomes. For example, individuals eating more fish can run faster than those eating meat.
- **Statistical Hypothesis:** This kind of hypothesis is most common in systematic investigations that involve a huge target audience. For example, in Louisiana, 45% of students have middle-income parents.

Four Steps of Hypothesis Testing

There are four main steps in **hypothesis testing in business analytics**:

Step 1: State the Null and Alternate Hypothesis

After the initial research hypothesis, it is essential to restate it as a null (H_0) hypothesis and an alternate (H_a) hypothesis so that it can be tested mathematically.

Step 2: Collate Data

For a test to be valid, it is essential to do some sampling and collate data in a manner designed to test the hypothesis. If your data are not representative, then statistical inferences cannot be made about the population you are trying to analyze.

Step 3: Perform a Statistical Test

Various statistical tests are present, but all of them depend on the contrast of within-group variance (how to spread out the data in a group) against between-group variance (how dissimilar the groups are from one another).

Step 4: Decide to Reject or Accept Your Null Hypothesis

Based on the result of your statistical test, you need to decide whether you want to accept or reject your null hypothesis.

1.7.1 Hypothesis Testing in Business

When we talk about data-driven decision-making, a specific amount of risk can deceive a professional. This could result from flawed observations or thinking inaccurate or incomplete information, or unknown variables. The threat over here is that if key strategic decisions are made on incorrect insights, it can lead to catastrophic outcomes for an organization.

Key Considerations for Hypothesis Testing

Let us look at the following key considerations of hypothesis testing:

- **Alternative Hypothesis and Null Hypothesis:** If a researcher wants to disapprove of a null hypothesis, then the researcher has to develop an opposite assumption—known as an alternative hypothesis. A null hypothesis is created when a researcher thinks that there is no connection between the variables that are being observed.
- **Significance Level and P-Value:** The statistical significance level is generally expressed as a p-value that lies between 0 and 1. The lesser the p-value, the more it suggests that you reject the null hypothesis. A p-value of less than 0.05 (generally ≤ 0.05) is significant statistically.
- **One-Sided vs. Two-Sided Testing:** One-sided tests suggest the possibility of an effect in a single direction only. Two-sided tests test for the likelihood of the effect in two directions—negative and positive. One-sided tests comprise more statistical power to identify an effect in a single direction than a two-sided test with the same significance level and design.
- **Sampling:** For **hypothesis testing**, you are required to collate a sample of data that has to be examined. In hypothesis testing, an analyst can test a statistical sample with the aim of providing proof of the credibility of the null hypothesis. Statistical analysts can test a hypothesis by examining and measuring a random sample of the population that is being examined.

1.7.2 What is Hypothesis Generation?

Hypothesis generation is an educated “*guess*” of various factors that are impacting the business problem that needs to be solved using machine learning. In framing a hypothesis, the data scientist must not know the outcome of the hypothesis that has been generated based on any evidence.

“A hypothesis may be simply defined as a guess. A scientific hypothesis is an intelligent guess.” – Isaac Asimov

Hypothesis Generation vs. Hypothesis Testing

This is a very common mistake data science beginners make.

Hypothesis generation is a process beginning with an educated guess whereas hypothesis testing is a process to conclude that the educated guess is true/false or the relationship between the variables is statistically significant or not.

How Does Hypothesis Generation Help?

Here are 5 key reasons why hypothesis generation is so important in data science:

- Hypothesis generation helps in comprehending the business problem as we dive deep in inferring the various factors affecting our target variable
- You will get a much better idea of what are the major factors that are responsible to solve the problem
- Data that needs to be collected from various sources that are key in converting your business problem into a data science-based problem
- Improves your domain knowledge if you are new to the domain as you spend time understanding the problem
- Helps to approach the problem in a structured manner

1. Distance/Speed based Features

Let us try to come up with a formula that would have a relation with trip duration and would help us in generating various hypotheses for the problem:

$\text{TIME} = \text{DISTANCE} / \text{SPEED}$
--

Distance and speed play an important role in predicting the trip duration.

We can notice that the trip duration is directly proportional to the distance traveled and inversely proportional to the speed of the taxi. Using this we can come up with a hypothesis based on distance and speed.

- **Distance**: More the distance traveled by the taxi, the more will be the trip duration.
- **Interior drop point**: Drop points to congested or interior lanes could result in an increase in trip duration
- **Speed**: Higher the speed, the lower the trip duration

2. Features based on Car

Cars are of various types, sizes, brands, and these features of the car could be vital for commute not only on the basis of the safety of the passengers but also for the trip duration. Let us now generate a few hypotheses based on the features of the car.

- **Condition of the car**: Good conditioned cars are unlikely to have breakdown issues and could have a lower trip duration
- **Car Size**: Small-sized cars (Hatchback) may have a lower trip duration and larger-sized cars (XUV) may have higher trip duration based on the size of the car and congestion in the city

3. Type of the Trip

Trip types can be different based on trip vendors – it could be an outstation trip, single or pool rides. Let us now define a hypothesis based on the type of trip used.

- **Pool Car**: Trips with pooling can lead to higher trip duration as the car reaches multiple places before reaching your assigned destination

4. Features based on Driver Details

A driver is an important person when it comes to commute time. Various factors about the driver can help in understanding the reason behind trip duration and here are a few hypotheses this.

- **Age of driver**: Older drivers could be more careful and could contribute to higher trip duration
- **Gender**: Female drivers are likely to drive slowly and could contribute to higher trip duration
- **Driver experience**: Drivers with very less driving experience can cause higher trip duration
- **Medical condition**: Drivers with a medical condition can contribute to higher trip duration

5. *Passenger details*

Passengers can influence the trip duration knowingly or unknowingly. We usually come across passengers requesting drivers to increase the speed as they are getting late and there could be other factors to hypothesize which we can look at.

- **Age of passengers**: Senior citizens as passengers may contribute to higher trip duration as drivers tend to go slow in trips involving senior citizens
- **Medical conditions or pregnancy**: Passengers with medical conditions contribute to a longer trip duration
- **Emergency**: Passengers with an emergency could contribute to a shorter trip duration
- **Passenger count**: Higher passenger count leads to shorter duration trips due to congestion in seating

6. Date-Time Features

The day and time of the week are important as New York is a busy city and could be highly congested during office hours or weekdays. Let us now generate a few hypotheses on the date and time-based features.

Pickup Day:

- Weekends could contribute to more outstation trips and could have a higher trip duration
- Weekdays tend to have higher trip duration due to high traffic
- If the pickup day falls on a holiday then the trip duration may be shorter
- If the pickup day falls on a festive week then the trip duration could be lower due to lesser traffic

Time:

- Early morning trips have a lesser trip duration due to lesser traffic
- Evening trips have a higher trip duration due to peak hours

7. Road-based Features

Roads are of different types and the condition of the road or obstructions in the road are factors that can't be ignored. Let's form some hypotheses based on these factors.

- **Condition of the road:** The duration of the trip is more if the condition of the road is bad
- **Road type:** Trips in concrete roads tend to have a lower trip duration
- **Strike on the road:** Strikes carried out on roads in the direction of the trip causes the trip duration to increase

8. Weather Based Features

Weather can change at any time and could possibly impact the commute if the weather turns bad. Hence, this is an important feature to consider in our hypothesis.

- **Weather at the start of the trip:** Rainy weather condition contributes to a higher trip duration

1.8 MODELING

A Business Model can be defined as a representation of a business or solution that often include a graphic component along with supporting text and relationships to other components. For example, if we have to understand a company's business model, then we would like to study the following areas like –

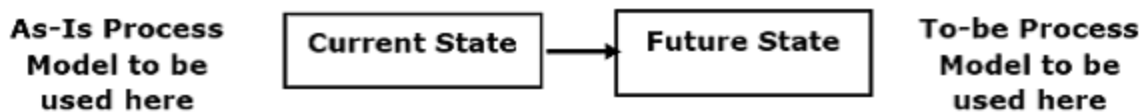
- Core values of the company
- What it serves?
- What is sets apart?

- Its key resources
- Major relationships
- Its delivery channels

1.8.1 Purpose of Business Modelling

Business modelling is used to design current and future state of an enterprise. This model is used by the Business Analyst and the stakeholders to ensure that they have an accurate understanding of the current “As-Is” model of the enterprise.

It is used to verify if, stakeholders have a shared understanding of the proposed “To-be of the solution.



1.8.2 9 essential business analysis models for the BA’s toolbox

The role of the business analyst (BA) requires not only sound data analysis but effective data modeling. The right visual model streamlines data and makes it easy for stakeholders at every level to identify and understand project strategies, relationships, and responsibilities.

In the past, data modeling meant writing out processes in plain text or sketching out a basic diagram.

1.8.3 What is a business analysis model?

Simply put, a business analysis model outlines the steps a business takes to complete a specific process, such as ordering a product or onboarding a new hire. Process modeling (or mapping) is key to improving process efficiency, training, and even complying with industry regulations.

Because there are many different kinds of processes, organizations, and functions within a business, BAs employ a variety of visual models to map and analyze data.

Take a look at these nine essential business analysis models to include in your toolbox.

1. Activity diagrams

An activity diagram is a type of UML behavioural diagram that describes what needs to happen in a system. They are particularly useful for communicating process and procedure to stakeholders from both the business and development teams.

A BA might use a UML diagram tool to create an activity diagram to map the process of logging in to a website or completing a transaction like withdrawing or depositing money.

ATM Withdrawal Activity Diagram

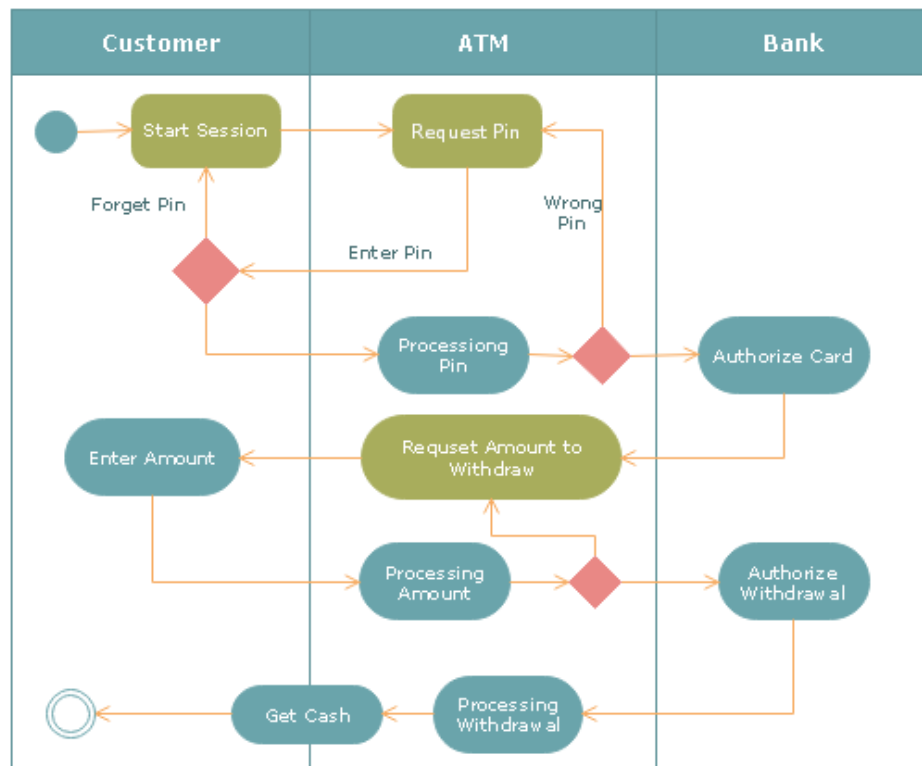


Figure 1.7: Activity Diagram for ATM

2. Feature mind maps

Business diagrams aren't just for late-stage analysis or documentation. They are also useful during a project's initial brainstorming phase. Feature mind maps help BAs organize the sometimes messy brainstorm process so that ideas, concerns, and requests are clearly captured and categorized.

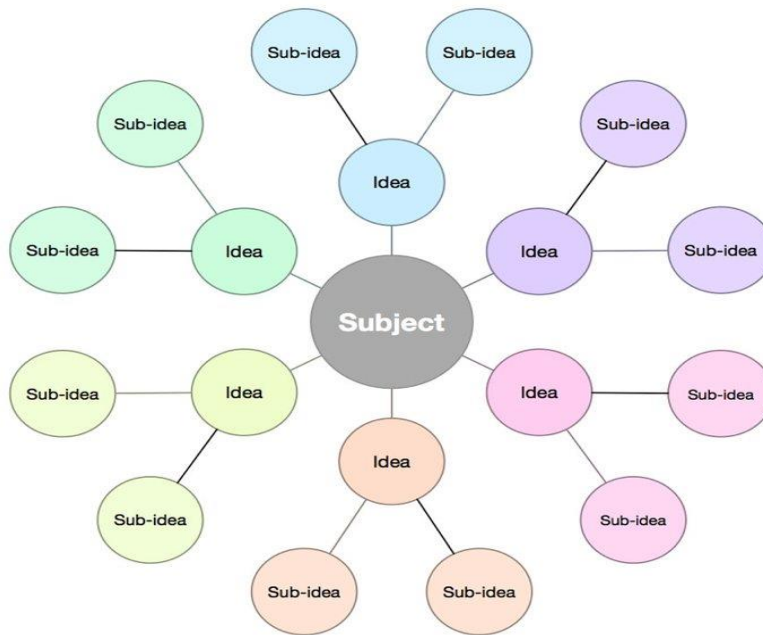


Figure 1.8: Basic Mind Map

3. Product roadmaps

A product (or feature) roadmap outlines the development and launches of a product and its features. They are a focused analysis of a product's evolution, which helps developers and other stakeholders focus on initiatives that add direct value to the user.

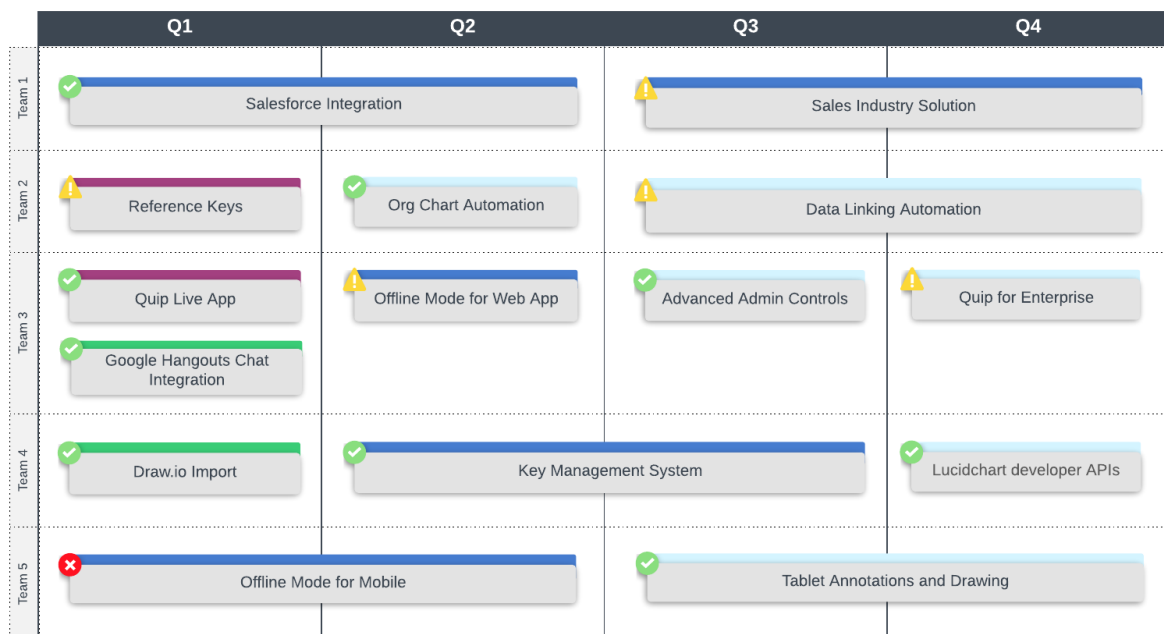


Figure 1.9: Product Roadmap Example

4. Organizational charts

An organizational chart outlines the hierarchy of a business or one of its departments or teams. They are especially helpful reference charts for employees to quickly understand how the company is organized and identify key stakeholders and points of contact for projects or queries.

Additionally, organizational charts prove useful for stakeholder analysis and modeling new groupings and teams following organizational shifts.

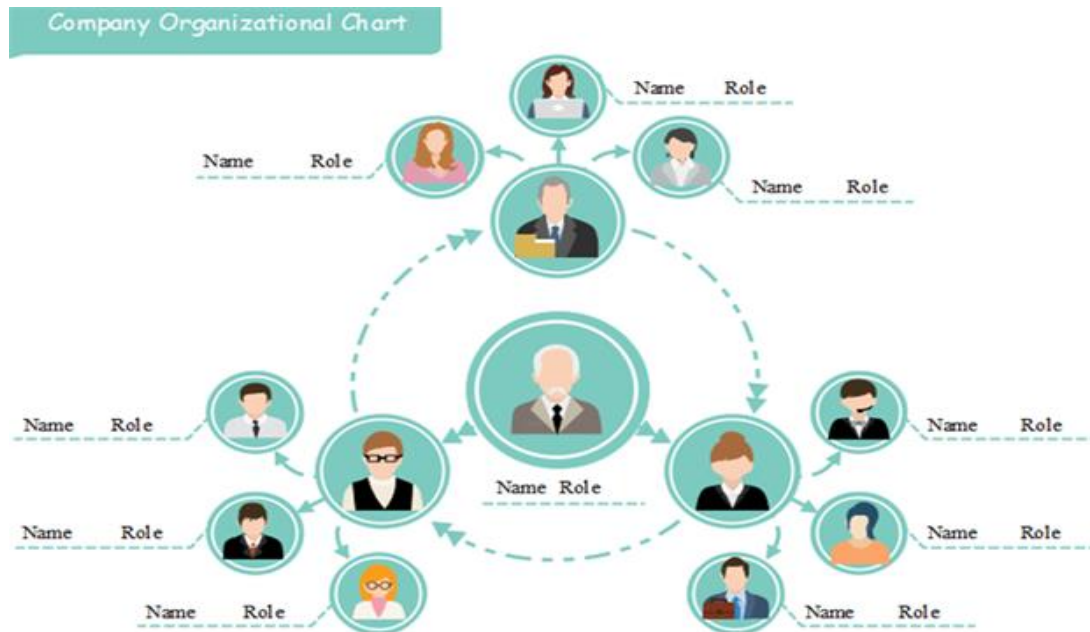


Figure 1.10: Org Chart

5. SWOT analysis

The SWOT analysis is a fundamental tool in a BA's arsenal. SWOT stands for strengths, weaknesses, opportunities, and threats. A SWOT analysis evaluates a business's strengths and weaknesses and identifies any opportunities or threats to that business.

SWOT analysis helps stakeholders make strategic decisions regarding their business. The goal is to capitalize on strengths and opportunities while reducing the impact of internal or external threats and weaknesses.



Figure 1.11: SWOT Analysis

6. User interface wireframe

Another essential business diagram is the UI wireframe. Software development teams use wireframes (also called mockups or prototypes) to visually outline and design a layout for a specific screen. In other words, wireframes are the blueprints for a website or software program. They help stakeholders assess navigational needs and experience for a successful practical application.

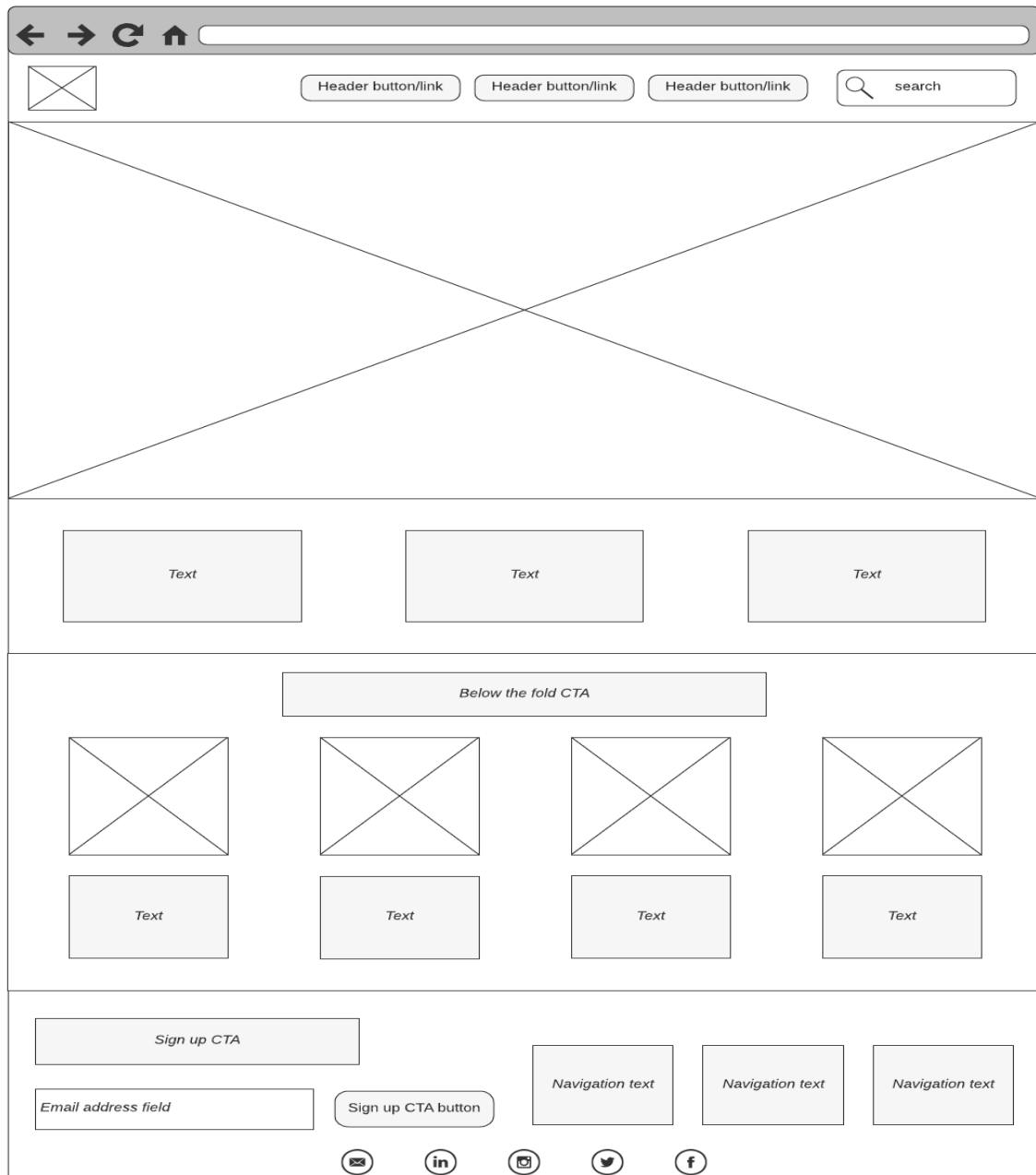


Figure 1.12: Basic Homepage Wireframe

7. Process flow diagram

A process flow diagram (PFD) is typically used in chemical and process engineering to identify the basic flow of processes

A PFD is best used to:

- Document a process.
- Study a process to make changes or improvements.
- Improve understanding and communication between stakeholders.

These diagrams focus on broad, high-level systems rather than annotating minor process details.

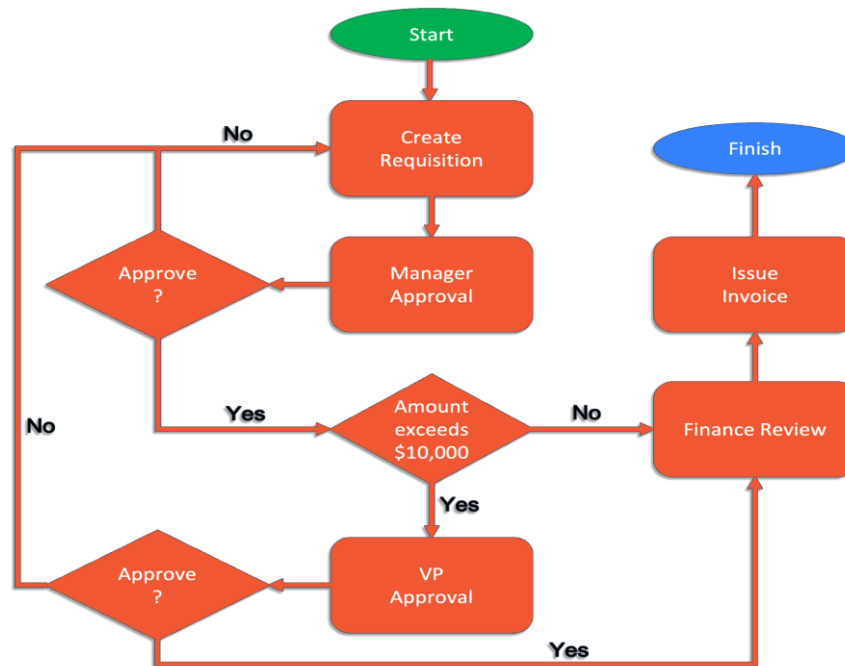


Figure 1.14: Business Process Flow Diagram

8. PESTLE analysis

A PESTLE analysis often goes hand-in-hand with a SWOT analysis. PESTLE evaluates external factors that could impact business performance. This acronym stands for six elements affecting business: political, economic, technological, environmental, legal, and sociological.

PESTLE analysis assesses the possible factors within each category, as well as their potential impact, duration of effect, type of impact (i.e., negative or positive), and level of importance.

This type of business analysis helps stakeholders manage risk, strategically plan and review business goals and performance, and potentially gain an advantage over competitors.

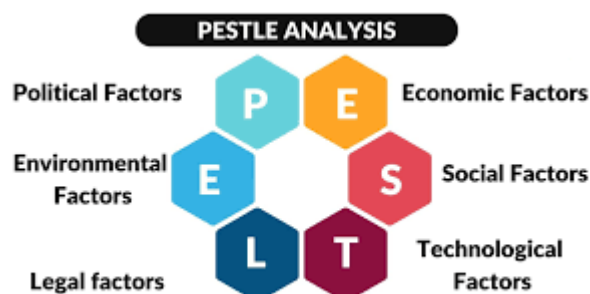


Figure 1.15: PESTLE Analysis Diagram

9. Entity-relationship diagram

An entity-relationship diagram (ER diagram) illustrates how entities (e.g., people, objects, or concepts) relate to one another in a system. For example, a logical ER diagram visually shows how the terms in an organization's business glossary relate to one another.

ER diagrams comprise three main parts:

- Entities
- Relationships
- Attributes

Attributes apply to the entities, describing further details about the concept. Relationships are where the key insights from ER diagrams arise. In a visual model, the relationships between entities are illustrated either numerically or via crow's foot notation.

These diagrams are most commonly used to model database structures in software engineering and business information systems and are particularly valuable tools for BAs in those fields.

you can automatically generate an ERD when you import your databases and schemas. Follow this guide to try it out or manually build your ERD with the template below!

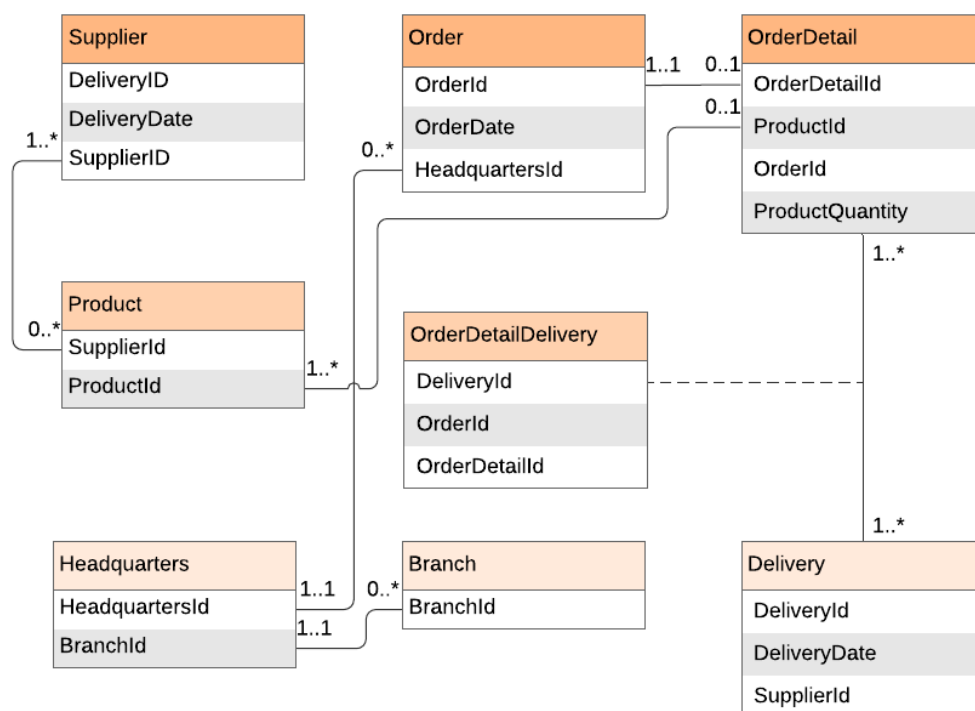


Figure 1.16: Entity-Relationship Diagram Example (UML Notation)

Some of the popular Business Modelling Tools used by large organizations in IT environments:

- ✓ Microsoft Visio
- ✓ Enterprise Architect
- ✓ Rational Requisite Pro

1.9 VALIDATION AND EVALUATION

VALIDATION

Validation is the process of confirming the completeness and correctness of requirements. Validation also ensures that the requirements:

- 1) achieve stated business objectives,
- 2) meet the needs of stakeholders, and
- 3) are clear and understood by the developers.

- ❖ Validation is essential to identification of missing requirements and to ensure that the requirements meet certain quality characteristics.

Validation addresses each individual requirement to ensure that it is:

Correct – accurately states a customer or external need

Clear – has only one possible meaning

Feasible – can be implemented within known constraints

Modifiable – can be easily changed, with history, when necessary

Necessary – documents something customers really need

Prioritized – ranked as to importance of inclusion in product

Traceable – can be linked to system requirements, and to designs, code, and tests

Verifiable – correct implementation can be determined by testing, inspection, analysis, or demonstration

1.9.1 The Four Elements

Model validation consists of four crucial elements which should be considered:

1. Conceptual Design

The foundation of any model validation is its conceptual design, which needs documented coverage assessment that supports the model's ability to meet business and regulatory needs and the unique risks facing a bank.

The design and capabilities of a model can have a profound effect on the overall effectiveness of a bank's ability to identify and respond to risks. For example, a poorly designed risk assessment model may result in a bank establishing relationships with clients that present a risk that is greater than its risk appetite, thus exposing the bank to regulatory scrutiny and reputation damage.

A validation should independently challenge the underlying conceptual design and ensure that documentation is appropriate to support the model's logic and the model's ability to achieve desired regulatory and business outcomes for which it is designed.

2. System Validation

All technology and automated systems implemented to support models have limitations. An effective validation includes: firstly, evaluating the processes used to integrate the model's conceptual design and functionality into the organisation's business setting; and, secondly, examining the processes implemented to execute the model's overall design. Where gaps or limitations are observed, controls should be evaluated to enable the model to function effectively.

3. Data Validation and Quality Assessment

Data errors or irregularities impair results and might lead to an organisation's failure to identify and respond to risks. Best practise indicates that institutions should apply a risk-based data validation, which enables the reviewer to consider risks unique to the organisation and the model.

4. Process Validation

To verify that a model is operating effectively, it is important to prove that the established processes for the model's ongoing administration, including governance policies and procedures, support the model's sustainability. A review of the processes also determines whether the models are producing output that is accurate, managed effectively, and subject to the appropriate controls.

1.9.2 Evaluation

- Evaluation is an integral part of the development process. It helps to find the best model that represents our data and how well the chosen model will work in the future. Evaluating model performance with the data used for training is not acceptable in data science because it can easily generate overoptimistic and overfitted models. There are two methods of evaluating models in data science, Hold-Out and Cross-Validation. To avoid overfitting, both methods use a test set (not seen by the model) to evaluate model performance.
- **Hold-Out:** In this method, the mostly large dataset is *randomly* divided to three subsets:
 1. **Training set** is a subset of the dataset used to build predictive models.
 2. **Validation set** is a subset of the dataset used to assess the performance of model built in the training phase. It provides a test platform for fine tuning model's parameters and selecting the best-performing model. Not all modelling algorithms need a validation set.
 3. **Test set** or unseen examples is a subset of the dataset to assess the likely future performance of a model. If a model fit to the training set much better than it fits the test set, overfitting is probably the cause.

- **Cross-Validation:** When only a limited amount of data is available, to achieve an unbiased estimate of the model performance we use k -fold cross-validation. In k -fold cross-validation, we divide the data into k subsets of equal size. We build models k times, each time leaving out one of the subsets from training and use it as the test set. If k equals the sample size, this is called “leave-one-out”.

Evaluation can be divided to **two sections:**

- **Classification Evaluation**
- **Regression Evaluation**

There have been four main spheres where business analytics has evolved greatly, these are:

- Artificial Intelligence and Automated Analytics
- Predictive Analytics
- Real-time Analytics
- Big Data

1.10 INTERPRETATION

1.10.1 What Is Data Interpretation?

Data interpretation refers to the process of using diverse analytical methods to review data and arrive at relevant conclusions. The interpretation of data helps researchers to categorize, manipulate, and summarize the information in order to answer critical questions.

The data interpretation definition in business terms is the implementation of different processes in which data is analysed and revised with the purpose of gaining insights and recognising emerging patterns and behaviours. These conclusions are helpful for you as a manager to make an informed decision, backed by numbers while having all the facts at your disposal. Data doesn't come from a single source but multiple diverse sources.

Data analysis is always subjective and connected to the goals and the nature of the industry. For example, even if you're using the same amount and the same type of data, a food chain and a fashion store will analyse it differently and gain a different awareness from it. Additionally, every business has its objective, and the interpretation of the results is going to produce a report that is uniquely qualified for them.

Data analysis has two main classifications, which we are going to elaborate on in this article: the “quantitative analysis” and the “qualitative analysis.”

However, before we continue, we also have to take a look at the scales of measurement because they decide the continuing effect the data is going to have on your decision making.

1. **Nominal scale** – It incorporates variables, which are exclusive and You can’t rank or compare with non-numeric classifications.
2. **Ordinal scale** – It incorporates variables which are exclusive and extensive but in a logical order (online quizzes and surveys are usually done in an ordinal scale).
3. **Interval scale** – With an arbitrary zero point, which systematically integrates the categories and groups them in equal distances.
4. **Ratio** – A combination of all of the above.

1.10.2 Qualitative data interpretation

A qualitative analysis is not done by running numbers but by classifying data in categories. It’s mostly used for observations, interviews, documents, surveys, etc. This type of research is intended to gain insights by using a person-to-person method for data gathering.

For example, when you complete an online survey, the answers are rated in the following manner:

- I strongly agree
- I agree
- I’m neutral
- I disagree
- I strongly disagree

The best feature about the qualitative data interpretation is that the findings are grouped into topics and categories, which makes it easier to notice trends and collect the data. A report created from a qualitative analysis tends to be more readable for the layman.

1.10.3 Quantitative data interpretation

If the qualitative analysis was contextual, the quantitative analysis is numerical. With quantitative data interpretation, you analyse the numbers in the data to gain insights, and that is achieved with statistical modelling. There are three types of statistical modelling:

- Mean
- Standard deviation
- Frequency distribution

The most common technique of quantitative analysis is running tests on two or more significant variables, which are later processed together or separately and in the end, are compared to one another to sum up a report.

There are four steps to data interpretation:

- 1) Gather Data: assemble the information you'll need,

- 2) Develop your discoveries
- 3) Draw Conclusions
- 4) Develop recommendations

Gather the data

The very first step in data interpretation is gathering all relevant data. You can do this by first visualizing it in a bar, graph, or pie chart. This step aims to analyze the data accurately and without bias. Now is the time to recall how you conducted your research.

Here are two question patterns that will help you to understand better.

- Were there any flaws or changes that occurred during the data collection process?
- Have you saved any observatory notes or indicators?

You can proceed to the next stage when you have all of your data.

Develop your discoveries

This is a summary of your findings. Here, you thoroughly examine the data to identify trends, patterns, or behavior. If you are researching a group of people using a sample population, this is the section where you examine behavioral patterns. You can compare these deductions to previous data sets, similar data sets, or general hypotheses in your industry. This step's goal is to compare these deductions before drawing any conclusions.

Draw Conclusions

After you've developed your findings from your data sets, you can draw conclusions based on your discovered trends. Your findings should address the questions that prompted your research. If they do not respond, inquire about why; it may produce additional research or questions.

Draw recommendations

The interpretation procedure of data comes to a close with this stage. Every research conclusion must include a recommendation. As recommendations are a summary of your findings and conclusions, they should be brief. There are only two options for recommendations; you can either recommend a course of action or suggest additional research.

1.11 DEPLOYMENT AND ITERATION

1.11.1 Deployment:

The use of something or someone, especially in order to achieve a particular effect.

The deployment phase is the final phase of any IT or SW development project, including those that are exploring Advanced Analytics capabilities.

The main objective is to ensure that the final solution is ready to be used within the operational environment and that end users have all the required tools to act upon the analytical insights discovered during the development phases of the project.

However, organisations, especially those that are deploying analytical initiatives for the first time, or that are still analytically “immature”, typically focus too much on building an IT infrastructure, instead of planning how to deliver actionable insights to their end users and integrate this intelligence into internal organisational processes.

The lack of vision regarding usability and access to analytical information to drive decision making, is a common mistake in Advanced Analytics projects.

At Presidion, we’ve seen organisations perform best when they take the before, during and after of the deployment of analytics initiatives into consideration. This will help to maximise return on investment and achieve a high impact on business and operations.

1.11.2 Things to consider BEFORE deployment

There is a strong need for a **pre-deployment, planning phase** to ensure that the final solution is ready to operate in production mode.

The key activities that should be performed before deploying the final solution are **Testing** and **Validation** activities and preparation of a well-defined **Transition period**.

Depending on the size and the scope of the analytical application, this can include a formal **User Acceptance Testing** (UAT) and/or a **Cutover** plan for the application, which will replace a legacy solution or existing manual processes and **Go-live**.

Before proceeding to deployment, Data scientists and Development teams should exhaustively evaluate analytical outputs to assess quality and accuracy, but most importantly to validate that business objectives are properly addressed and success criteria, that were set during project initiation, are fully met.

During the pre-deployment phase, end users should be placed at the centre of attention. **Training** and formal **Knowledge Transfer** sessions can significantly help them learn how to operate the new solution, validate the usability of the new system and ensure a smooth transition period.

Instead of deploying the technology, pre-deployment activities in Advanced Analytics projects should focus on the business problems that the new solution addresses. Data Scientists and Developers should be aware that they are deploying business, analytical insights and not technical, statistical outputs or model results. The terminology used in front end applications

should be appropriate for non-technical users to enable them to adopt and quickly use the solution.

1.11.3 Things to consider DURING deployment

At this stage, technical teams can focus on deploying the technology and integrating the solution into an operational environment to automate decision making process.

The **Technical Infrastructure** and the **Production environment**, which will host the solution, and any required **Integration Interfaces** should by now be already developed, tested and ready in order to initiate deployment and successfully incorporate analytical results into an organisation's daily operations.

Typically, deployment of Advanced Analytics insights includes all operations to generate reports and recommendations for end users, visualisation of key findings, self-service and data discovery functionalities for business users, and finally, depending on the size and scope of the analytical application, implementation of a scoring process or workflows that integrate analytical outputs (in real time or not) with custom, operational and core systems.

During deployment, many iterations, enhancements and fine-tuning activities might be necessary to finalise the deployment of the system. Other activities necessary during deployment include **Administration**, **Security** and **Authorisation**, as well as finalising **Documentation** and **Transferring Ownership** to business and operations.

1.11.4 Things to consider AFTER deployment

Market conditions, trends, policies and regulations, all change over time. **Monitoring** Advanced Analytics insights is essential to ensure that performance and accuracy is maintained over longer periods.

The goal of a **post-deployment** monitoring phase is to create the strategy and the foundations to continually monitor the solution's performance, **review** outputs, collect **feedback** from business users and address issues detected on an ongoing basis in a timely manner, without creating operational disruptions.

Model management is a very important concept towards that objective, as it is possible to systematically compare and assess analytical outputs, detect decrease in performance, and promote the best possible analytical results.

Finally, do not forget to **learn** from your previous mistakes and incorporate your end users feedback into this monitoring process to address issues either in future enhancements of the application or during regular updates to overall improve the accuracy of analytical outputs.

1.11.5 ITERATION

Iterative Model

In this Model, you can start with some of the software specifications and develop the first version of the software. After the first version if there is a need to change the software, then a new version of the software is created with a new iteration. Every release of the Iterative Model finishes in an exact and fixed period that is called iteration.

The Iterative Model allows the accessing earlier phases, in which the variations made respectively. The final output of the project renewed at the end of the Software Development Life Cycle (SDLC) process.

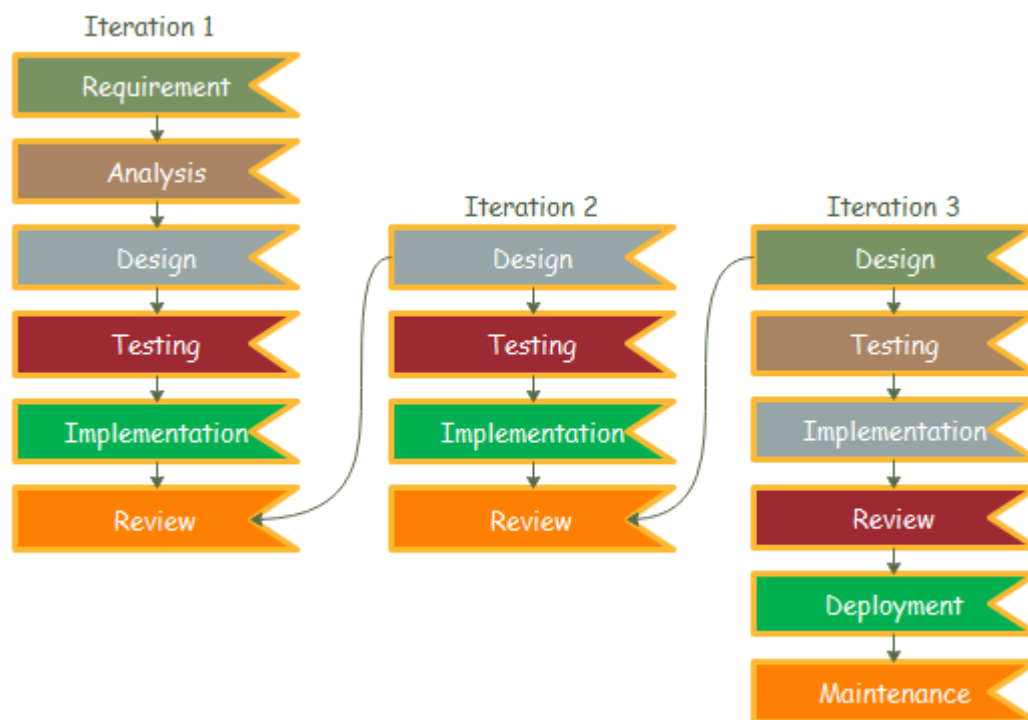


Figure 1.17: Iterative Model

The various phases of Iterative model are as follows:

- 1. Requirement gathering & analysis:** In this phase, requirements are gathered from customers and check by an analyst whether requirements will fulfil or not. Analyst checks that need will achieve within budget or not. After all of this, the software team skips to the next phase.
- 2. Design:** In the design phase, team design the software by the different diagrams like Data Flow diagram, activity diagram, class diagram, state transition diagram, etc.
- 3. Implementation:** In the implementation, requirements are written in the coding language and transformed into computer programmes which are called Software.

4. Testing: After completing the coding phase, software testing starts using different test methods. There are many test methods, but the most common are white box, black box, and grey box test methods.

5. Deployment: After completing all the phases, software is deployed to its work environment.

6. Review: In this phase, after the product deployment, review phase is performed to check the behaviour and validity of the developed product. And if there are any error found then the process starts again from the requirement gathering.

7. Maintenance: In the maintenance phase, after deployment of the software in the working environment there may be some bugs, some errors or new updates are required. Maintenance involves debugging and new addition options.

When to use the Iterative Model?

1. When requirements are defined clearly and easy to understand.
2. When the software application is large.
3. When there is a requirement of changes in future.

Advantage(Pros) of Iterative Model:

1. Testing and debugging during smaller iteration is easy.
2. A Parallel development can plan.
3. It is easily acceptable to ever-changing needs of the project.
4. Risks are identified and resolved during iteration.
5. Limited time spent on documentation and extra time on designing.

Disadvantage(Cons) of Iterative Model:

1. It is not suitable for smaller projects.
2. More Resources may be required.
3. Design can be changed again and again because of imperfect requirements.
4. Requirement changes can cause over budget.
5. Project completion date not confirmed because of changing requirements.

UNIT – 2

BUSINESS INTELLIGENCE

Data Warehouses and Data Mart – Knowledge Management – Types of Decisions – Decision Making Process – Decision Support Systems – Business Intelligence – OLAP – Analytic functions.

2. INTRODUCTION

What is Business Intelligence?

- BI(Business Intelligence) is a set of processes, architectures, and technologies that convert raw data into meaningful information that drives profitable business actions. It is a suite of software and services to transform data into actionable intelligence and knowledge.

- BI represents the technical infrastructure that collects, stores, and analyses company data.
- BI parses data and produces reports and information that help managers to make better decisions.

How Business Intelligence Works?

Companies gather and analyze the necessary data to find answers to questions and track the progress of their business goals. This gives them the insights needed to determine action points to achieve their business objectives.

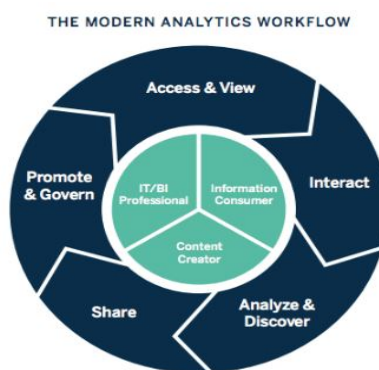


Figure 2.1 Modern Analytics workflow

How Business Intelligence systems are implemented?

Here are the steps:

Step 1) Raw Data from corporate databases is extracted. The data could be spread across multiple systems heterogeneous systems.

Step 2) The data is cleaned and transformed into the data warehouse. The table can be linked, and data cubes are formed.

Step 3) Using BI system the user can ask queries, request ad-hoc reports or conduct any other analysis.

Examples of Business Intelligence System used in Practice

Example 1:

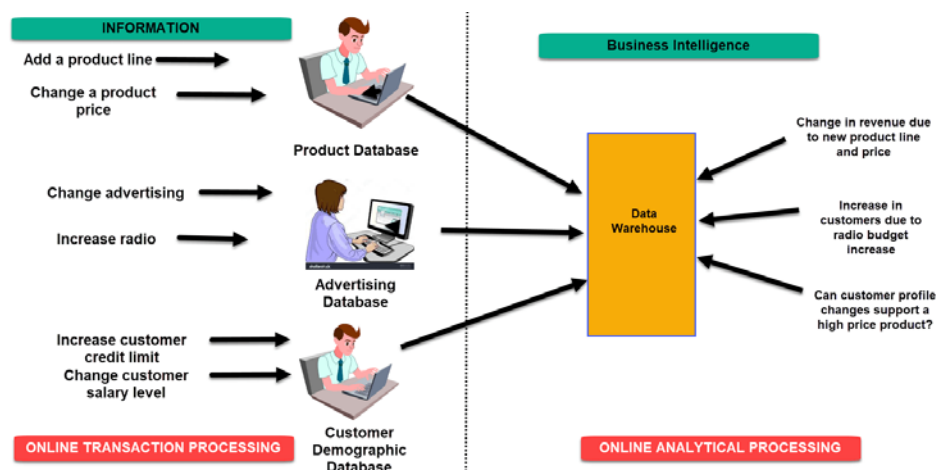


Figure 2.2 Business Intelligence System

Four types of BI users

Following given are the four key players who are used Business Intelligence System:

1. The Professional Data Analyst:

The data analyst is a statistician who always needs to drill deep down into data. BI system helps them to get fresh insights to develop unique business strategies.

2. The IT users:

The IT user also plays a dominant role in maintaining the BI infrastructure.

3. The head of the company:

CEO or CXO can increase the profit of their business by improving operational efficiency in their business.

4. The Business Users

Business intelligence users can be found from across the organization. There are mainly two types of business users

1. Casual business intelligence user
2. The power user.

Advantages of Business Intelligence

Here are some of the advantages of using Business Intelligence System:

1. Boost productivity

With a BI program, It is possible for businesses to create reports with a single click thus saves lots of time and resources. It also allows employees to be more productive on their tasks.

2. To improve visibility

BI also helps to improve the visibility of these processes and make it possible to identify any areas which need attention.

3. Fix Accountability

BI system assigns accountability in the organization as there must be someone who should own accountability and ownership for the organization's performance against its set goals.

4. It gives a bird's eye view:

BI system also helps organizations as decision makers get an overall bird's eye view through typical BI features like dashboards and scorecards.

5. It streamlines business processes:

BI takes out all complexity associated with business processes. It also automates analytics by offering predictive analysis, computer modeling, benchmarking and other methodologies.

6. It allows for easy analytics.

BI software has democratized its usage, allowing even nontechnical or non-analysts users to collect and process data quickly. This also allows putting the power of analytics from the hand's many people.

BI System Disadvantages

1. Cost:

Business intelligence can prove costly for small as well as for medium-sized enterprises. The use of such type of system may be expensive for routine business transactions.

2. Complexity:

Another drawback of BI is its complexity in implementation of datawarehouse. It can be so complex that it can make business techniques rigid to deal with.

3. Limited use

Like all improved technologies, BI was first established keeping in consideration the buying competence of rich firms. Therefore, BI system is yet not affordable for many small and medium size companies.

4. Time Consuming Implementation

It takes almost one and half year for data warehousing system to be completely implemented. Therefore, it is a time-consuming process.

Trends in Business Intelligence

The following are some business intelligence and analytics trends that you should be aware of.

Artificial Intelligence: Gartner' report indicates that AI and machine learning now take on complex tasks done by human intelligence. This capability is being leveraged to come up with real-time data analysis and dashboard reporting.

Collaborative BI: BI software combined with collaboration tools, including social media, and other latest technologies enhance the working and sharing by teams for collaborative decision making.

Embedded BI: Embedded BI allows the integration of BI software or some of its features into another business application for enhancing and extending it's reporting functionality.

Cloud Analytics: BI applications will be soon offered in the cloud, and more businesses will be shifting to this technology. As per their predictions within a couple of years, the spending on cloud-based analytics will grow 4.5 times faster..

Types of BI Tools and Software

BI tools and software come in a wide variety of forms. Let's take a quick look at some common types of BI solutions.

- **Spreadsheets:** Spreadsheets like Microsoft Excel and Google Docs are some of the most widely used BI tools.
- **Reporting software:** Reporting software is used to report, organize, filter, and display data.
- **Data visualization software:** Data visualization software translates datasets into easy-to-read, visually appealing graphical representations to quickly gain insights.
- **Data mining tools:** Data mining tools "mine" large amounts of data for patterns using things like artificial intelligence, machine learning, and statistics.
- **Online analytical processing (OLAP):** OLAP tools allow users to analyze datasets from a wide variety of angles based on different business perspectives.

2.1 DATA WAREHOUSES AND DATA MART

Business intelligence, as we know it today, would not be possible without the data warehouse.

At its core, business intelligence is the ability to answer complex questions about your data and use those answers to make informed business decisions. In order to do this well, you need a data warehouse, which not only provides a safe way to centralize and store all your data but also a method to quickly find the answers you need, when you need them.

2.1.1 What Is a Data Warehouse?

A data warehouse is a data management system that stores large amounts of data for later use in processing and analysis. You can think of it as a large warehouse where trucks (i.e., source data) unload their data. That data is then sorted into rows and rows of well-organized shelves that make it easy to find exactly what you're looking for later.

In business intelligence, data warehouses serve as the backbone of data storage. Business intelligence relies on complex queries and comparing multiple sets of data to inform everything from everyday decisions to organization-wide shifts in focus.

2.1.3 When Should I Use a Data Warehouse for Business Intelligence?

As we explain in our Cloud Data Management eBook (a super easy — and dare we say fun — read), there are generally four stages of data sophistication: source data, data lakes, data warehouses, and data marts. Knowing when to invest in a data warehouse requires you to know each stage, but at the end of the day, the data warehouse stage is what unlocks the true power of your data.

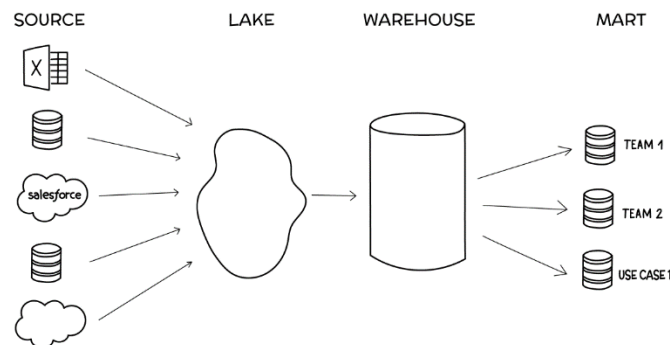


Figure 2.3 Data Warehouse

SOURCE DATA

Source data is any individual set of data like databases, Excel spreadsheets, individual application reports, etc. It's structured (i.e., organized) yet siloed data that works fine alone but does not provide a larger picture of your organization's data as a whole.

DATA LAKE

For teams who have graduated to a need to centralize their source data into one place, a data lake is increasingly becoming the next step. A data lake serves as a central repository for all raw, unstructured (i.e., not organized) data.

DATA WAREHOUSE

Like a data lake, a data warehouse centralizes your data, but as we've established, it's well-organized and set up for efficient analysis. It's a single source of truth for all data that's easier to understand and navigate.

DATA MART

Using a data warehouse for some projects can be like swatting a fly with a sledgehammer. If, for instance, the marketing team returns time and time again to the warehouse to make similar queries, you can set up a data mart.

2.1.4 Business Intelligence vs Data Warehouse

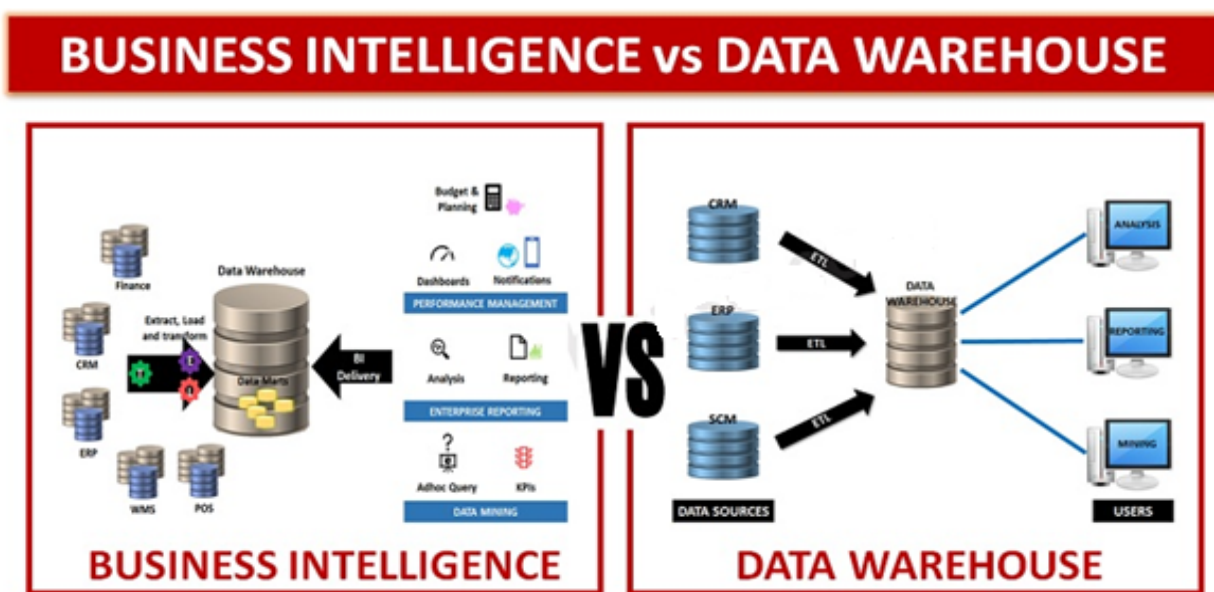


Figure 2.3 Business Intelligence vs Data Warehouse

Applying intelligence to businesses is one of key factors for success to take a leap over others, be agile, more informed and reach customers faster in today's scenario. Agility of businesses depends on how quickly organizations respond to changing demands of its customers. This helps businesses to make better informed decisions which improves performance and creates new strategic opportunities for growth.

Today we look more in detail about two important terminologies Business Intelligence and Data warehouses which are helping businesses to take more informed and better decisions to shape up the right direction for their businesses and place themselves ahead over their competitors in the market. This article will give insight into the major differences between the two, its purpose for which they are deployed and use cases.

About Business Intelligence

Business intelligence is a set of tools and methods which are used by organizations for exploration and accessing data from multiple varied sources to better understand how their businesses are performing and let them make better and more informed strategic decisions for the betterment of the organization as a whole and also bring in new business opportunities.

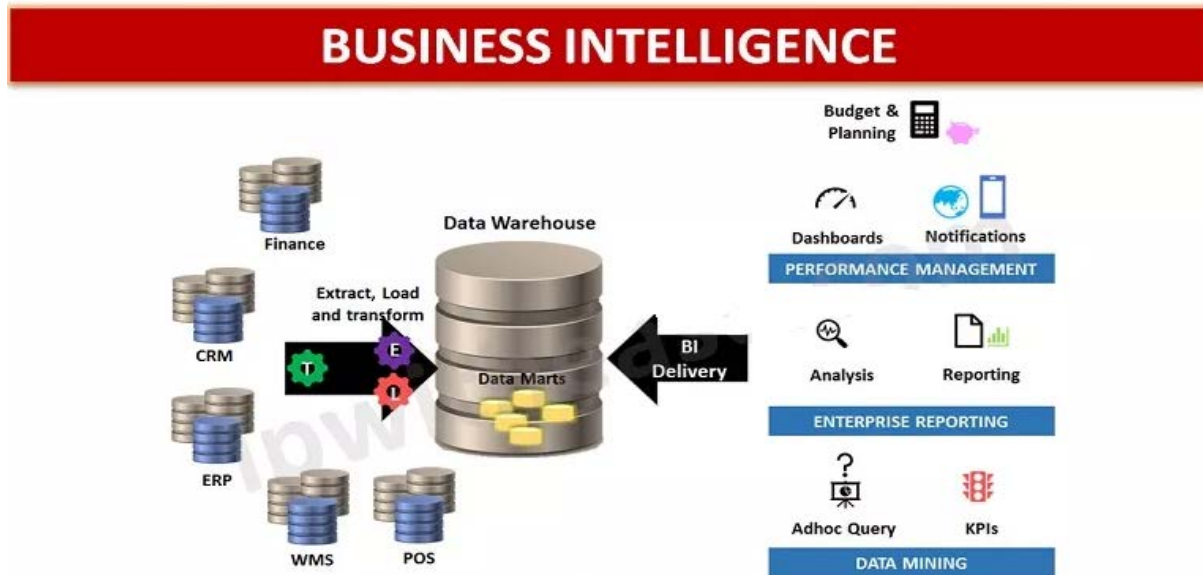


Figure 2.4 Business intelligence

Business intelligence gives insight into what is happening or happened in business and gives actual description of the situation currently more like a real time scenario. Data is stored in Data Warehouse (DDs, cubes) and Business intelligence systems make use of Data Warehouse data and you can apply metrics of your choice to huge unstructured data sets and query or perform mining , online analytical processing and generate reports as well as business performance monitoring, predictive and prescriptive analytics.

Applications of Business Intelligence

- Sales intelligence
- Visualization
- Reporting
- Performance management

About Data Warehouse

Data Warehouse is consolidation of data from multiple sources which actually gives foundation to derive business intelligence from it. It helps organizations in making better strategic and tactical decisions. We can say in other words Data Warehouse is the backbone for business intelligence. Databases store data of different sources , types in a common format and Warehouse is like a Godown for data where many things can be stored but with intelligent algorithms like indexing, pattern matching it makes it easier to locate and retrieve data faster.

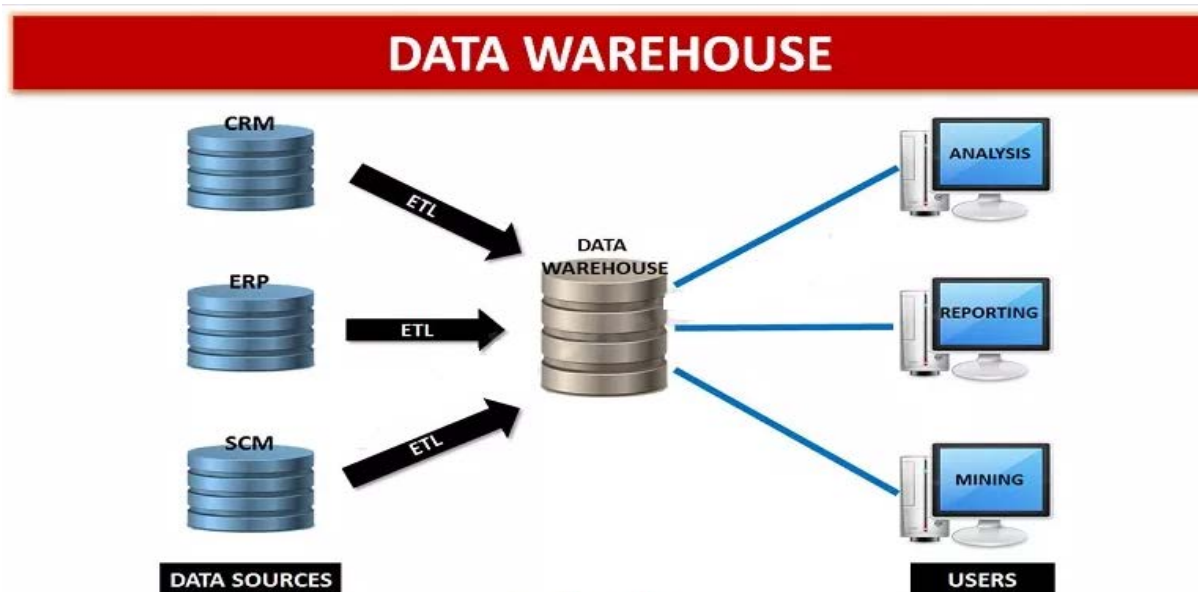


Figure 2.5 Data Warehouse

Data Warehouse is like a relational database which is aimed at querying and analysing the data instead of transaction processing. It contains historical data derived from transactional data but might have data from various data sources. Data Warehouses hold data in Fact tables and dimensions.

Applications of Data Warehouse

- Used by varied industries such as Banking, finance, education, Healthcare, manufacturing and distribution, retail, insurance etc.
- High performance queries on large data volumes
- In depth data exploration
- Historical data collection across organizations
- Easy interface for business analysts and ready data for them

Comparison Table: Business Intelligence vs Data Warehouse

Below table summarizes the differences between the two terms:

<i>FUNCTION</i>	<i>BUSINESS INTELLIGENCE</i>	<i>DATA WAREHOUSE</i>
Definition	Strategy and technology used by businesses for analysis of data to get insights into businesses.	Historical data and current data repository collected by an enterprise from various operational systems.
Source	Business intelligence uses data from Data Warehouse or data marts as a source to perform analytics.	Data Warehouse may obtain data from multiple sources.

Outcome	Business reports, charts graphs, data visualization , dashboards etc.	Fact and dimension tables storing data which is ingested into business applications and tools.
Used by	C-level executives , managers and data analysts	Data engineers, data analysts, business analysts, Backend developers.
Tools used	MSBI, QlikView, Cognos , Clear Analytics, SAP Business intelligence Oracle Hyperion system Pentaho BI etc.	Ab Initio Software, Amazon Redshift, Informatica etc.
Characteristics	Helps to identify, develop and create new business strategies and opportunities.	Helps to store data in a centralized repository for analysis.

Table 2.1

2.2 DATA MART

2.2.1 What is Data Mart?

A **Data Mart** is a subset of a directorial information store, generally oriented to a specific purpose or primary data subject which may be distributed to provide business needs. Data Marts are analytical record stores designed to focus on particular business functions for a specific community within an organization. Data marts are derived from subsets of data in a data warehouse, though in the bottom-up data warehouse design methodology, the data warehouse is created from the union of organizational data marts.

The fundamental use of a data mart is **Business Intelligence (BI)** applications. **BI** is used to gather, store, access, and analyse record. It can be used by smaller businesses to utilize the data they have accumulated since it is less expensive than implementing a data warehouse.

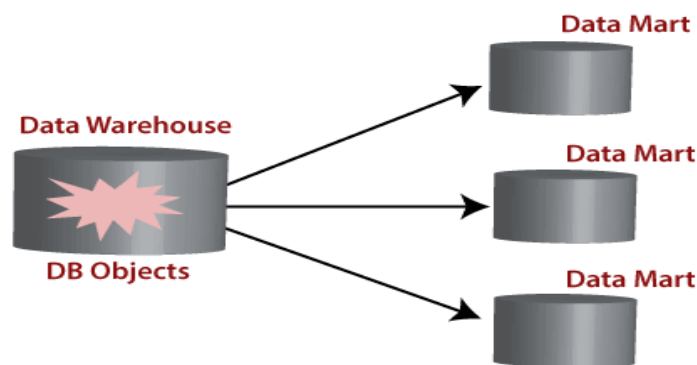


Figure 2.6 data mart

2.2.2 Reasons for creating a data mart

- o Creates collective data by a group of users
- o Easy access to frequently needed data
- o Ease of creation
- o Improves end-user response time
- o Lower cost than implementing a complete data warehouses
- o Potential clients are more clearly defined than in a comprehensive data warehouse
- o It contains only essential business data and is less cluttered.

2.2.3 Types of Data Marts

There are mainly two approaches to designing data marts. These approaches are

- o Dependent Data Marts
- o Independent Data Marts

Dependent Data Marts

A dependent data marts is a logical subset of a physical subset of a higher data warehouse. According to this technique, the data marts are treated as the subsets of a data warehouse. In this technique, firstly a data warehouse is created from which further various data marts can be created. These data mart are dependent on the data warehouse and extract the essential record from it. In this technique, as the data warehouse creates the data mart; therefore, there is no need for data mart integration. It is also known as a **top-down approach**.

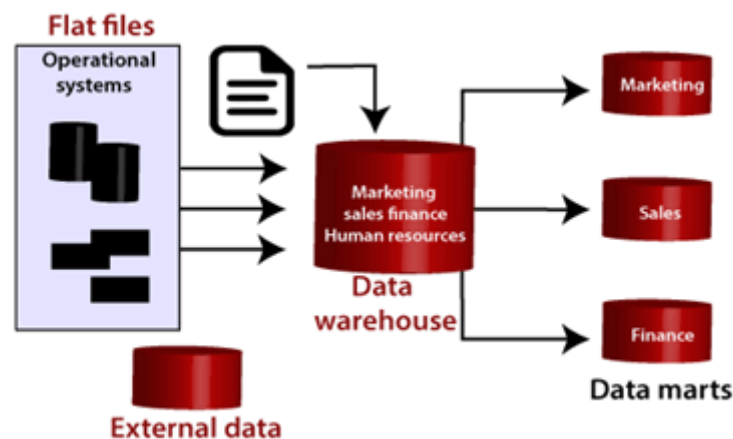


Figure 2.7 dependent data marts

Independent Data Marts

The second approach is Independent data marts (IDM) Here, firstly independent data marts are created, and then a data warehouse is designed using these independent multiple data marts. In this approach, as all the data marts are designed independently; therefore, the integration of data marts is required. It is also termed as a **bottom-up approach** as the data marts are integrated to develop a data warehouse.

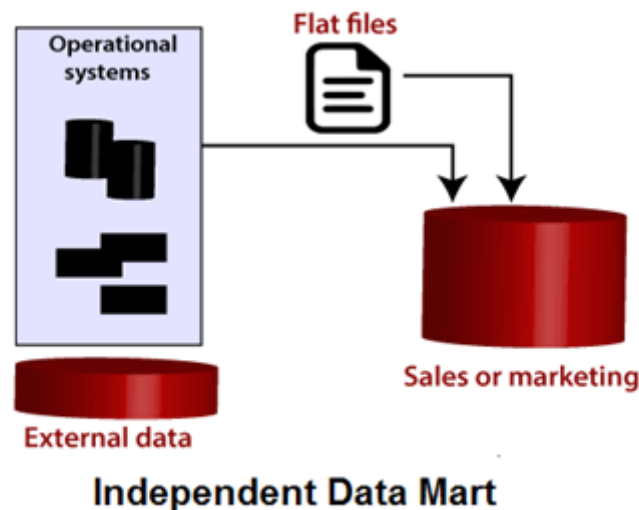


Figure 2.8 Independent data marts

Other than these two categories, one more type exists that is called "**Hybrid Data Marts.**"

Hybrid Data Marts

It allows us to combine input from sources other than a data warehouse. This could be helpful for many situations; especially when Adhoc integrations are needed, such as after a new group or product is added to the organizations.

2.2.4 Steps in Implementing a Data Mart

The significant steps in implementing a data mart are to design the schema, construct the physical storage, populate the data mart with data from source systems, access it to make informed decisions and manage it over time. So, the steps are:

Designing

The design step is the first in the data mart process. This phase covers all of the functions from initiating the request for a data mart through gathering data about the requirements and developing the logical and physical design of the data mart.

It involves the following tasks:

1. Gathering the business and technical requirements
2. Identifying data sources
3. Selecting the appropriate subset of data
4. Designing the logical and physical architecture of the data mart.

Constructing

This step contains creating the physical database and logical structures associated with the data mart to provide fast and efficient access to the data.

It involves the following tasks:

1. Creating the physical database and logical structures such as tablespaces associated with the data mart.
2. creating the schema objects such as tables and indexes describe in the design step.
3. Determining how best to set up the tables and access structures.

Populating

This step includes all of the tasks related to the getting data from the source, cleaning it up, modifying it to the right format and level of detail, and moving it into the data mart.

It involves the following tasks:

1. Mapping data sources to target data sources
2. Extracting data
3. Cleansing and transforming the information.
4. Loading data into the data mart
5. Creating and storing metadata

Accessing

This step involves putting the data to use: querying the data, analyzing it, creating reports, charts and graphs and publishing them.

It involves the following tasks:

1. Set up and intermediate layer (Meta Layer) for the front-end tool to use. This layer translates database operations and objects names into business conditions so that the end-clients can interact with the data mart using words which relates to the business functions.
2. Set up and manage database architectures like summarized tables which help queries agree through the front-end tools execute rapidly and efficiently.

Managing

This step contains managing the data mart over its lifetime. In this step, management functions are performed as:

1. Providing secure access to the data.
2. Managing the growth of the data.
3. Optimizing the system for better performance.
4. Ensuring the availability of data event with system failures.

2.2.5 Difference between Data Warehouse and Data Mart

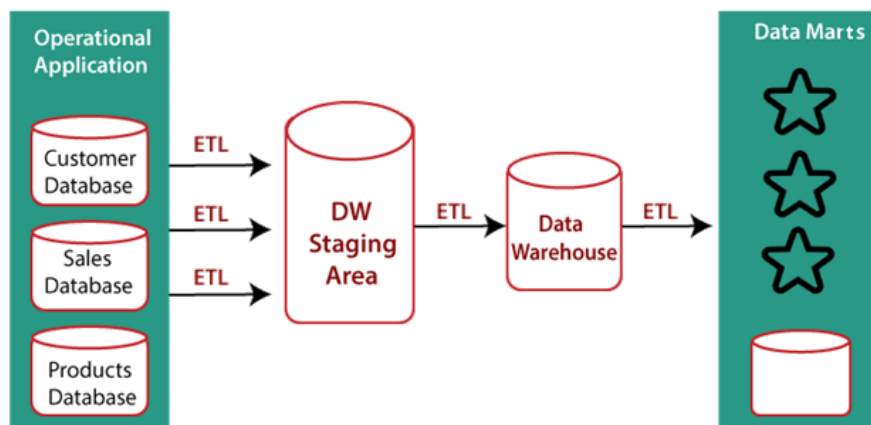


Figure 2.8 Difference between Data Warehouse and Data Mart

Data Warehouse	Data Mart
A Data Warehouse is a vast repository of information collected from various organizations or departments within a corporation.	A data mart is an only subtype of a Data Warehouses. It is architecture to meet the requirement of a specific user group.
It may hold multiple subject areas.	It holds only one subject area. For example, Finance or Sales.
It holds very detailed information.	It may hold more summarized data.

Works to integrate all data sources	It concentrates on integrating data from a given subject area or set of source systems.
In data warehousing, Fact constellation is used.	In Data Mart, Star Schema and Snowflake Schema are used.
It is a Centralized System. It is a Decentralized System.	
Data Warehousing is the data-oriented.	Data Marts is a project-oriented.

Table 2.2

2.2.6 Data Mart and Data Warehouse Comparison

Data Mart

Focus: A single subject or functional organization area

Data Sources: Relatively few sources linked to one line of business

Size: Less than 100 GB

Normalization: No preference between a normalized and denormalized structure

Decision Types: Tactical decisions pertaining to particular business lines and ways of doing things

Cost: Typically from \$10,000 upwards

Setup Time: 3-6 months

Data Held: Typically summarized data

Data Warehouse

Focus: Enterprise-wide repository of disparate data sources

Data Sources: Many external and internal sources from different areas of an organization

Size: 100 GB minimum but often in the range of terabytes for large organizations

Normalization: Modern warehouses are mostly denormalized for quicker data querying and read performance

Decision Types: Strategic decisions that affect the entire enterprise

Cost: Varies but often greater than \$100,000; for cloud solutions costs can be dramatically lower as organizations pay per use

Setup Time: At least a year for on-premise warehouses; cloud data warehouses are much quicker to set up

Data Held: Raw data, metadata, and summary data

2.2.7 Inmon vs. Kimball

Two data warehouse pioneers, Bill Inmon and Ralph Kimball differ in their views on how data warehouses should be designed from the organization's perspective.

Bill Inmon's approach favours a top-down design in which the data warehouse is the centralized data repository and the most important component of an organization's data systems.

The Inmon approach first builds the centralized corporate data model, and the data warehouse is seen as the physical representation of this model. Dimensional data marts related to specific business lines can be created from the data warehouse when they are needed.

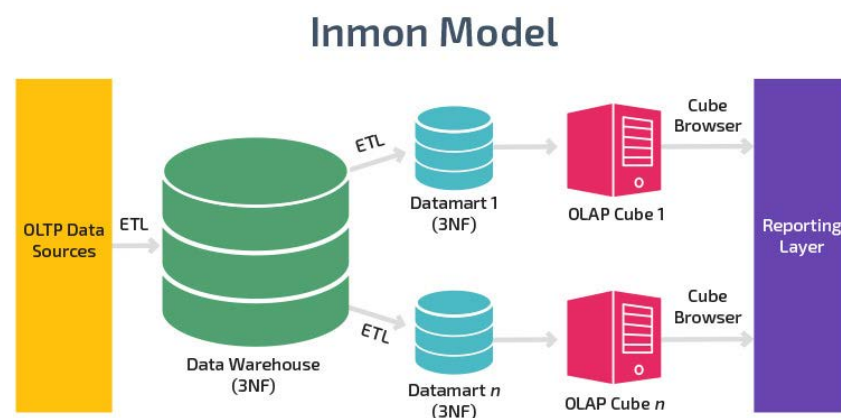


Figure 2.8 Inmon model

In the Inmon model, data in the data warehouse is integrated, meaning the data warehouse is the source of the data that ends up in the different data marts. This ensures data integrity and consistency across the organization.

Ralph Kimball's data warehouse design starts with the most important business processes. In this approach, an organization creates data marts that aggregate relevant data around subject-specific areas. The data warehouse is the combination of the organization's individual data marts.

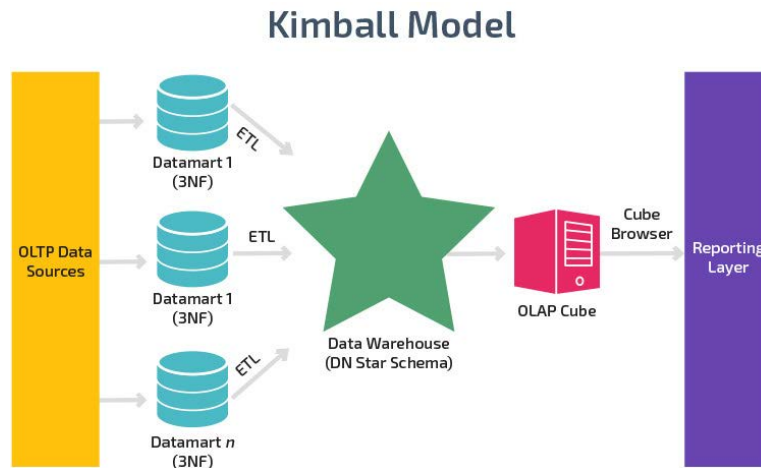


Figure 2.8 Kimball model

With the Kimball approach, the data warehouse is the conglomerate of a number of data marts. This is in contrast to Inmon's approach, which creates data marts based on information in the warehouse. As Kimball said in 1997, “the data warehouse is nothing more than the union of all data marts.”*

2.3 KNOWLEDGE MANAGEMENT

Generally, BI is considered to be *a set of tools and techniques applied to gather data and transform it into information that can be used in business analysis for the purposes of business development.*

Every company gathers, collects, or to say more accurately, **deals with a large amounts of data**, including various business documents, emails, newspaper articles, web pages, reports, contracts, technical journals and reviews, spreadsheets, graphs and charts and other relevant sources of business data.

BI technologies usually deal with large amounts of unstructured data via the use of data warehousing and online analytical processing (OLAP).

All these data needs to be organized and validated – prepared for business analytics.

Thus,

“BI is about providing the right data at the right time to the right people so that they can take the right decisions.” (Nic Smith)

2.3.1 What is Knowledge Management (KM)?

Knowledge Management can be defined in many ways as it **spans many multi-disciplinary approaches** – content management, collaboration, the science of organizational behavior, analyses like observation of trends and appearance of anomalies, clustering, classification, summarization, taxonomy building and so on.

This is probably one of the widely quoted definitions (Davenport, 1994), yet simple and to the point:

“Knowledge management is the process of capturing, distributing, and effectively using knowledge.”

Knowledge management (KM) is the process of identifying, organizing, storing and disseminating information within an organization.

KM refers to a set of techniques used to capture, share, and use the information available in order to achieve business objectives and to aid in business decision making based on business analytics.

2.3.2 Similarities and Differences between BI and KM

Confusion between these two technologies comes from the fact that they deal with many similar processes.

Both business intelligence and knowledge management **capture, collect, organize, analyze and aggregate data** in order to find the best solutions regarding business decision making processes.

“Knowledge management will never work until corporations realize it’s not about how you capture knowledge but how you create and leverage it.” Etienne Wenger

We could say that knowledge management draws on business intelligence in order to produce new knowledge, organize it into knowledge networks and utilize it for achieving specific goals.

But, these concepts promote decision-making, learning, and understanding:

“KM and BI, while differing, need to be considered together as necessarily integrated and mutually critical components in the management of intellectual capital.”

2.3.3 Types of knowledge

The definition of knowledge management also includes three types of knowledge—tacit, implicit, and explicit knowledge. These types of knowledge are largely distinguished by the codification of the information.

- **Tacit knowledge:** This type of knowledge is typically acquired through experience, and it is intuitively understood. As a result, it is challenging to articulate and codify, making it difficult to transfer this information to other individuals. Examples of tacit knowledge can include language, facial recognition, or leadership skills.
- **Implicit knowledge:** While some literature equivocates implicit knowledge to tacit knowledge, some academics break out this type separately, expressing that the definition of tacit knowledge is more nuanced. While tacit knowledge is difficult to codify, implicit knowledge does not necessarily have this problem. Instead, implicit information has yet to be documented. It tends to exist within processes, and it can be referred to as “know-how” knowledge.

- **Explicit knowledge:** Explicit knowledge is captured within various document types such as manuals, reports, and guides, allowing organizations to easily share knowledge across teams. This type of knowledge is perhaps the most well-known and examples of it include knowledge assets such as databases, white papers, and case studies. This form of knowledge is important to retain intellectual capital within an organization as well as facilitate successful knowledge transfer to new employees.

2.3.4 Knowledge management process

The knowledge management process as involving knowledge acquisition, creation, refinement, storage, transfer, sharing and utilization. This process can be synthesized this a little further. Effective knowledge management system typically goes through three main steps:

1. **Knowledge Creation:** During this step, organizations identify and document any existing or new knowledge that they want to circulate across the company.
2. **Knowledge Storage:** During this stage, an information technology system is typically used to host organizational knowledge for distribution. Information may need to be formatted in a particular way to meet the requirements of that repository.
3. **Knowledge Sharing:** In this final stage, processes to share knowledge are communicated broadly across the organization. The rate in which information spreads will vary depending on organizational culture. Companies that encourage and reward this behavior will certainly have a competitive advantage over other ones in their industry.

2.3.5 Knowledge Management (KM) with Business Intelligence (BI)

In a traditional Business Intelligence (BI), the system must provide capabilities such as business process management, collaborative portal, and business planning software, portals and content management systems and be able to support more timely data feeds.

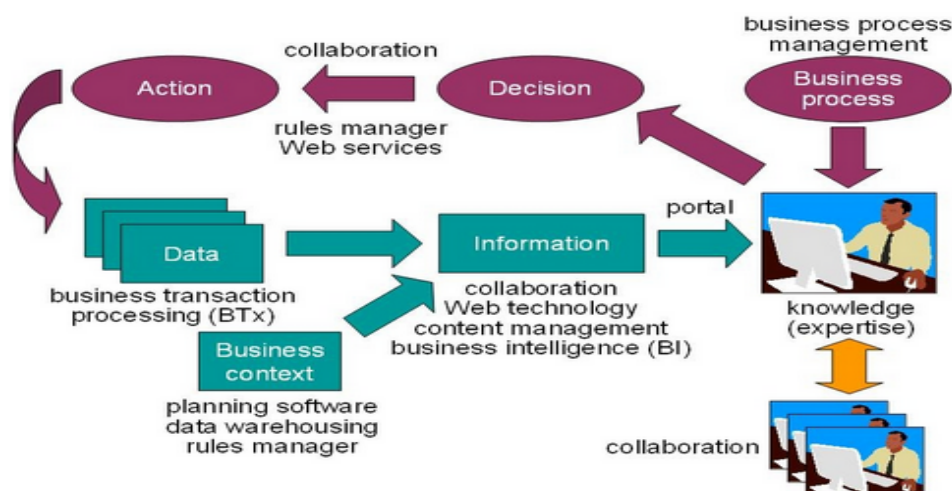


Figure 2.9 Knowledge Management (KM) with Business Intelligence (BI)

The 2.9 illustrates the knowledge cycle on how the Knowledge Management (KM) can help the business to improve their processes. The figure shows how the Business Intelligence which contains information becomes the central role in knowledge management. It has data, business context, decision, action and the collaboration of experts.

There are some limitations and issues in decision-making based on static reports from different systems:

1. Information overload,
2. Lack of information,
3. No interactivity,
4. Lack of unified cross-database analysis tools.

Implementing a Data Warehouse with modern Business Intelligence (BI) software can be the solution to the problem of a poor analytical environment in a company. The poor analytical environment will have multiple data sources, different report writers and lack of analytical tools.

It is finally becoming feasible for financial institution to implement Data Warehouse that can be updated and maintained with relative ease by utilizing modern ETL (Extraction Transformation Load).

There are some key features that offered by many Business Intelligence tools. The some key features are:

1. Drill-down,
2. Graphs, charting and trees,
3. Exception highlighting,
4. Pivot rows and columns,
5. Drag and drop dimension into the current view or to the background,
6. Custom calculations,
7. Queries,
8. Comments,
9. Combo views,
10. Dashboards,
11. Business Intelligence (BI) and web portals,

12. Distribution of cubes/reports,

13. Other popular analytical features (ranking, filtering, sorting, etc).

2.4 DECISION MAKING BUSINESS INTELLIGENCE

Because of the importance of high-quality decision making, firms are investing heavily in **business intelligence** systems, which consist of technologies and applications designed to help users make better business decisions. The major applications and technologies used for business intelligence include supply chain management, customer relationship management, and enterprise systems; systems for knowledge management; and technologies such as data mining and online analytical processing (OLAP)

Simply put, decision-making is the process of deciding something, especially with a group of people. From a business decision perspective, the aim is to achieve business objectives to satisfy stakeholder requirements, needs, and expectations.

For the decision to be effective, however, decision makers must forecast the outcome of each option and determine which is best for a particular situation. That makes decision support systems (DSS) like decision intelligence and business intelligence absolute essentials.

2.4.1 Three primary types of business intelligence decisions

Business intelligence supports the three types of decision-making mentioned above: strategic, tactical, and operational. Its frequency and organizational impact characterize each.

Strategic decisions

Strategic decisions comprise the highest level of organizational business decisions and are usually less frequent and made by the organization's executives. Yet, their impact is enormous and far-reaching.

Some types of strategic decisions include selecting a particular market to penetrate, a company to acquire, or whether to hire additional staff.

Decisions made at this level usually involve significant expenditure. However, they are generally non-repetitive in nature and are taken only after careful analysis and evaluation of many alternatives.

Tactical decisions

Tactical decisions (or semi structured decisions) occur with greater frequency (e.g., weekly or monthly) and fall into the mid-management level. Often, they relate to the implementation of strategic decisions.

Examples of tactical decisions include product price changes, work schedules, departmental reorganization, and similar activities.

The impact of these types of decisions is medium regarding risk to the organization and impact on profitability.

Operational decisions

Operational decisions (or structured decisions) usually happen frequently (e.g., daily or hourly), relate to day-to-day operations of the enterprise, and have a lesser impact on the organization. Operational decisions determine the day-to-day profitability of the business, how effectively it retains customers, or how well it manages risk.

Answering a sales inquiry, approving a quotation, or calculating employee bonuses may be examples of this decision type.

Business intelligence decisions typically fall into three categories: strategic, tactical, and operational.

You can summarize these types of decisions in business intelligence this way:

- Strategic: Long-term, complex, made by senior managers
- Tactical: Medium-term, less complex, made by mid-level managers
- Operational: Day-to-day, simple, routine, made by junior manager

2.4.2 Types of Decisions

Decisions are made at all levels of the firm. Some decisions are very common and routine, but exceptionally valuable. Although the value of improving any single one of these decisions may be small, improving hundreds of thousands of these small decisions adds up to a large annual value. There are four different decision-making constituencies in a firm:

- (1) senior management;
- (2) middle management and project teams;
- (3) operational management and project teams; and
- (4) individual employees.

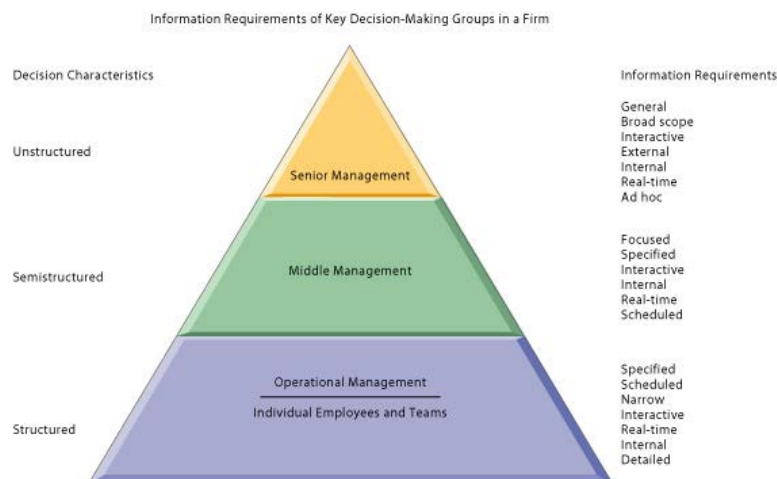


Figure 2.10 Information requirements of key decision-making groups

Unstructured decisions are those in which the decision maker must provide judgment, evaluation, and insights into the problem definition.

Structured decisions, by contrast, are repetitive and routine, and decision makers can follow a definite procedure for handling them to be efficient.

Semi structured decisions are those in which only part of the problem has a clear-cut answer provided by an accepted procedure. In general, structured decisions are more prevalent at lower organizational levels, and unstructured decision making is more common at higher levels.

- ✓ There are four kinds of systems used to support the different levels and types of decisions: Management Information systems (MIS),
- ✓ decision-support systems (DSS);
- ✓ executive support systems (ESS); and
- ✓ group decision-support systems (GDSS).

2.4.3 Decision Making Process

There are four different stages in decision making: intelligence, design, choice, and implementation.

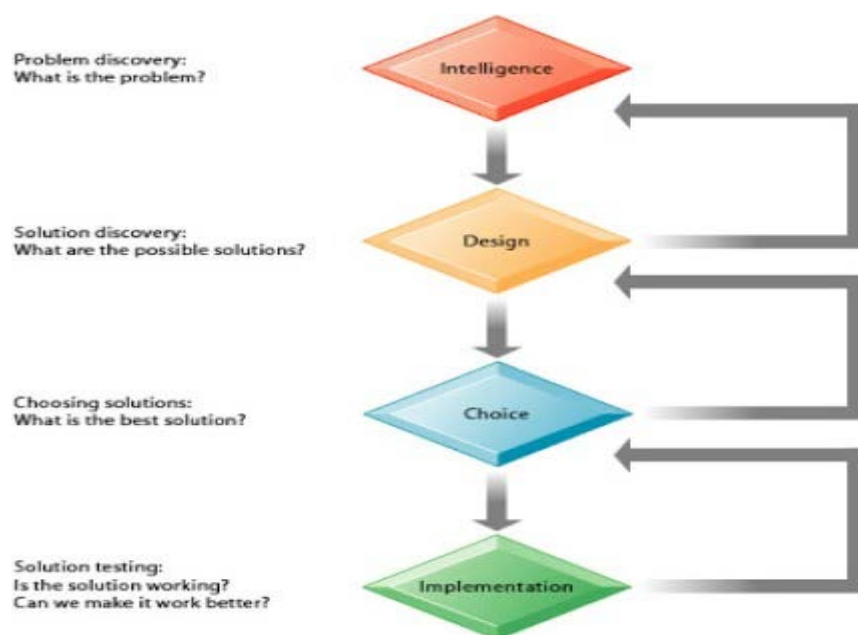


Figure 2.11 Stages in Decision process

- Intelligence consists of identifying and understanding a problem;
- design involves exploring various solutions; c
- hoice consists of choosing among available solutions; and
- implementation involves making the chosen alternative work and monitoring how the solution is working.

There are many different decision-making methodologies, but most share at least five steps in common:

- Identify a business problem.

- Seek information about different possible decisions and their likely effect.
- Evaluate the alternatives and choose one of them.
- Implement the decision in business operations.
- Monitor the situation, gather data about the decision's impact and make changes if necessary.

Data-driven decision making

Traditionally, decisions were made by business managers or corporate executives using their intuitive understanding of the situation at hand. However, intuitive decision-making has several drawbacks.

Challenges in the decision-making process

Balancing data-driven and intuitive approaches to decision-making is a difficult proposition. Managers and executives may be sceptical about relying on data that goes against their intuition in making decisions or feel that their experience and knowledge is being discounted or ignored completely. As a result, they may push back against the findings of BI and analytics tools during the decision-making process.

2.5 DECISION SUPPORT SYSTEMS

2.5.1 What Is a Decision Support System (DSS)?

A decision support system (DSS) is a computerized program used to support determinations, judgments, and courses of action in an organization or a business. A DSS sifts through and analyzes massive amounts of data, compiling comprehensive information that can be used to solve problems and in decision-making.

KEY TAKEAWAYS

- A decision support system (DSS) is a computerized system that gathers and analyzes data, synthesizing it to produce comprehensive information reports.
- A decision support system differs from an ordinary operations application, whose function is just to collect data.
- Decision support systems allow for more informed decision-making, timely problem-solving, and improved efficiency in dealing with issues or operations, planning, and even management.

2.5.2 Understanding a Decision Support System (DSS)

- A decision support system gathers and analyzes data, synthesizing it to produce comprehensive information reports. In this way, as an informational application, a DSS differs from an ordinary operations application, whose function is just to collect data.
- The DSS can either be completely computerized or powered by humans. In some cases, it may combine both. The ideal systems analyze information and actually make decisions for the user. At the very least, they allow human users to make more informed decisions at a quicker pace.

Typical information that a decision support application might gather and present would be,

- (a) Accessing all information assets, including legacy and relational data sources;
- (b) Comparative data figures;
- (c) Projected figures based on new data or assumptions;
- (d) Consequences of different decision alternatives, given past experience in a specific context.

2.5.3 Types of DSS:

There are a number of Decision Support Systems. These can be categorized into five types:

- **Communication-driven DSS**

Most communications-driven DSSs are targetted at internal teams, including partners. Its purpose are to help conduct a meeting, or for users to collaborate. The most common technology used to deploy the DSS is a web or client server. Examples: chats and instant messaging softwares, online collaboration and net-meeting systems.

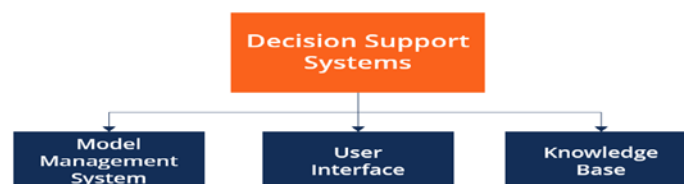
- **Data-driven DSS**

Most data-driven DSSs are targeted at managers, staff and also product/service suppliers. It is used to query a database or data warehouse to seek specific answers for specific purposes. It is deployed via a main frame system, client/server link, or via the web. Examples: computer-based databases that have a query system to check (including the incorporation of data to add value to existing databases.

- **Document-driven DSS**
- **Knowledge-driven DSS**
- **Model-driven DSS**

2.5.4 Components of a Decision Support System

The three main components of a DSS framework are:



1. Model Management System

The model management system stores models that managers can use in their decision-making. The models are used in decision-making regarding the financial health of the organization and forecasting demand for a good or service.

2. User Interface

The user interface includes tools that help the end-user of a DSS to navigate through the system.

3. Knowledge Base

The knowledge base includes information from internal sources (information collected in a transaction process system) and external sources (newspapers and online databases).

2.5.5 Advantages of a Decision Support System

- A decision support system increases the speed and efficiency of decision-making activities. It is possible, as a DSS can collect and analyze real-time data.
- It promotes training within the organization, as specific skills must be developed to implement and run a DSS within an organization.
- It automates monotonous managerial processes, which means more of the manager's time can be spent on decision-making.
- It improves interpersonal communication within the organization.

2.5.6 Disadvantages of a Decision Support System

- The cost to develop and implement a DSS is a huge capital investment, which makes it less accessible to smaller organizations.
- A company can develop a dependence on a DSS, as it is integrated into daily decision-making processes to improve efficiency and speed. However, managers tend to rely on the system too much, which takes away the subjectivity aspect of decision-making.
- A DSS may lead to information overload because an information system tends to consider all aspects of a problem. It creates a dilemma for end-users, as they are left with multiple choices.
- Implementation of a DSS can cause fear and backlash from lower-level employees. Many of them are not comfortable with new technology and are afraid of losing their jobs to technology.

2.5.7 Decision support system examples

Organizations use decision support systems in several different contexts, including the following:

- **GPS routing.**

- **ERP dashboards.**
- **Clinical decision support system.**

2.6 OLAP (Online Analytical Processing)

OLAP stands for **On-Line Analytical Processing**. OLAP is a classification of software technology which authorizes analysts, managers, and executives to gain insight into information through fast, consistent, interactive access in a wide variety of possible views of data that has been transformed from raw information to reflect the real dimensionality of the enterprise as understood by the clients.

OLAP implement the multidimensional analysis of business information and support the capability for complex estimations, trend analysis, and sophisticated data modelling. It is rapidly enhancing the essential foundation for Intelligent Solutions containing Business Performance Management, Planning, Budgeting, Forecasting, Financial Documenting, Analysis, Simulation-Models, Knowledge Discovery, and Data Warehouses Reporting. OLAP enables end-clients to perform ad hoc analysis of record in multiple dimensions, providing the insight and understanding they require for better decision making.

2.6.1 How OLAP systems work

To facilitate this kind of analysis, data is collected from multiple data sources and stored in data warehouses then cleansed and organized into data cubes. Each OLAP cube contains data categorized by dimensions (such as customers, geographic sales region and time period) derived by dimensional tables in the data warehouses. Dimensions are then populated by members (such as customer names, countries and months) that are organized hierarchically.

Analysts can then perform five types of OLAP analytical operations against these multidimensional databases:

- **Roll-up.** Also known as *consolidation*, or *drill-up*, this operation summarizes the data along the dimension.
- **Drill-down.** This allows analysts to navigate deeper among the dimensions of data, for example drilling down from "time period" to "years" and "months" to chart sales growth for a product.
- **Slice.** This enables an analyst to take one level of information for display, such as "sales in 2017."
- **Dice.** This allows an analyst to select data from multiple dimensions to analyze, such as "sales of blue beach balls in Iowa in 2017."

- **Pivot.** Analysts can gain a new view of data by rotating the data axes of the cube.

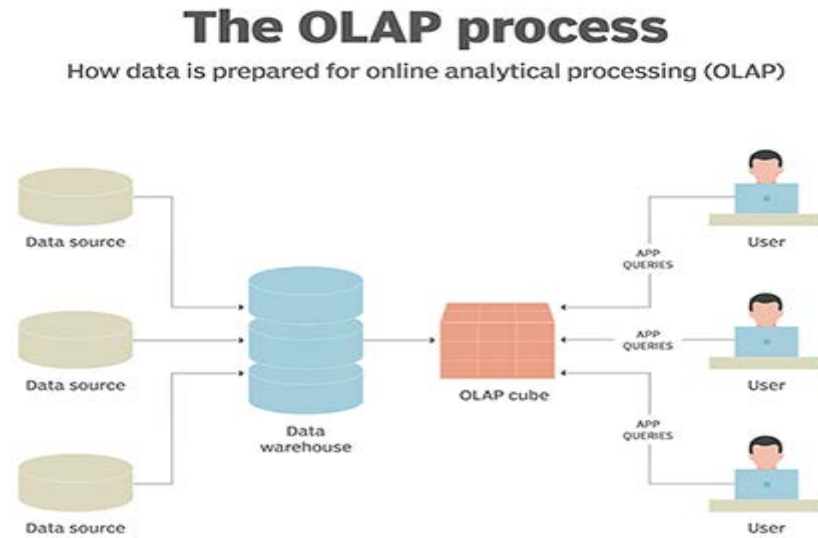


Figure 2.12 OLAP Process

OLAP begins with data accumulated from multiple sources and stored in a data warehouse. The data is then cleansed and stored in OLAP cubes, which users run queries against.

2.6.2 Who uses OLAP and Why?

OLAP applications are used by a variety of the functions of an organization.

Finance and accounting:

- o Budgeting
- o Activity-based costing
- o Financial performance analysis
- o And financial modelling

Sales and Marketing

- o Sales analysis and forecasting
- o Market research analysis
- o Promotion analysis
- o Customer analysis
- o Market and customer segmentation

Production

- o Production planning

- o Defect analysis

OLAP cubes have two main purposes. The first is to provide business users with a data model more intuitive to them than a tabular model. This model is called a Dimensional Model.

The second purpose is to enable fast query response that is usually difficult to achieve using tabular models.

2.6.3 How OLAP Works?

Fundamentally, OLAP has a very simple concept. It pre-calculates most of the queries that are typically very hard to execute over tabular databases, namely aggregation, joining, and grouping. These queries are calculated during a process that is usually called 'building' or 'processing' of the OLAP cube. This process happens overnight, and by the time end users get to work - data will have been updated.

OLAP Guidelines (Dr.E.F.Codd Rule)

Dr E.F. Codd, the "father" of the relational model, has formulated a list of 12 guidelines and requirements as the basis for selecting OLAP systems:



Figure 2.13 Guidelines and Requirements

1) Multidimensional Conceptual View: This is the central features of an OLAP system. By needing a multidimensional view, it is possible to carry out methods like slice and dice.

2) Transparency: Make the technology, underlying information repository, computing operations, and the dissimilar nature of source data totally transparent to users. Such transparency helps to improve the efficiency and productivity of the users.

3) Accessibility: It provides access only to the data that is actually required to perform the particular analysis, present a single, coherent, and consistent view to the clients. The OLAP system must map its own logical schema to the heterogeneous physical data stores and perform any necessary transformations. The OLAP operations should be sitting between data sources (e.g., data warehouses) and an OLAP front-end.

4) Consistent Reporting Performance: To make sure that the users do not feel any significant degradation in documenting performance as the number of dimensions or the size of the database increases. That is, the performance of OLAP should not suffer as the number of

dimensions is increased. Users must observe consistent run time, response time, or machine utilization every time a given query is run.

5) Client/Server Architecture: Make the server component of OLAP tools sufficiently intelligent that the various clients to be attached with a minimum of effort and integration programming. The server should be capable of mapping and consolidating data between dissimilar databases.

6) Generic Dimensionality: An OLAP method should treat each dimension as equivalent in both its structure and operational capabilities. Additional operational capabilities may be allowed to selected dimensions, but such additional tasks should be grantable to any dimension.

7) Dynamic Sparse Matrix Handling: To adapt the physical schema to the specific analytical model being created and loaded that optimizes sparse matrix handling. When encountering the sparse matrix, the system must be easy to dynamically assume the distribution of the information and adjust the storage and access to obtain and maintain a consistent level of performance.

8) Multiuser Support: OLAP tools must provide concurrent data access, data integrity, and access security.

9) Unrestricted cross-dimensional Operations: It provides the ability for the methods to identify dimensional order and necessarily functions roll-up and drill-down methods within a dimension or across the dimension.

10) Intuitive Data Manipulation: Data Manipulation fundamental the consolidation direction like as reorientation (pivoting), drill-down and roll-up, and another manipulation to be accomplished naturally and precisely via point-and-click and drag and drop methods on the cells of the scientific model. It avoids the use of a menu or multiple trips to a user interface.

11) Flexible Reporting: It implements efficiency to the business clients to organize columns, rows, and cells in a manner that facilitates simple manipulation, analysis, and synthesis of data.

12) Unlimited Dimensions and Aggregation Levels: The number of data dimensions should be unlimited. Each of these common dimensions must allow a practically unlimited number of customer-defined aggregation levels within any given consolidation path.

2.6.4 The main characteristics of OLAP are as follows:

- 1. Multidimensional conceptual view:** OLAP systems let business users have a dimensional and logical view of the data in the data warehouse. It helps in carrying slice and dice operations.
- 2. Multi-User Support:** Since the OLAP techniques are shared, the OLAP operation should provide normal database operations, containing retrieval, update, adequacy control, integrity, and security.

3. **Accessibility:** OLAP acts as a mediator between data warehouses and front-end. The OLAP operations should be sitting between data sources (e.g., data warehouses) and an OLAP front-end.
4. **Storing OLAP results:** OLAP results are kept separate from data sources.
5. **Uniform documenting performance:** Increasing the number of dimensions or database size should not significantly degrade the reporting performance of the OLAP system.

2.6.5 Benefits of OLAP

OLAP holds several benefits for businesses: -

1. OLAP helps managers in decision-making through the multidimensional record views that it is efficient in providing, thus increasing their productivity.
2. OLAP functions are self-sufficient owing to the inherent flexibility support to the organized databases.
3. It facilitates simulation of business models and problems, through extensive management of analysis-capabilities.
4. In conjunction with data warehouse, OLAP can be used to support a reduction in the application backlog, faster data retrieval, and reduction in query drag.

2.6.6 Difference between OLTP and OLAP

OLTP (On-Line Transaction Processing) is featured by a large number of short on-line transactions (INSERT, UPDATE, and DELETE). The primary significance of OLTP operations is put on very rapid query processing, maintaining record integrity in multi-access environments, and effectiveness consistent by the number of transactions per second. In the OLTP database, there is an accurate and current record, and schema used to save transactional database is the entity model (usually 3NF).

OLAP (On-line Analytical Processing) is represented by a relatively low volume of transactions. Queries are very difficult and involve aggregations. For OLAP operations, response time is an effectiveness measure. OLAP applications are generally used by Data Mining techniques. In OLAP database there is aggregated, historical information, stored in multi-dimensional schemas (generally star schema).

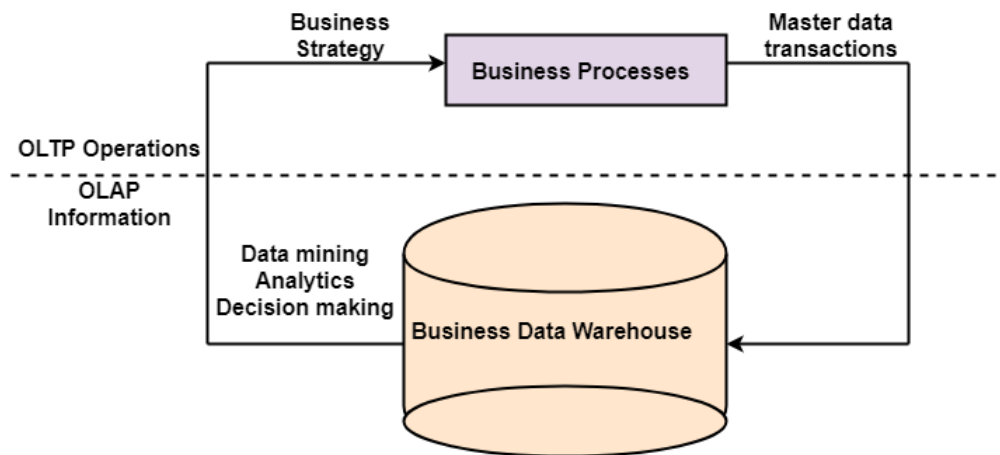


Figure 2.14 OLAP

Following are the difference between OLAP and OLTP system.

1) Users: **OLTP** systems are designed for office worker while the OLAP systems are designed for decision-makers. Therefore while an OLTP method may be accessed by hundreds or even thousands of clients in a huge enterprise, an OLAP system is suitable to be accessed only by a select class of manager and may be used only by dozens of users.

2) Functions: **OLTP** systems are mission-critical. They provide day-to-day operations of an enterprise and are largely performance and availability driven. These operations carry out simple repetitive operations. **OLAP** systems are management-critical to support the decision of enterprise support tasks using detailed investigation.

3) Nature: Although **SQL** queries return a set of data, OLTP methods are designed to step one record at the time, for example, a data related to the user who may be on the phone or in the store. OLAP system is not designed to deal with individual customer records. Instead, they include queries that deal with many data at a time and provide summary or aggregate information to a manager. OLAP applications include data stored in a data warehouses that have been extracted from many tables and possibly from more than one enterprise database.

4) Design: **OLTP** database operations are designed to be application-oriented while **OLAP** operations are designed to be subject-oriented. OLTP systems view the enterprise record as a collection of tables (possibly based on an entity-relationship model). **OLAP** operations view enterprise information as multidimensional).

5) Data: OLTP systems usually deal only with the current status of data. For example, a record about an employee who left three years ago may not be feasible on the Human Resources System. The old data may have been achieved on some type of stable storage media and may not be accessible online. On the other hand, OLAP systems needed historical data over several years since trends are often essential in decision making.

6) Kind of use: **OLTP** methods are used for reading and writing operations while OLAP methods usually do not update the data.

7) View: An **OLTP** system focuses primarily on the current data within an enterprise or department, which does not refer to historical data or data in various organizations. In contrast, an **OLAP** system spans multiple version of a database schema, due to the evolutionary process of an organization. OLAP system also deals with information that originates from different organizations, integrating information from many data stores. Because of their huge volume, these are stored on multiple storage media.

8) Access Patterns: The access pattern of an OLTP system consist primarily of short, atomic transactions. Such a system needed concurrency control and recovery techniques. However, access to OLAP systems is mostly read-only operations because these data warehouses store historical information.

The biggest difference between an OLTP and OLAP system is the amount of data analyzed in a single transaction. Whereas an OLTP handles many concurrent customers and queries touching only a single data or limited collection of records at a time, an OLAP system must have the efficiency to operate on millions of data to answer a single query.

2.6.7 OLAP Operations in the Multidimensional Data Model

Roll-Up

The roll-up operation (also known as **drill-up** or **aggregation operation**) performs aggregation on a data cube, by climbing down concept hierarchies, i.e., dimension reduction. Roll-up is like **zooming-out** on the data cubes. Figure shows the result of roll-up operations performed on the dimension location. The hierarchy for the location is defined as the Order Street, city, province, or state, country. The roll-up operation aggregates the data by ascending the location hierarchy from the level of the city to the level of the country.

The roll-up operation groups the information by levels of temperature.

The following diagram illustrates how roll-up works.

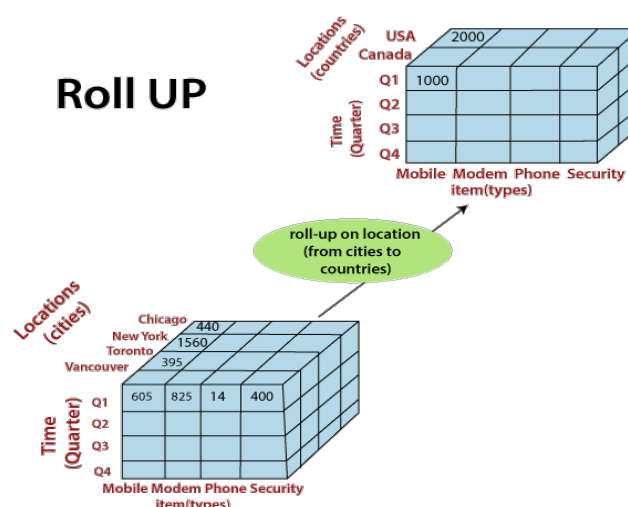


Figure 2.15 Roll UP

Drill-Down

The drill-down operation (**also called roll-down**) is the reverse operation of **roll-up**. Drill-down is like **zooming-in** on the data cube. It navigates from less detailed record to more detailed data. Drill-down can be performed by either **stepping down** a concept hierarchy for a dimension or adding additional dimensions.

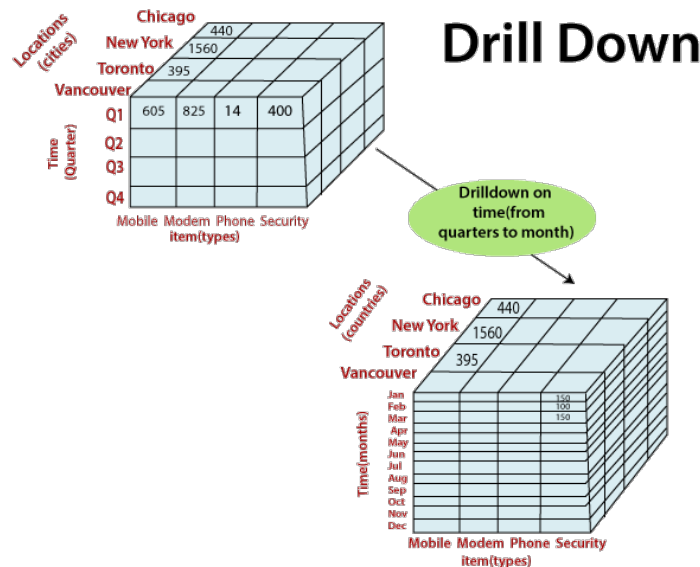


Figure 2.16 Drill Down

Slice

A **slice** is a subset of the cubes corresponding to a single value for one or more members of the dimension. For example, a slice operation is executed when the customer wants a selection on one dimension of a three-dimensional cube resulting in a two-dimensional site. So, the Slice operations perform a selection on one dimension of the given cube, thus resulting in a sub cube.

The following diagram illustrates how Slice works.

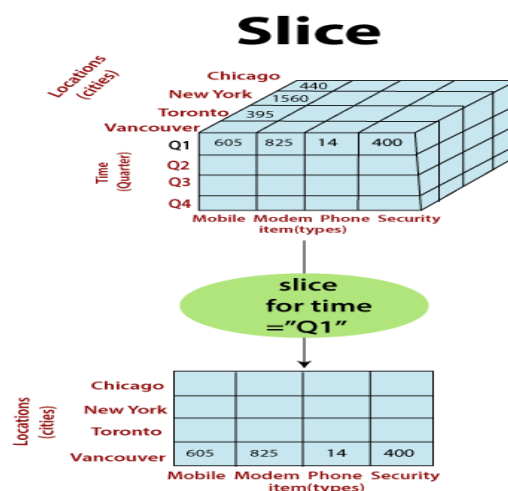


Figure 2.17 Slice

Here Slice is functioning for the dimensions "time" using the criterion time = "Q1".

It will form a new sub-cubes by selecting one or more dimensions.

Dice

The dice operation describes a subcube by operating a selection on two or more dimension.

Consider the following diagram, which shows the dice operations.

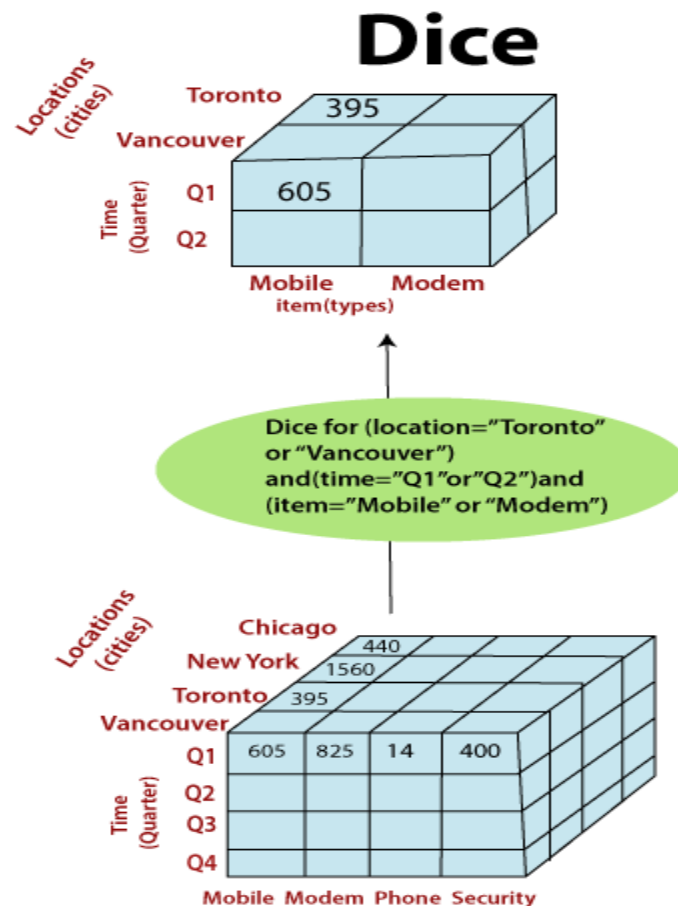


Figure 2.18 Dice

Pivot

The pivot operation is also called a rotation. Pivot is a visualization operations which rotates the data axes in view to provide an alternative presentation of the data. It may contain swapping the rows and columns or moving one of the row-dimensions into the column dimensions.

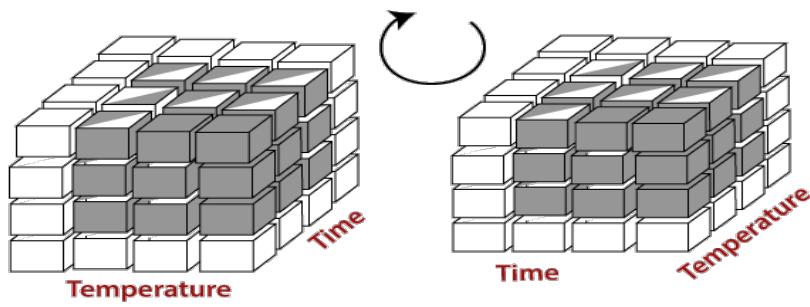


Figure 2.19 Pivot

Consider the following diagram, which shows the pivot operation.

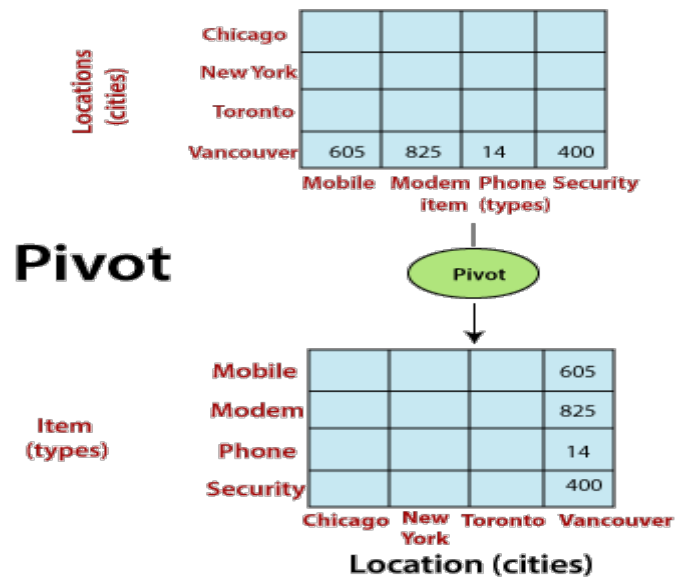


Figure 2.19 Pivot operation

2.6.8 Types of OLAP

There are three main types of OLAP servers are as following:

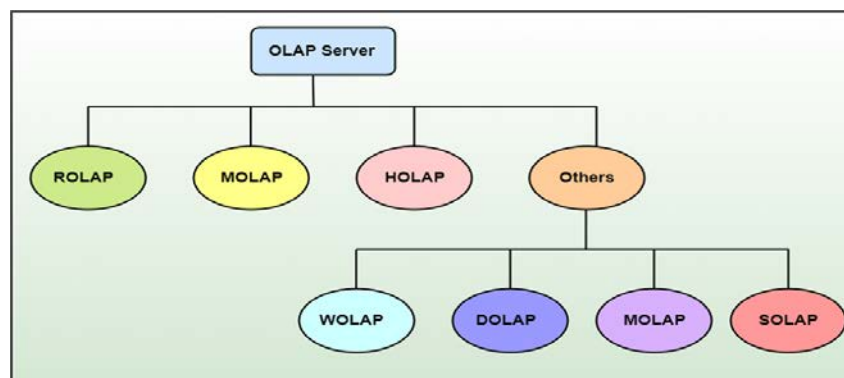


Figure 2.19 Types of OLAP

ROLAP stands for Relational OLAP, an application based on relational DBMSs.

MOLAP stands for Multidimensional OLAP, an application based on multidimensional DBMSs.

HOLAP stands for Hybrid OLAP, an application using both relational and multidimensional techniques.

Relational OLAP (ROLAP) Server

These are intermediate servers which stand in between a relational back-end server and user frontend tools.

They use a relational or extended-relational DBMS to save and handle warehouse data, and OLAP middleware to provide missing pieces.

ROLAP servers contain optimization for each DBMS back end, implementation of aggregation navigation logic, and additional tools and services.

ROLAP technology tends to have higher scalability than MOLAP technology.

ROLAP systems work primarily from the data that resides in a relational database, where the base data and dimension tables are stored as relational tables. This model permits the multidimensional analysis of data.

This technique relies on manipulating the data stored in the relational database to give the presence of traditional OLAP's slicing and dicing functionality. In essence, each method of slicing and dicing is equivalent to adding a "WHERE" clause in the SQL statement.

Relational OLAP Architecture

ROLAP Architecture includes the following components

- o Database server.
- o ROLAP server.
- o Front-end tool.

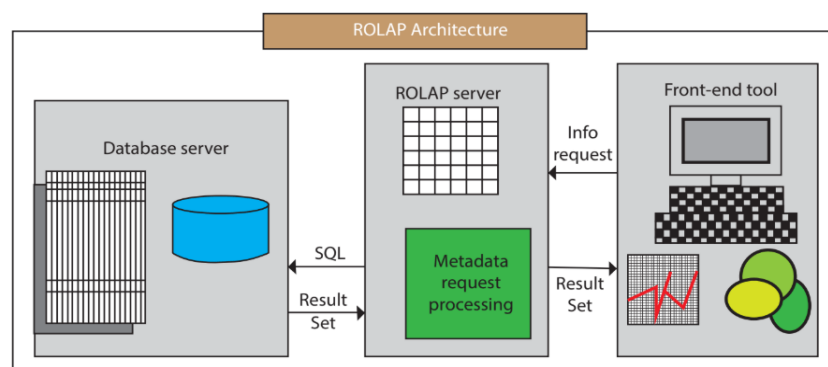


Figure 2.20 ROLAP Architecture

Relational OLAP (ROLAP) is the latest and fastest-growing OLAP technology segment in the market. This method allows multiple multidimensional views of two-dimensional relational tables to be created, avoiding structuring record around the desired view.

Some products in this segment have supported reliable SQL engines to help the complexity of multidimensional analysis. This includes creating multiple SQL statements to handle user requests, being 'RDBMS' aware and also being capable of generating the SQL statements based on the optimizer of the DBMS engine.

Advantages

Can handle large amounts of information: The data size limitation of ROLAP technology is depends on the data size of the underlying RDBMS. So, ROLAP itself does not restrict the data amount.

RDBMS already comes with a lot of features. So ROLAP technologies, (works on top of the RDBMS) can control these functionalities.

Disadvantages

Performance can be slow: Each ROLAP report is a SQL query (or multiple SQL queries) in the relational database, the query time can be prolonged if the underlying data size is large.

Limited by SQL functionalities: ROLAP technology relies on upon developing SQL statements to query the relational database, and SQL statements do not suit all needs

Multidimensional OLAP (MOLAP) Server

A MOLAP system is based on a native logical model that directly supports multidimensional data and operations. Data are stored physically into multidimensional arrays, and positional techniques are used to access them.

One of the significant distinctions of **MOLAP** against a **ROLAP** is that data are summarized and are stored in an optimized format in a multidimensional cube, instead of in a relational database. In MOLAP model, data are structured into proprietary formats by client's reporting requirements with the calculations pre-generated on the cubes.

MOLAP Architecture

MOLAP Architecture includes the following components

- o Database server.
- o MOLAP server.
- o Front-end tool.

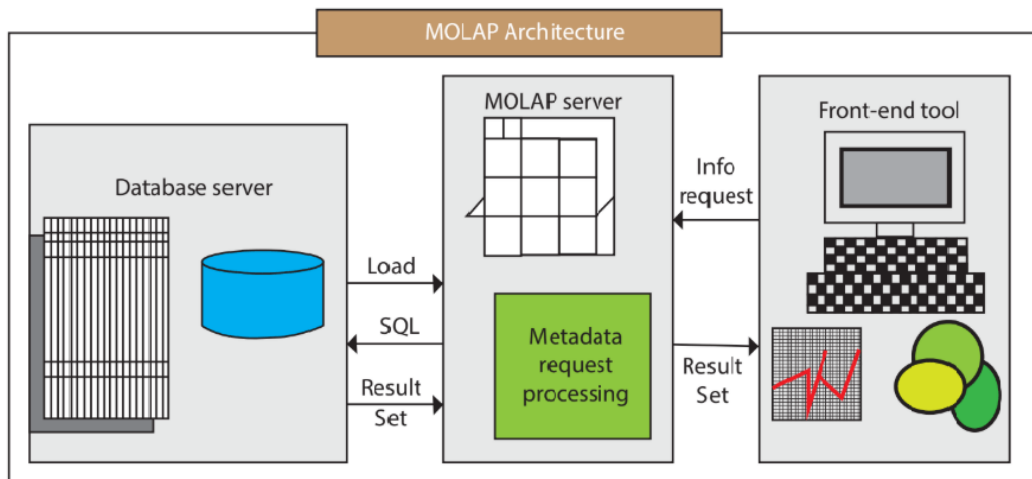


Figure 2.21 MOLAP Architecture

MOLAP structure primarily reads the precompiled data. MOLAP structure has limited capabilities to dynamically create aggregations or to evaluate results which have not been pre-calculated and stored.

Applications requiring iterative and comprehensive time-series analysis of trends are well suited for MOLAP technology (e.g., financial analysis and budgeting).

Examples include Arbor Software's Essbase, Oracle's Express Server, Pilot Software's Lightship Server, Sniper's TM/1, Planning Science's Gentium and Kenan Technology's Multiway.

Some of the problems faced by clients are related to maintaining support to multiple subject areas in an RDBMS. Some vendors can solve these problems by continuing access from MOLAP tools to detailed data in and RDBMS.

This can be very useful for organizations with performance-sensitive multidimensional analysis requirements and that have built or are in the process of building a data warehouse architecture that contains multiple subject areas.

An example would be the creation of sales data measured by several dimensions (e.g., product and sales region) to be stored and maintained in a persistent structure. This structure would be provided to reduce the application overhead of performing calculations and building aggregation during initialization. These structures can be automatically refreshed at predetermined intervals established by an administrator.

Advantages

Excellent Performance: A MOLAP cube is built for fast information retrieval, and is optimal for slicing and dicing operations.

Can perform complex calculations: All evaluation have been pre-generated when the cube is created. Hence, complex calculations are not only possible, but they return quickly.

Disadvantages

Limited in the amount of information it can handle: Because all calculations are performed when the cube is built, it is not possible to contain a large amount of data in the cube itself.

Requires additional investment: Cube technology is generally proprietary and does not already exist in the organization. Therefore, to adopt MOLAP technology, chances are other investments in human and capital resources are needed.

Hybrid OLAP (HOLAP) Server

HOLAP incorporates the best features of **MOLAP** and **ROLAP** into a single architecture. HOLAP systems save more substantial quantities of detailed data in the relational tables while the aggregations are stored in the pre-calculated cubes. HOLAP also can drill through from the cube down to the relational tables for delineated data. The **Microsoft SQL Server 2000** provides a hybrid OLAP server.

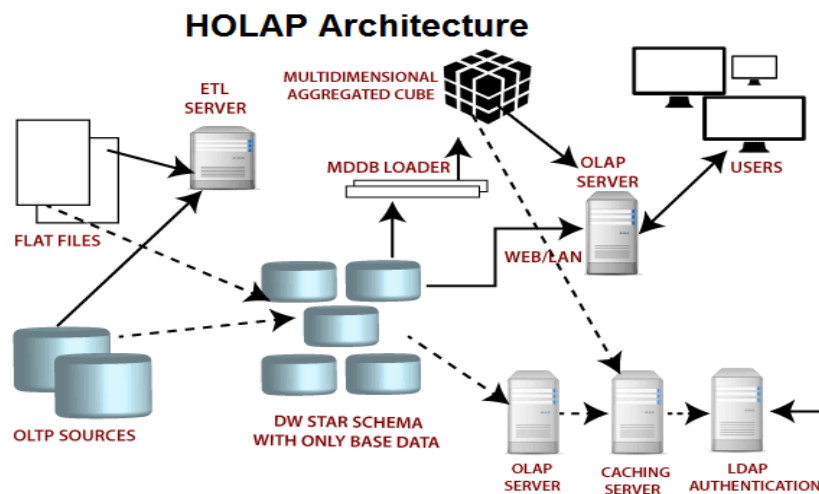


Figure 2.22 HOLOP Architecture

Advantages of HOLAP

1. HOLAP provide benefits of both MOLAP and ROLAP.
2. It provides fast access at all levels of aggregation.
3. HOLAP balances the disk space requirement, as it only stores the aggregate information on the OLAP server and the detail record remains in the relational database. So no duplicate copy of the detail record is maintained.

Disadvantages of HOLAP

1. HOLAP architecture is very complicated because it supports both MOLAP and ROLAP servers.

Other Types

There are also less popular types of OLAP styles upon which one could stumble upon every so often. We have listed some of the less popular brands existing in the OLAP industry.

Web-Enabled OLAP (WOLAP) Server

WOLAP pertains to OLAP application which is accessible via the web browser. Unlike traditional client/server OLAP applications, WOLAP is considered to have a three-tiered architecture which consists of three components: a client, a middleware, and a database server.

Desktop OLAP (DOLAP) Server

DOLAP permits a user to download a section of the data from the database or source, and work with that dataset locally, or on their desktop.

Mobile OLAP (MOLAP) Server

Mobile OLAP enables users to access and work on OLAP data and applications remotely through the use of their mobile devices.

Spatial OLAP (SOLAP) Server

SOLAP includes the capabilities of both Geographic Information Systems (GIS) and OLAP into a single user interface. It facilitates the management of both spatial and non-spatial data.

2.7 ANALYTIC FUNCTIONS

The overall functionality for retrieving, processing, and formatting data requested by a query is provided by Analytic Manager.

Analytic Manager, a component of the SAP BW/4HANA server, lies between the user and the database. It makes the multidimensionally formatted data available to both the front end and, using special interfaces (Open Analysis Interfaces), to third-party administrator front ends. Analytic Manager is therefore optimized for analyzing and reporting large datasets.

The table below provides an overview of the OLAP functions and services implemented in Analytic Manager in SAP BW/4HANA.

UNIT – 3 BUSINESS FORECASTING

Introduction to Business Forecasting and Predictive analytics – Logic and Data Driven Models – Data Mining and Predictive Analysis Modelling – Machine Learning for Predictive analytics.

BUSINESS FORECASTING

- ▣ Business forecasting refers to the process of predicting future market conditions by using business intelligence tools and forecasting methods to analyze historical data.

Forecasting is a technique for



METHODS OF FORECASTING

Companies choose between two basic methods when they want to predict what can happen in the future

- ▣ 1. Qualitative methods
- ▣ 2. Quantitative methods

Qualitative methods:

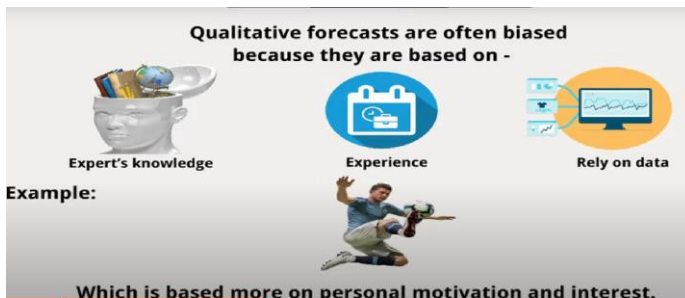
- ▣ Qualitative models have typically been successful with short-term predictions, where the scope of the forecast was limited.
- ▣ Qualitative models can be useful in predicting the short-term success of companies, products, and services, but they have limitations due to their reliance on opinion over measurable data.

Qualitative models include:

1. **Market research:** Polling a large number of people on a specific product or service to predict how many people will buy or use it once launched.
2. **Delphi method:** Asking field experts for general opinions and then compiling them into a forecast.

Quantitative methods:

- ▣ Also called Judgmental methods



Quantitative models include

1. [Indicator](#) approach
2. [Econometric](#) modelling
3. [Time series](#) methods

Process of Forecasting:

- ▣ Forecasters need to follow a careful process in order to generate accurate results.

Step 1: Develop the basis of forecasting

The first step in the process is developing the basis of the analysis of the company's condition and determining where the business is currently standing in the market.



- ▣ Depending upon the research organized during the first step, the second part of forecasting is concerned with estimating the future conditions of the industry how the business operates and predicting how the company will handle it.

Step 3. Regulate the forecast



Different forecasts in the past



Comparing them with the actual impact



Analyzed differences in previous results and current forecasts



Reasons for the deviations are considered

Step 4. Review the process



Every step is checked



Made refinements and modifications

Benefits of forecasting:

Helps to Predict the Future

It gives management a general idea of what to expect.



Which will allow it to function better in the marketplace.



Sense of direction

Good for Customers



Predict demand



Products are always available

There is a greater chance of



Meeting orders



Delivering on time

Keeps a Company Up-to-date

Businesses that forecast regularly must think ahead all the time.



Anticipate changing market trends

Learn from Past Experience



Helps people remember -
what worked previously?
what didn't?



Receiving Financing






Business Forecasting Methods:

1. Delphi Method
2. Market Research
3. Time Series Analysis(also as Trend analyze method)
4. The Average Approach
5. The Naïve Approach(forecast based on last observed value)

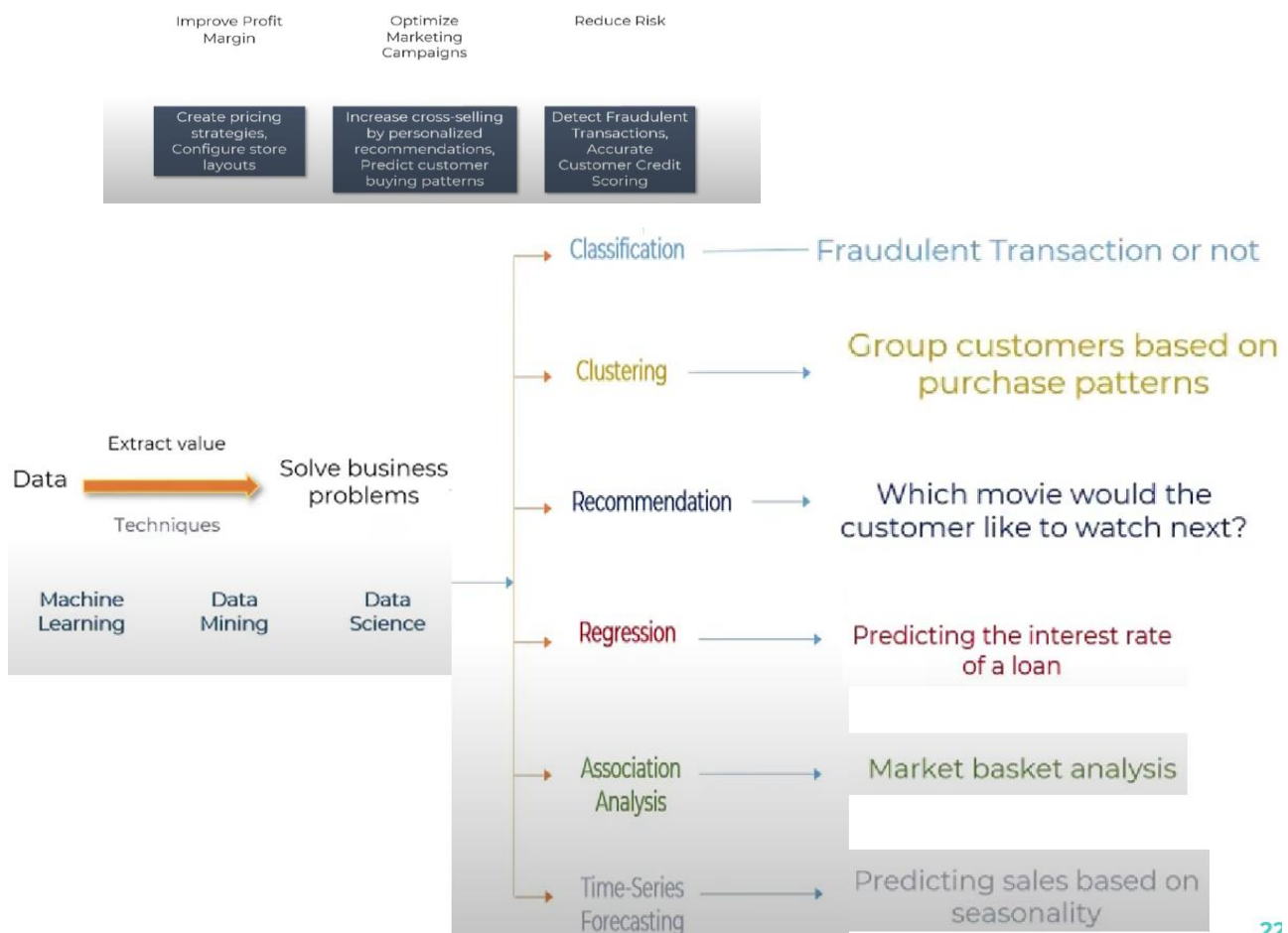
Elements of Business Forecasting:

- ▣ Develop the Basis
 - investigate the current economic situation around you
- ▣ Estimating Future Business Operations
- ▣ Regulating Forecasts
- ▣ Reviewing Forecasting Process

PREDICTIVE ANALYTICS

- ▣ What is going to happen?

How Does Predictive Analytics
Help Organizations?

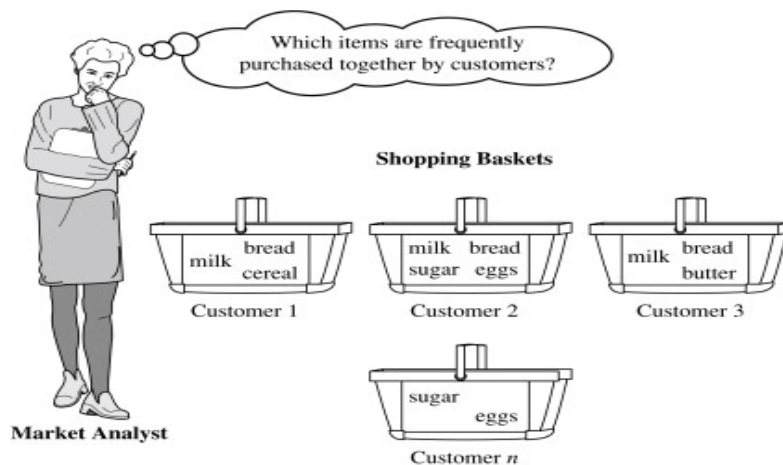


Types of predictive modelling:

- ▣ Classification models
 - [logistic regression](#), decision trees, random forest, neural networks, and Naïve Bayes.
- ▣ Clustering model
 - Common clustering algorithms include k-means clustering, mean-shift clustering, density-based spatial clustering of applications with noise (DBSCAN), expectation-maximization (EM) clustering using Gaussian Mixture Models (GMM), and hierarchical clustering.
- ▣ Time series models
 - Autoregressive (AR), moving average (MA), ARMA, and ARIMA models are all frequently used time series models.

Market Basket Analysis(Association Analysis):

- ❖ data mining technique used by retailers to increase sales by better understanding customer purchasing patterns.



REGRESSION:

- ▣ Regression analysis is a statistical process that helps assess the relationships between a dependent variable and one or more independent variables.

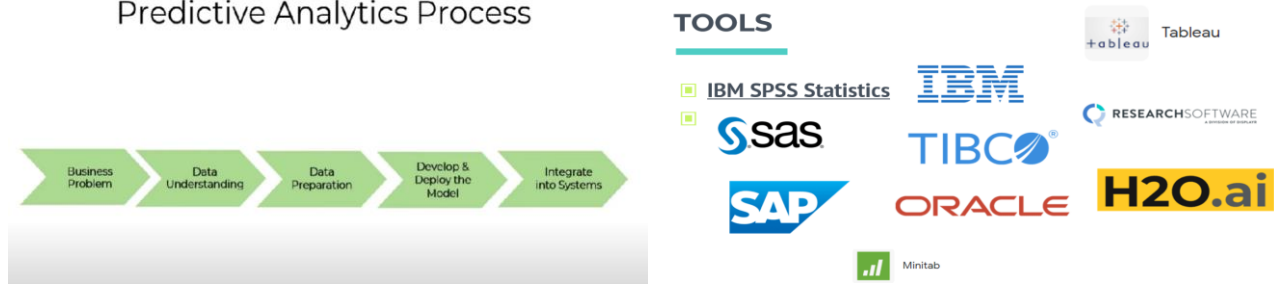
Predictive Analytics Examples

- Financial services use machine learning to predict credit risk and detect fraud.
- Identify and hire employees and predict an employee's performance level.
- Identify product recommendations, forecast sales, analyze markets and manage seasonal inventory.

Examples:

- ▣ 1. Finance: Forecasting Future Cash Flow
- ▣ 2. Entertainment & Hospitality: Determining Staffing Needs
- ▣ 3. Marketing: Behavioural Targeting
- ▣ 4. Manufacturing: Preventing Malfunction
- ▣ 5. Health Care: Early Detection of Allergic Reactions

Predictive Analytics Process



Predictive Analytics: Advantages & Disadvantages

ADVANTAGES

- Gives you an advantage in the market by revealing future insights.
- Saves time, money, energy and other resources.
- Enables better business decision-making and growth.

DISADVANTAGES

- Setting up entire projects requires extensive planning.
- Getting the correct team onboarded might pose a problem.
- You need to spend time to let machine learning learn the business.

Predictive Analytics Techniques

- ▣ Decision Trees
- ▣ Neural Networks
- ▣ Text Analytics
- ▣ Regression Model

LOGIC AND DATA DRIVEN MODELS

- ▣ **Logic driven models** remain based on experience, knowledge and logical relationships of variables and constants connected to the desired business performance outcome situation.
- ▣ **Data-driven Models** refers to the models in which data is collected from many sources to qualitatively establish model relationships. Data driven models include sampling and estimation, regression analysis, correlation analysis, forecasting models and stimulation.

Logic Driven Model:

- ▣ A logic-driven model refers to a system or approach that relies on logical reasoning, rules, and principles to draw conclusions, make decisions, or solve problems.

- ▣ In Business Forecasting:

A logic-driven model in business analytics refers to a data-driven approach that uses logical reasoning and mathematical principles to analyze and solve business problems.

This approach typically involves the use of various analytical techniques, algorithms, and statistical methods to derive insights from data and make informed business decisions

Key components of a logic-driven model in business analytics include:

1. Data Collection:

- from databases, spreadsheets, and other sources, as well as unstructured data from social media, text documents, and more.

2. Data Preprocessing:

- Once the data is collected, it needs to be cleaned, transformed, and organized to ensure its quality and suitability for analysis. This step involves removing duplicates, handling missing values, and converting data into a suitable format.

3. Data Exploration:

- In this stage, analysts visualize and explore the data to understand patterns, relationships, and potential insights.

4. Model Selection:

- These can include linear regression, logistic regression, decision trees, random forests, support vector machines, and neural networks, among others.

5. Model Building:

- the chosen model is trained on the prepared data to learn patterns and relationships between variables. The model undergoes an iterative process to optimize its performance and generalizability.

6. Model Evaluation:

- After training, the model's performance is evaluated using metrics such as accuracy, precision, recall, and F1-score. The goal is to ensure the model is accurate and reliable in making predictions.

7. Model Deployment:

- Once the model is considered satisfactory, it is deployed in the business environment to assist in decision-making. This can involve integrating it into existing systems or using it as a standalone tool.

8. Monitoring and Maintenance:

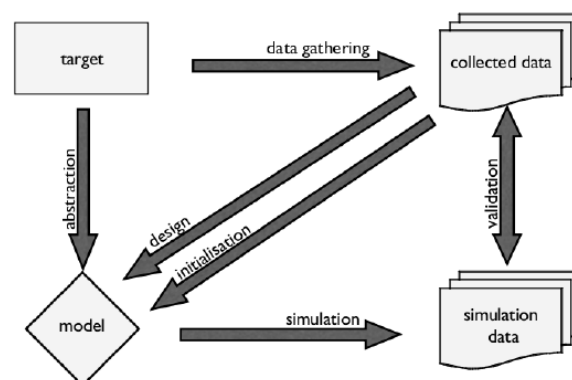
- Business analytics is an ongoing process, and it is crucial to monitor the model's performance in real-world scenarios continuously. Regular maintenance and updates are necessary to keep the model relevant and effective over time.

Key characteristics of a logic-driven model include:

1. **Rules and Conditions**
2. **Deterministic Behavior:** Logic-driven models are deterministic, meaning that given the same inputs and conditions, they will always produce the same output or conclusion. There is no randomness or probability involved in their decision-making process.
3. **Formal Representation:** The rules and logic used in these models are typically represented in a formal and structured way. This allows for clear and unambiguous communication of the model's behavior.
4. **Inference and Deduction:** Logic-driven models use inference and deduction to draw conclusions based on the given rules and facts. They can derive new information from existing knowledge.
5. **Transparent and Explainable**
6. **Limitations**

Examples of logic-driven models include:

1. **Expert Systems:** Expert systems are computer programs that use a knowledge base of rules and facts to mimic the decision-making capabilities of human experts in a specific domain. These systems are often used in fields like medicine, finance, and engineering.
2. **Rule-based Systems:** Rule-based systems are used in various applications where decision-making can be explicitly defined using rules and conditions. For example, they are widely used in business processes and workflow automation.
3. **Logic Programming:** Logic programming languages, like Prolog, are based on formal logic and are used for solving problems using a set of rules and facts represented as logical clauses.



Example 1:

- ▣ A restaurant customer dines 6 times a year and spends an average of \$50 per visit. The restaurant realizes a 40% margin on the average bill for food and drinks.

Annual gross profit on a customer = $\$50(6)(0.40) = \120

30% of customers do not return each year.

Average lifetime of a customer = $1/.3 = 3.33$ years.

Average gross profit for a customer = $\$120(3.33) = \400

OR Average gross profit for a customer = $\$120/.3 = \400

$$V = \frac{R \times F \times M}{D}$$

▣ Thus, the economic value of a customer is

- V = value of a loyal customer
- R = revenue per purchase
- F = purchase frequency (number visits per year)
- M = gross profit margin
- D = defection rate (proportion customers not returning each year)

Example 2: Profit Model

- Develop a decision model for predicting profit in face of uncertain demand.

P = profit
 R = revenue
 C = cost

p = unit price
 c = unit cost
 F = fixed cost
 S = quantity sold
 D = demand
 Q = quantity produced

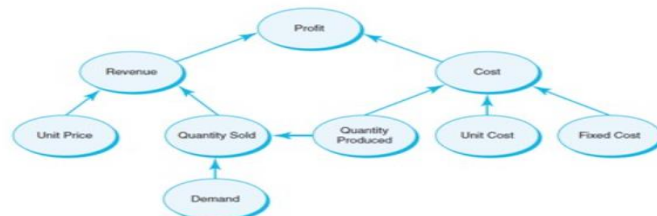


Figure 8.1

- Cost = fixed cost + variable cost
 $C = F + cQ$
- Revenue = price times quantity sold
 $R = pS$
- Quantity sold = Minimum{demand, quantity sold}
 $S = \min\{D, Q\}$
- Profit = Revenue – Cost
 $P = p \cdot \min\{D, Q\} - (F + cQ)$

- $p = \$40$
- $c = \$24$
- $F = \$400,000$
- $D = 50,000$
- $Q = 40,000$

Compute:

- $R = p * \min\{D, Q\}$
 $= 40(40,000) = 1,600,000$
- $C = F + cQ = 1,360,000$
 $= 400,000 + 24(40,000)$
- $P = R - C = 1,600,000 - 1,360,000 = \$240,000$

	A	B	C
1	Profit Model		
2			
3	Data		
4			
5	Unit Price	\$40.00	
6	Unit Cost	\$24.00	
7	Fixed Cost	\$400,000.00	
8	Demand	50000	
9			
10			
11	Model		
12			
13	Unit Price	\$40.00	
14	Quantity Sold	40000	
15	Revenue		\$1,600,000.00
16			
17	Unit Cost	\$24.00	
18	Quantity Produced	40000	
19	Variable Cost		\$960,000.00
20	Fixed Cost		\$400,000.00
21			
22	Profit		\$240,000.00

	A	B	C
1	Profit Model		
2			
3	Data		
4			
5	Unit Price	\$40.00	
6	Unit Cost	\$24.00	
7	Fixed Cost	\$400,000.00	
8	Demand	50000	
9			
10			
11	Model		
12			
13	Unit Price	\$40.00	
14	Quantity Sold	40000	
15	Revenue		\$1,600,000.00
16			
17	Unit Cost	\$24.00	
18	Quantity Produced	40000	
19	Variable Cost		\$960,000.00
20	Fixed Cost		\$400,000.00
21			
22	Profit		\$240,000.00

Figure 8.2a

	A	B	C
1	Profit Model		
2			
3	Data		
4			
5	Unit Price	40	
6	Unit Cost	24	
7	Fixed Cost	400000	
8	Demand	50000	
9			
10			
11	Model		
12			
13	Unit Price	=B5	
14	Quantity Sold	=MIN(B8,B18)	
15	Revenue		=B13*B14
16			
17	Unit Cost	=B6	
18	Quantity Produced	40000	
19	Variable Cost		=B17*B18
20	Fixed Cost		=B7
21			
22	Profit		=C15-C19-C20

Figure 8.2b

Data Driven Model

Logic-drive modeling is usually the first step to establish relationships through data driven models (using data collected from many sources to quantitatively establish model relationships).

- ❑ The data driven models build relationships between input and output data, without worrying too much about the underlying processes, using statistical/machine learning techniques.
- ❑ On the other hand, the process driven models are based on well established mathematical/physical laws.
- ❑ A **linear regression** model is an example of a data driven model that, for example, builds a relationship between a dependent variable and a set of independent variables.
- ❑ **Language Translators** are built purely by a data-driven approach. For example, Google Translate does not know anything about any language.
- ❑ is built using Deep Neural Networks based on statistics and probabilities rather than grammar rules of the language.

Example

In the spring, a department store introduces a new line of bathing suits that sells for \$70. The store purchases 1000 of these bathing suits. During the prime selling season, the store sells an average of 7 units per day at full price (40 days). On 10 sale days, the price is discounted 30% and sales increase to 32.2 units per day. Around July 4th, the price is marked down 70% to sell off remaining inventory. Determine total revenue from the bathing suits.

Assume a linear trend model between sales and price:

$$\text{daily sales} = a - b(\text{price})$$

$$7 = a - b(70)$$

$$32.2 = a - b(49)$$

$$\text{Daily sales} = 91 - 1.2(\text{price})$$

Revenue from full retail sales

$$= \text{units sold} * \text{days} * \text{price}$$

$$= (7)*(40)*(70)$$

$$= \$19,600$$

Revenue from sale weekends

$$= (32.2)*(10)*(49)$$

$$= \$15,778$$

Revenue from clearance sales

$$= \text{leftovers} * \text{price}$$

$$= (1000 - 7(40) - 32.2(10))*(21)$$

$$= (398)(21)$$

$$= \$8,358$$

$$\text{Total revenue} = \$43,736$$

DATA MINING

- ▣ Data mining is the process of searching and analyzing a large batch of raw data in order to identify patterns and extract useful information.
- ▣ Companies use data mining software to learn more about their customers. It can help them to develop more effective marketing strategies, increase sales, and decrease costs. Data mining relies on [effective data collection](#), [warehousing](#), and computer processing.

How Data Mining Works

The data mining process breaks down into four steps:

- Data is collected and loaded into data warehouses on-site or on a cloud service.
- Business analysts, management teams, and information technology professionals access the data and determine how they want to organize it.

- Custom application software sorts and organizes the data.
- The end user presents the data in an easy-to-share format, such as a graph or table.

Data Mining and Predictive Analysis Modelling

- ▣ Businesses prefer data mining because it aims to predict. Predictive analyses, on the other hand, refine data resources, in particular, to extract hidden value from those newly discovered patterns.

“Data mining + Domain knowledge => predictive analytics => Business Value”

Overall, predictive analysis and data mining, both make use of algorithms to discover knowledge and find the best possible solutions around.

Data Mining Techniques:

- ▣ Data mining uses algorithms and various other techniques to convert large collections of data into useful output.

The most popular types of data mining techniques include:

- Association rules
- Classification
- Clustering
- Decision Trees
- K-Nearest neighbor (KNN)
- Neural networks Predictive analysis



The Data Mining Process:

- ▣ Step 1: Understand the Business
- ▣ Step 2: Understand the Data
- ▣ Step 3: Prepare the Data
- ▣ Step 4: Build the Model

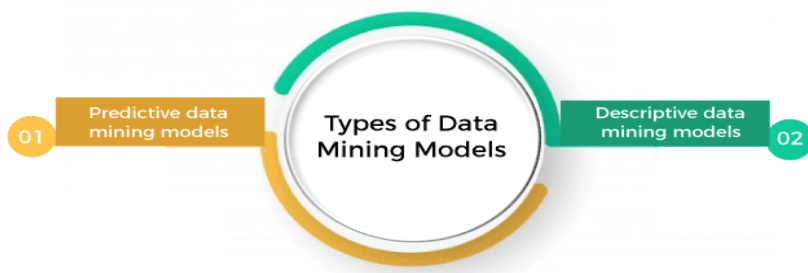
- ▣ Step 5: Evaluate the Results
- ▣ Step 6: Implement Change and Monitor

DATA MINING MODELS

- ▣ A [Data mining](#) model refers to a method that usually use to present the information and various ways in which they can apply information to specific questions and problems.

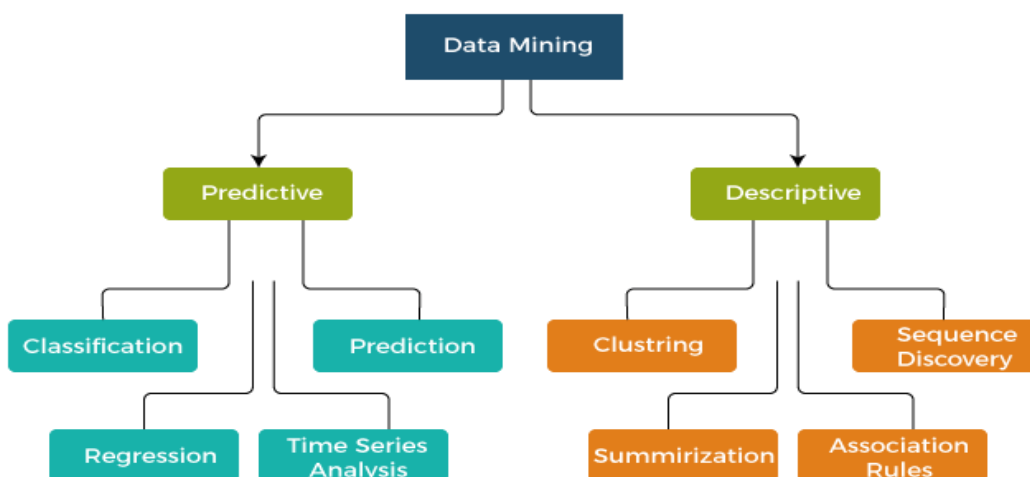
As per the specialists, the data mining regression model is the most commonly used data mining model.

Types of Data Mining Models



Predictive data mining models:

- ▣ A predictive data mining model predicts the values of data using known results gathered from the different data sets.
- ▣ The main objective of predictive data mining models is to predict the future based on the past data, generally but not always on the statistical modeling.
- ▣ Predictive modeling is used in healthcare industries to identify high-risk patients with congestive heart failures, high blood pressure, diabetes, infection, cancer, etc.
- ▣ It is also used in the vehicle insurance company to assign the risk of accidents to the policyholder with congestive heart failures, high blood pressure, diabetes, infection, cancer, etc. It is also used in the vehicle insurance company to assign the risk of accidents to the policyholder.



- ▣ A predictive model of a data mining task comprises classification, regression, prediction, and time series analysis. The predictive model of data mining is also called statistical regression.

Classification:

- ▣ classification refers to a form of data analysis where a machine learning model assigns a specific category to a new observation.
- ▣ One example of classification in the banking and financial services industry is identifying whether transactions are fraudulent or not. In the same way, machine learning can also be used to predict whether a loan application would be approved or not.

Regression:

- ▣ Regression refers to a method that verifies the value of data for a function. Generally, it is used for appropriate data.
- ▣ A linear regression model in the context of machine learning or statistics is basically a linear approach for modeling the relationships between the dependent variable known as the result and your independent variable is known as features.

▣ Types of regression

1. Linear Regression

2. Multi-linear regression

- ▣ If your model has only one independent variable, it is called simple linear regression, and else it is called multiple linear regression.

Prediction:

- ▣ In data mining, prediction is used to identify data value based on the description of another corresponding data value. The prediction in data mining is known as Numeric Prediction.
- ▣ Generally, regression analysis is used for prediction. For example, in credit card fraud detection, data history for a particular person's credit card usage has to be analyzed. If any abnormal pattern was detected, it should be reported as 'fraudulent action'.

Time series analysis:

- ▣ Time series analysis refers to the data sets based on time. It serves as an independent variable to predict the dependent variable in time.

Clustering:

- ▣ Clustering is grouping a set of objects so that objects in the same group called a cluster are more similar than those in other groups clusters.

Association rules:

- ▣ Association rules determine a causal relationship between huge sets of data objects. The way the algorithm works is that you have.
- ▣ For example, a list of items you purchase at the grocery store for the past six months data, and it calculates a percentage at which items are purchased together.
- ▣ For example, what are the chances of you buying milk with cereal?

Sequence:

- ▣ Sequence refers to the discovery of useful patterns in the data is in relation to some objective of how it is interesting.

Summarization:

- ▣ Summarization holds a data set in more depth which is easy to understand form.

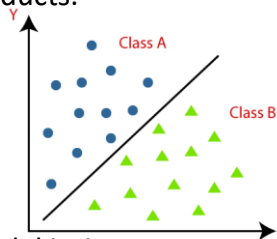
Predictive Analysis Modelling:

- ▣ *Classification Model*
- ▣ *Clustering Model*
- ▣ *Forecast Model*
- ▣ *Outliers Model*
- ▣ *Time Series Model*

Predictive modeling techniques break down into a few key steps:

1. First, all previous data collected is **analyzed** to determine what patterns or parameters the customers you already have followed, and thus what patterns both they and your future customers will follow.
2. Next, You can use this predictive model to see which marketing campaigns in the past have **seen success** with each different segment of your customer base.
3. Finally, you can **determine** which products that were advertised during each campaign did or did not successfully see a rise in sales, which in turn translates to whether those products have a reasonable chance of success if advertised; after all, if a product gets advertised for your marketing but does not see valuable results, it would be better to emphasize other products.

Classification Model



- ▣ The classification model is, in some ways, the simplest of the several types of predictive analytics. It puts data in categories based on what it learns from historical data.

- ▣ Classification models are best to answer yes or no questions, providing broad analysis that's helpful for guiding decisive action. These models can answer questions such as:
 - For a loan provider, “Will this loan be approved?” or “Is this applicant likely to default?”
 - For an online banking provider, “Is this a fraudulent transaction?”

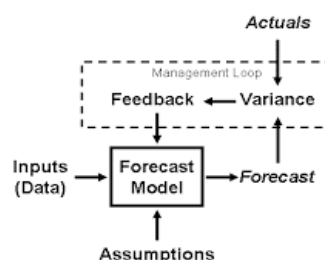
Clustering Model

- ▣ The clustering model sorts data into separate, nested smart groups based on similar attributes.
- ▣ Using the clustering model, they can quickly separate customers into similar groups based on common characteristics and devise strategies for each group at a larger scale.



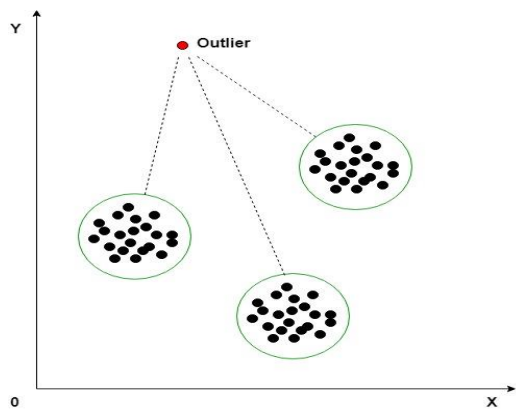
Forecast Model

- ▣ forecast model deals in metric value prediction, estimating numeric value for new data based on learnings from historical data.
- ▣ This model can be applied wherever historical numerical data is available. Scenarios include:
 - A call center can predict how many support calls they will receive per hour.
 - If a restaurant owner wants to predict the number of customers she is likely to receive in the following week
 - What is the weather forecast?



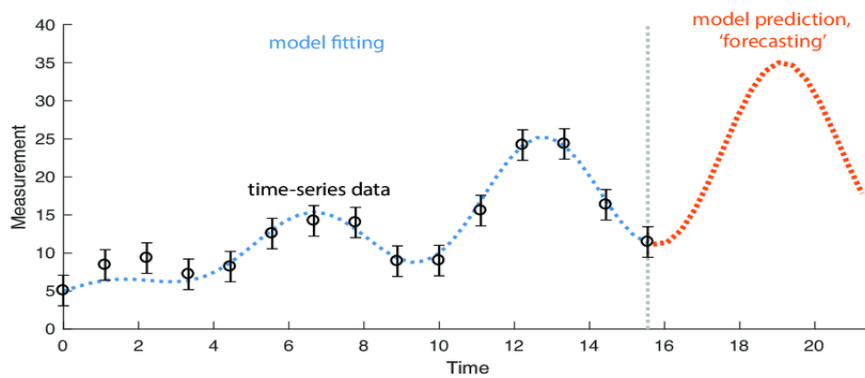
Outliers Model:

- ▣ The outliers model is oriented around anomalous data entries within a dataset. It can identify anomalous figures either by themselves or in conjunction with other numbers and categories.
- ▣ The analysis of outlier data is referred to as outlier analysis or outlier mining.



Time Series Model:

The time series model comprises a sequence of data points captured, using time as the input parameter. It uses the last year of data to develop a numerical metric and predicts the next three to six weeks of data using that metric.



UNIT – 4

HR & SUPPLY CHAIN ANALYTICS

Human Resources - Planning and Recruitment - Training and Development - Supply chain network - Planning Demand, Inventory and Supply - Logistics - Analytics applications in HR & Supply Chain.

4.1 HUMAN RESOURCES

In an organization, Human Resources is the department in charge of all employees and employee-related operations. As a term, we also use it to describe the entire workforce of an organization. In this HR definition, we'll focus on human resources as a business function.

4.1.1 What are HR Analytics?

Definition: Human Resource analytics (HR Analytics) is defined as the area in the field of analytics that deals with people analysis and applying analytical process to the human capital within the organization to improve employee performance and improving employee retention.

4.1.2 The 4 types of HR analytics:

There is a specific purpose for the 4 types of data in HR analytics: Each one offers unique insights and tells a part of the story.

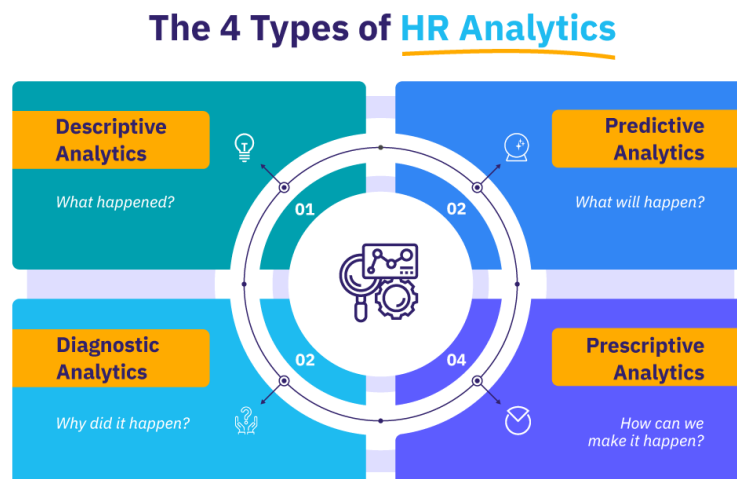


Figure 4.1 Types of HR Analytics

1. Descriptive analytics

The first type of HR analytics on the list is descriptive analytics. This is the most basic type that analyzes data patterns to gain insight into the past.

It is known as decision analytics, and uses statistical analysis techniques to explain or summarize a particular set of historical, raw data. It focuses on past data to account for what happened but doesn't make predictions for the future.

Descriptive analytics advantages	Descriptive analytics disadvantages
– The simplest form of data analysis. – Requires only basic math skills, and it allows you to present complex data in an easy-to-digest format	– Limited to a simple analysis of a few variables after the fact. – For instance, an employee headcount summary captures a time period and reports the “what” but not the “why” or “how.”

Descriptive analytics examples

Efficiency metrics that HR has traditionally tracked fall under the descriptive analytics category. Here are two examples:

PTO: Using descriptive analytics, HR can analyze the average number of paid time off days that employees use in one year.

Turnover: Descriptive analytics could be used to analyze employee turnover rates to compare the annual turnover between two teams or two departments.

2. Diagnostic analytics

Diagnostic analytics takes descriptive analytics to the next level by providing an explanation for what has been revealed. It aims to determine the underlying reasons for what the data exposes.

Diagnostic analytics process:

Conducting a diagnostic analysis typically involves the following steps:

- ✓ Identifying the patterns and anomalies within the data that raise questions and need to be studied further.
- ✓ Discovering what factors could be contributing to the patterns and anomalies to identify the relevant data.
- ✓ Determining causal connections by analyzing the data with various methods.

There are multiple diagnostic analytics techniques, including:

Data drilling: Taking information from a more general overview and providing a more granular view of the data.

Data mining: Extracting patterns from data to help predict future events

Probability theory: Quantifying uncertain measures of random events

Regression analysis: Determining which variables will impact an outcome

Correlation analysis: Tests the relationships between variables

Statistical analysis: Collecting and interpreting data to determine underlying patterns

Diagnostic analytics advantages	Diagnostic analytics disadvantages
– Shows a more comprehensive interpretation of the data for informed decision-making.	– Focuses on past occurrences which makes it very reactive. – Can't provide actionable insights to support your planning process.

3. Predictive analytics

Estimates what might happen in the future; it forecasts future outcomes. The process involves categorizing past and present data to isolate patterns, correlations, and irregularities followed by estimating a model to predict what will occur in the future. Then the model's accuracy is evaluated by applying it to new data.

Predictive analytics advantages	Predictive analytics disadvantages
– It can reduce human error, help you avoid risks, improve operational efficiencies, and refine the forecasting for your organization.	– It requires substantial and relevant data (big data sets). – It's also challenging to ensure that all of the variables are considered, and the model must be updated as data changes.

Predictive analytics is a valuable tool in many HR functions. Here are two predictive analytics use cases:

1. Recruitment

Predictive analytics can analyze data from the hiring process (resumes, job descriptions, etc.) to narrow in on the desired skill sets. Certain elements of social media profiles and answers to automated application questions can also reveal the key attributes that indicate a candidate is a right fit for long-term success with the organization. Then you can tailor your recruitment strategy to attract and engage this type of applicant.

2. Retention

With predictive analytics, you can forecast various talent management outcomes, such as who will quit. The HR department at consumer credit institution Experian developed a predictive model that can identify who's at risk of leaving and what the surrounding factors are.

4. Prescriptive analytics

Prescriptive analytics is the final and most complex stage of the analytics journey that transfers predictive analytics into ideas for what to do next.

A general prescriptive analytics definition would be *the targeted recommendation for decision options and actions based on the findings of predictive analytics*. It offers options for where and how to act to achieve success.

Prescriptive analytics relies on big data and uses an assortment of technical tools, including:

- ✓ Machine learning
- ✓ Algorithms
- ✓ Artificial intelligence
- ✓ Pattern recognition

How prescriptive analytics works

You can think of prescriptive analytics like Netflix for business. It works in the same way that Netflix suggests films based on viewing behaviors. Prescriptive analytics goes beyond predictive analytics with a more pre-emptive approach to looking at the future.

Predictive analytics advantages	Predictive analytics disadvantages
– Equips HR leaders to make informed, real-time decisions to improve performance, solve complicated problems, and take advantage of opportunities. – For example, it can recommend strategies for training that will boost	– An iterative process that requires time. Also, the quality of recommendations is dependent on the quality of the data, so it won't be effective if your data is incomplete or unreliable. – You must also be careful about weighing the options presented and ensure that taking the recommended action is reasonable from an HR perspective. – Algorithms can't always reflect the diverse intricacies of dealing with human beings.

Prescriptive analytics examples:

Here are two prescriptive analytics use cases related to HR.

1. Staffing

Prescriptive analytics can help you prepare for upcoming staffing needs. Data surrounding employees' interactions with digital benefit options can reflect potential openings. An uptick in activity surrounding retirement planning or medical and family leave policies can lead to staffing recommendations that will address departures and long-term absences.

2. Attrition

As mentioned above, Experian is using AI to predict high-flight-risk employees. However, they are also taking the next step with prescriptive analytics to prevent the contributing factors to flight risk from happening.

4.1.3. Examples of HR Analytics

To better understand the use of analytics in HR activities, let us consider the below real-life examples:

- So, an HR manager will use HR analytics if they want to have deeper insights into the contribution of employee engagement in the overall financial performance of the organization. For the analysis, data regarding the performance of the employees for a given period will be collected; the organization's financial performance data will also be collected for that period. Many other types of data will be collected, and all this data will be analyzed. Not only will this help in identifying the underlying reasons, but it will also help the HR department to formulate better employee engagement and performance management policies in the future.

4.1.4 Benefits of HR Analytics

HR management is one of the key concerns for any organization as human resources form the backbone of the entire system. Hence, the usage of HR analytics has increased over the last few years for its varied benefits and outstanding results.

A few of the important benefits of HR analytics are discussed below:

Quality Recruitment

One of the important aspects of HR management is hiring good quality talent. However, the hiring process can be very long and the talent acquisition team has to take care of a multitude of different aspects, which keeps them busy.

Reduced Attrition Levels

These days, employees are preferring to leave organizations within a short span of time. Why? There can be many reasons behind this, but it is crucial for you to know the reason behind employee attrition in your organization.

Builds a Productive Workforce

It is widely known that productivity levels can never be constant given that all employees are different from one other.

Achieves Employee Trust

For example, you implement a new strategy and wish to know the employee reaction; HR analytics can help you with this by collecting the data of each employee's reaction.

Starting with Predictive Data Analytics

Predictive Analytics as we all know analyzes past and historical data to make predictions and forecasting for the future. Predictive analytics is used in HR analytics to get actionable insights on improving human resources related processes and data-driven decision making.

4.1.5 Popular HR Analytics Market Trends

So far, we have discussed how HR analytics has helped a majority of organizations. Hence, at this point, it becomes important to know what are the future market trends in HR analytics. The following are some of the popular HR analytics market trends:

- The global HR analytics market is expected to grow at a compound annual growth rate of 14.2 percent, from 2020 to 2027, to reach US\$6.29 billion by 2027.
- Millions of organizations around the world are expected to witness an increase in artificial intelligence (AI) and machine learning (ML) in HR analytics, thus paving a

way for even more job opportunities for data analytics professionals and HR analytics solution vendors

- The banking, financial services, and insurance (BFSI) sector is expected to be the highest market shareholder in the global HR analytics market.

Top HR Analytics Organizations

With the increase in the opportunities in the HR analytics market, many HR analytics organizations have been founded in the last few years. However, the top ones on the list are SAP, Oracle, Infor, Visier, TALENTSOFT, Kronos, MicroStrategy, IBM, Tableau, Zoho, Workday, Sage Software, Crunchr, GainInsights, and Sisense.

4.1.6 Who is an HR Analyst?

An HR analyst is just like a data analyst but is called a functional data analyst. An HR analyst majorly focuses on the HR data of an organization and helps in improving the overall organization's workforce performance by collecting and analyzing HR data. A majority of an HR analyst's time and performance involves talent analytics, people analytics, or even workforce analytics.

Roles and Responsibilities of an HR Analyst

Now, let us understand in detail what are the roles and responsibilities performed by an HR analyst in an organization:

- To collect and analyze past data for identifying underlying causes and trends in employee attrition
- To collate data regarding HR productivity and engagement
- To understand new employee behavior through data analysis
- To draw insights out of the data and report the findings to the management
- To correlate various types of data and understand the patterns behind important concerns such as low employee performance, compensation revision, etc.
- To create predictive models for better understanding employee behavior, employee performance, etc.

- To develop HR strategies and make decisions to improve the overall performance and employee satisfaction level.

HR business analytics uses the following data-based methods

- Fact-based data
- Statistics
- Quantitative Analysis
- Business modelling
- Data visualisation
- Software such as Python, SQL, R

4.2 PLANNING AND RECRUITMENT IN HR

4.2.1 HUMAN RESOURCE PLANNING

What is human resource planning?

Human resource planning (HRP) describes an ongoing, data-driven process in which a company systematically plans for the future in terms of human resources to ensure that available jobs are suited with appropriately skilled employees. It identifies key HR initiatives for the time period ahead that will help the organization achieve its strategic goals and maintain its competitive advantage without staffing shortages or excesses.

5 steps in the human resource planning process

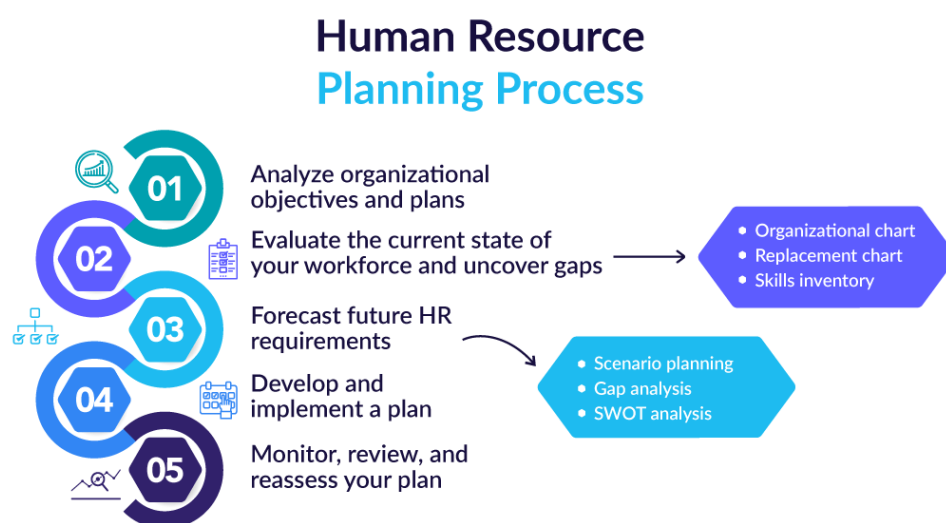


Figure 4.2.1 Human resource planning

The actual process of human resource planning involves five general phases. Listed below is a summary of each step to help you navigate the process:

1. Analyze organizational objectives and plans

Success in HR planning relies on its connection to business goals, so you must start with a final purpose in mind. Determine what the organization wants to achieve in the future, how it intends to accomplish this, and how HR efforts will contribute to it.

2. Evaluate the current state of your workforce and uncover gaps

Before you go forward with plans for the future, you need to assess where the organization's workforce currently stands by assessing its strengths and weaknesses. Do you have the right number of experienced employees?

An **organizational chart** is a visual description of a company's staff structure that designates roles and reporting relationships. Below you can see an example of an HR organization chart in a mid-sized organization:

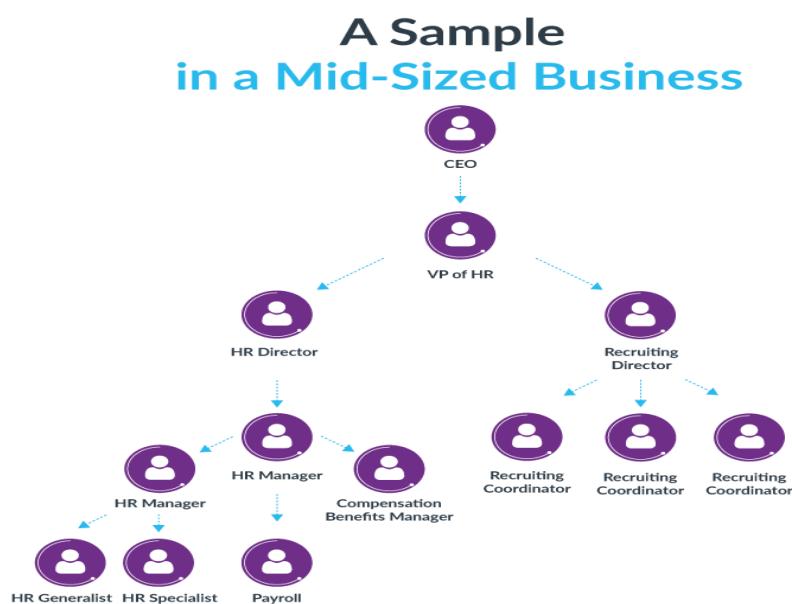


Figure 4.2.2 Mid-Sized organization

A **replacement chart** is a diagram of potential candidates and their readiness to step into certain roles upon employee departures.

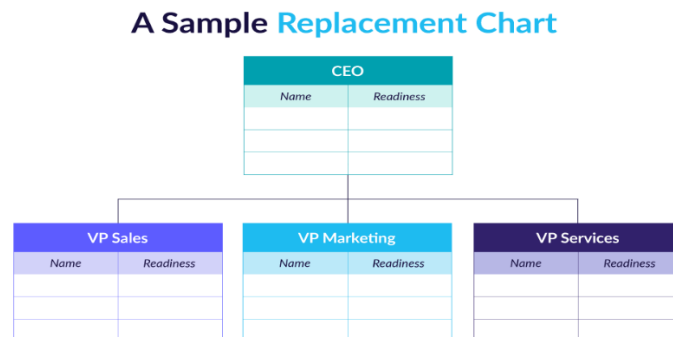


Figure 4.2.3 Replacement chart

A **skills inventory** looks at the availability and preparedness of current employees to move into either lateral or higher-level roles. It allows you to identify which employees have the ability to take on new positions as the needs of the company change. It also helps you uncover any skills deficiencies.

Here is how you can use an employee's skills inventory to identify opportunities for development:

Skill	Opportunity for Development	Formal Training	Coaching and Mentoring	On the Job
Advanced Excel	Work, on an upcoming project, on training via LinkedIn Learning.	x		X
Digital Marketing	A book on digital marketing, learn from a colleague, get involved in social media activities at work.	x	x	X
Video editing	Edit a video for the workplace onboarding programme, attend the 'introduction to video editing seminar.'	x		X

3. Forecast future HR requirements

Forecasting demand is the critical stage of the HR planning process. Bearing in mind the organization's future goals, gauge the coming demand and supply of qualified employees. Demand forecasting involves determining the number of people needed and the level of talent required. Supply forecasting estimates future internal (promotion, transfers, position expansion) and external sources and their ability to meet your needs.

You accomplish this forecasting using internal data and market and industry trends. Be sure to include factors such as these:

- New products or services on the horizon
- Prospective mergers or acquisitions
- Labor costs
- Projected employee retirements/vacancies and turnover rates
- Technological advances and automation

Scenario planning is a way to brainstorm situations that may affect the direction of your organization and labor requirements in the future. Consider technological advancements, economic changes, new government regulations, etc., and consider how you can be flexible and which precautionary measures to take.



Figure 4.2.4 Scenario Planning

A **gap analysis** allows you to compare current employee quantity and skill levels with what will be needed to meet the organization's goals. This method involves:

- Recognizing the current workforce status.
- Ascertaining what the ideal situation looks like.
- Identifying where the current lapses exist.

You can use a SWOT (strengths, weaknesses, opportunities, threats) analysis to compile your findings and see which elements need the most focus.

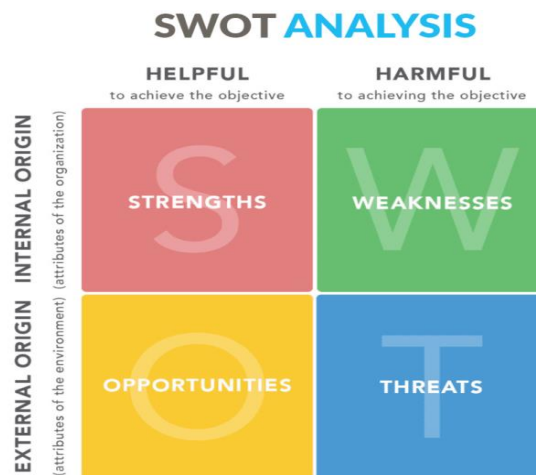


Figure 4.2.5 SWOT Analysis

4. Develop and implement a plan

Now it's time to formulate a human resource action plan that aligns with your organization's overall strategy. It should take into account all the analyses you've completed and include talent strategies to match the supply and demand to get your organization ready for the future.

Your plan can start with the theoretical concept of transforming the company from X to Y and then name the step-by-step approach that HR will facilitate. This will include developing methods to enhance specific strategies to take you to the ideal stage. Typical areas of focus include recruiting, onboarding, training, benefits, performance management, remote/flexible work options, and company culture.

Here is a template of what an action plan could look like:

Action Plan

GOAL 1	ACTION STEP DESCRIPTIONS	PARTY/DEPT RESPONSIBLE	DATE DUE
Write your goal statement here.			
List Resources & Desired Outcomes			

GOAL 2	ACTION STEP DESCRIPTIONS	PARTY/DEPT RESPONSIBLE	DATE DUE
Write your goal statement here.			
List Resources & Desired Outcomes			

GOAL 3	ACTION STEP DESCRIPTIONS	PARTY/DEPT RESPONSIBLE	DATE DUE
Write your goal statement here.			
List Resources & Desired Outcomes			

GOAL 4	ACTION STEP DESCRIPTIONS	PARTY/DEPT RESPONSIBLE	DATE DUE
Write your goal statement here.			
List Resources & Desired Outcomes			

Figure 4.2.6 Action Plan

5. Monitor, review, and reassess your plan

After you put your human resource plan in place, you must determine whether it is yielding results. Review how well it is helping you execute the organizational strategy and achieve goals in areas such as productivity or employee retention and satisfaction, etc. Monitor the new practices and look for areas where the plan might be hindered.

4.2.2 HUMAN RESOURCE RECRUITMENT:

Recruitment refers to the process of identifying, attracting, interviewing, selecting, hiring and onboarding employees. In other words, it involves everything from the identification of a staffing need to filling it.

Full cycle recruiting can benefit the candidate, the recruiter, and the organization. In this article, we'll take a look at full cycle recruiting, its benefits and drawbacks, and the six main stages of this recruitment process.

What is full cycle recruiting?

Full cycle recruiting, also referred to as full life cycle recruiting or end-to-end recruiting, is a term used to describe a holistic recruitment process with a full cycle recruiter involved in every step of the process. A full cycle recruitment process involves six main stages: preparing, sourcing, screening, selecting, hiring, and onboarding.

Companies can, of course, also decide to outsource their entire recruitment process.



Figure 4.2.6 Full Cycle Recruitment

1. Preparing
2. Sourcing
3. Screening
4. Selecting
5. Hiring
6. Onboarding

4.3 TRAINING AND DEVELOPMENT IN HUMAN RESOURCE

Training and development is a systematic process to enhance an employee's skills, knowledge, and competency, resulting in better performance in a work setting. Specifically, *learning* is concerned with the acquisition of knowledge, skills, and attitudes. *Development* is the broadening and deepening of knowledge in line with one's development goals.

Learning, training, and development are often used interchangeably. However, there are subtle differences between these concepts, which are shown in the table below.

Concept	Description
Learning	The acquisition of knowledge, skills, or attitudes through experience, study, or teaching. Training, development, and education all involve learning.
Training	Training is aimed at teaching immediately applicable knowledge, skills, and attitudes to be used in a specific job. Training may focus on delivering better performance in the current role or to overcome future changes.
Development	Development is aimed at the long term. It revolves around the broadening or deepening of knowledge. This has to fit within one's personal development goals and the (future) goals of the organization. Development usually happens voluntarily.
Education	Education is a more formal way to broaden one's knowledge. Education is often non-specific and applicable for a long time and is especially relevant when a person has little experience in a certain area.

Training and development Strategies:

A useful model that guides a learning and development strategy is created by van Gelder and colleagues (ENG). Its original name translates to 'Pedagogical Analysis'. The model starts with the organizational starting situation and prior knowledge based on which learning goals and objectives are defined. This information is used as input for the subject matter, teaching methods, and learning methods and activities. These lead to a certain result, which is monitored and evaluated. Based on this evaluation, the goals and objectives are updated.

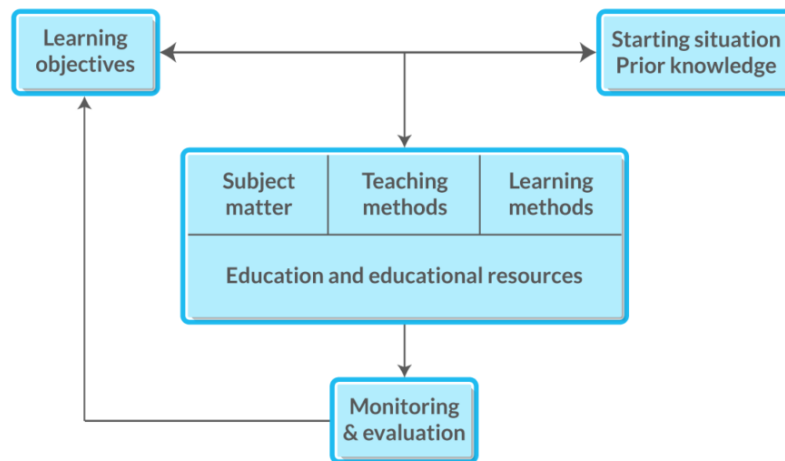


Figure 4.3.1 Training and development

Based on this model, we identify four phases required to create an effective learning and development process.

1. An analysis of training needs (starting situation)
2. Specification of learning objectives
3. Design of training content and method
4. Monitoring and evaluation

An effective learning and development strategy relies on a process in which one continually moves through these four phases. Let's examine them one by one.

Phase 1. Analysis of training needs

The first step is an analysis of the starting situations and prior knowledge to identify training needs. We don't want employees to learn for the sake of learning. Otherwise, we would be happy to send them on a pottery course. Instead, we want employees to acquire new knowledge, skills, and attitudes that are relevant for their (future) function. This way learning is a way to create new business capabilities.

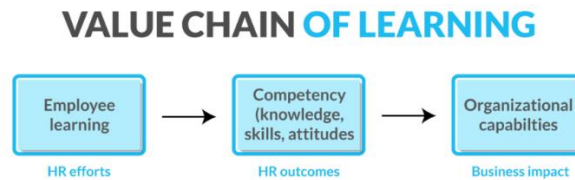


Figure 4.3.2 Value Chain of Learning

Identifying the learning goal requires you to analyze where the organization wants to go and what skills are missing to get there. This happens in three parts.

1. **Organizational analysis.**
2. **Function, task, or competency analysis.**
3. **Personal analysis.**

Using these three analyses, training goals can be specified. However, it is important to ensure there is sponsorship and support within the organization for the initiative.

Phase 2. Specification of learning objectives

The training needs need to be translated into learning objectives. These objectives serve as the starting point for the design of the training's content and method.

According to Lievens (2011), a training objective consists of three elements.

1. **The ability to realize specific objectives.**
2. **The conditions required for effective behavior.**
3. **A specific and measurable training goal.**

Phase 3. Design of the training material and method

In this phase, the teaching material and learning method are determined. This is where the choices about the training material, teaching method, and learning activities are made. This is often done together with an external trainer or training provider, and ideally also with involvement from the trainee.

In addition to learning methods, techniques, pacing, setting, and many more factors are determined.

Phase 4. Monitoring and evaluation

The last phase of the learning process is monitoring and evaluation. In this phase, the learning objectives are evaluated and learning effectiveness is assessed. A very useful model for evaluating learning effectiveness is Bloom's taxonomy, which we will explain later in this article.

Methods of learning

We already mentioned some methods of learning – but there are many more. We will list a number of them below. However, this list is far from comprehensive.

- **Lectures and seminars.** This is a more formal setting often used in universities with a lecturer and students. The setting inhibits interaction.
- **Discussion groups.** Highly interactive setting aimed at sharing viewpoints.
- **Debate.** Highly interactive setting aimed at convincing others of one's viewpoints.
- **Case study and projects.** These actively involve the participant and activate them to come up with solutions and answers.
- **Experiential activities.** These involve active participation and are often used in team building
- **Role Play.** A role is acted out or performed, for example as a technique to train customer interaction.

- **Simulation/Games.** An increasingly popular and highly interactive way of experimental learning. With the rise of virtual and augmented reality, this can be made very realistic.
- **Job shadowing.** Working with another employee who has a different experience to learn from them. This is a good way to learn and exchange ideas.
- **Outdoor management development (OMD).** A form of experiential activities. A 2001 study by Hamilton & Cooper showed that this could be effective.
- **Coaching.** Coaching focuses on hands-on skill development. The coach is often allocated and is the driving force. The coachee follows and learns.
- **Mentoring.** Mentoring is more strategic. The mentor is chosen by the mentee and the process is also driven by the mentee. Mentoring goes beyond skills.

Learning and development effectiveness

One of the key themes when it comes to learning and development is learning effectiveness. A key question often asked to the L&D professional is: “what is the return on learning?”, or “how effective are our learning programs?”. These questions are hard to answer.

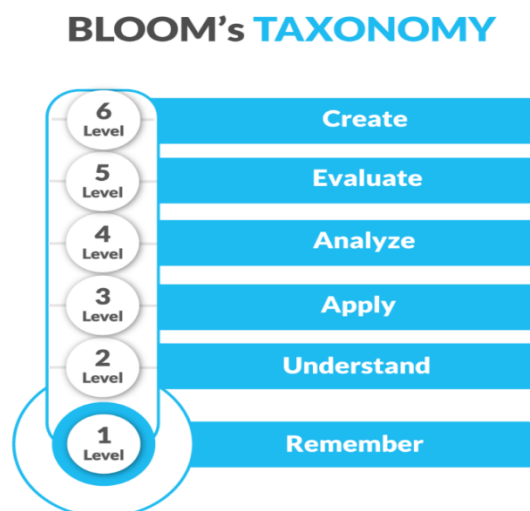


Figure 4.3.3 BLOOM'S TAXANOMY

This taxonomy is often used to specify what level of information processing is relevant to do a job, for example in training development, and to evaluate learning effectiveness. If someone has to be able to create or synthesize knowledge (e.g., an academic writing a paper on a topic), the approach to mastering the relevant information will be different than if someone only needs to understand (e.g., remembering Latin words) or apply the knowledge (e.g., conjugate Latin verbs).

Learning and development jobs

Let's conclude this guide on learning and development with the different job roles that are part of the learning and development team. Please note that the exact responsibility per role will differ between organizations. Typical learning and development jobs include:

- **L&D specialist.** The L&D specialist often occupies an operational role, focusing on analyzing learning needs, specifying role competencies, L&D budget distribution, and providing learning advice to employees.
- **L&D manager.** The Learning and Development Manager has a more tactical role, focusing on analyzing learning needs at a higher level, specifying core organizational competencies, L&D budget allocation, and distribution between departments and teams.
- **L&D director.** The L&D director has a strategic role, focusing on analyzing organizational needs for development, aligning L&D activities with organizational strategy, drafting the L&D strategy, and ensuring budget to execute this strategy.
- **L&D consultant.** The L&D consultant does all of the above in a consulting capacity. Depending on the role and seniority of the consultant, these activities can be operational or strategic.

4.4 SUPPLY CHAIN NETWORK

4.4.1 What is a supply chain?

A supply chain is the network of all the individuals, organizations, resources, activities and technology involved in the creation and sale of a product. A supply chain encompasses everything from the delivery of source materials from the supplier to the manufacturer through to its eventual delivery to the end user. The supply chain segment involved with getting the finished product from the manufacturer to the consumer is known as the distribution channel.

Steps in the supply chain

The fundamental steps of a supply chain in order are as follows:

- Sourcing raw materials.
- Refining those materials into basic parts.
- Combining those basic parts to create a product.
- Order fulfilment/Sales.
- Product delivery.
- Customer support and return services.

4.4.2 Supply chain management

Supply chain management (SCM) is the oversight of materials, information and finances as they move in a process from supplier to manufacturer to wholesaler to retailer and then to the consumer. The three main flows of the supply chain are the product flow, the information flow and the finances flow. These occur across three main stages: strategy, planning and operation. SCM involves coordinating and integrating these flows both within and among companies.



Figure 4.4.1 Supply Chain Management

4.4.3 Supply chain models

There are several common supply chain business models that supply chains fit into. The models have two main focuses: responsiveness and efficiency. Each model strives for some combination of both but approaches those goals differently. In addition, models tend to favor one over the other. Organizations can evaluate the value proposition of each in relation to their goals and constraints, and choose which suits them best.

The model types are:

- Continuous flow model -- works best for mature industries with a degree of stability.
- Agile model -- works best for industries with unpredictable demand and products that are made to order.
- Fast chain model -- works best for products with a short lifecycle, such as fashion items.
- Flexible model -- works best for industries with a level of stability and a few relatively predictable demand peaks.
- Custom configured model -- focuses on customizing.
- Efficient chain model -- works best for highly competitive markets in which pricing plays a large part.

The models are subject to overlap and should be designed by the supply chain manager to fit the unique supply chain.

4.4.4 The Importance of Business Analytics

Every organization with an online presence is now in a position to cultivate big data to improve business performance. Companies hire experts to analyze business information using sophisticated technology and statistical methods. This helps to find sustainable solutions to business problems, impacts growth and generates impactful insights.

Supply chain-related data is based on information recorded in the following categories:

- Products
- Customers
- Orders
- Sales
- Shipping
- Delivery

4.4.5 Key Benefits Of Using Business Analytics In Supply Chain Operations

1. Analytics in supply chain can forecast future demands
2. Business analytics in supply chain helps keep track of inventory
3. Supply chain management analytics improves logistics
4. Analytics in supply chain reduces costs and wastage
5. Business analytics in supply chain can offer real-time solutions
6. Supply chain management analytics helps with supplier relations

4.4.6 Five Types of Supply Chain Analytics

1. Descriptive Analytics
2. Predictive Analytics
3. Prescriptive Analytics

4. Diagnostic Analytics

5. Cognitive Analytics

4.4.7 Role of Business Analytics in Supply Chain Management

Analytics has become an integral tool for all kinds of businesses.

If you operate a business that involves a supply chain, the more you use analytical tools to get a better understanding of your past and current data, the more you can come up with effective ways of managing your supply chain.

Tracking Materials Becomes Easier

With companies like RTS Labs, RTS logistics involving integrations, applications, and data analysis enables you to better understand your suppliers and make improvements for forecasting your raw material needs.

Capacity Planning Becomes More Effective

Market demand will often fluctuate. When you use business analysis tools, you can better determine the needs of your production capacity and plan according to changing market demands.

With effective capacity planning, manufacturers can maintain the ideal balance between raw materials procurement and manufacturing capacity in order to meet your sales demand.

You Can Improve Your Sales And Operations Planning

By utilizing business analytics, the efficiency of your sales and operations planning can be majorly improved.

You can pin down your financial challenges while overseeing the bottom line of your operations. In turn, that enables you to streamline your financial management and enhance production planning.

Analyzing Historical Data Enables You To Come Up With Better Strategies For The Future

When you make the most of analyzing your historical data, you can gain better insights into the past performance of your supply chain. That includes parameters like your stock levels, sales ratio, past consignments, and inventory turnover.

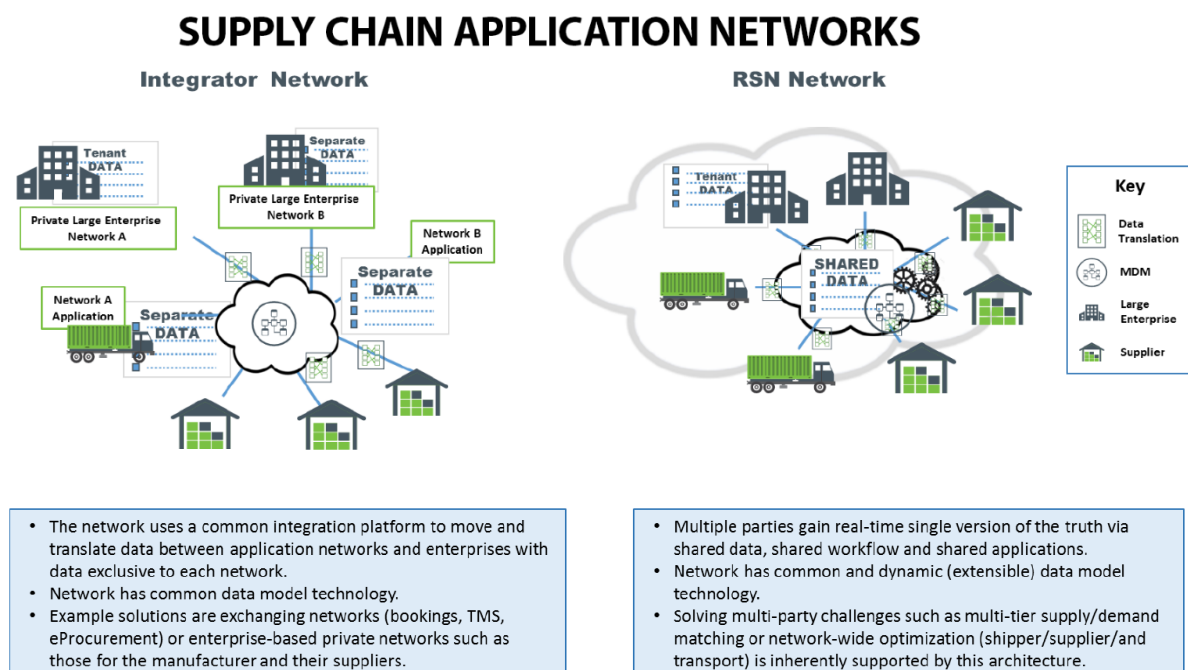
Demand Forecasting Enables You To Make More Intelligent Business Decisions

You can also come up with better strategies for the future when you utilize demand forecasting. This specific type of analysis involves incorporating advanced techniques like predictive analysis and data mining in order to predict future events with a high level of accuracy.

4.4.8 Two Types of Supply Chain Networks

In the marketplace there are two basic types of supply chain application network offerings. We call them Integrator Networks (IN) and Real-time Single Version of the Truth Networks (RSN).

Real-time SVoT (Single Version of the Truth) Network—These are networks that have generally been organically developed with a single database and single processing engine. This is a many-to-many approach in which all tenants share the same ledger-like, multi-enterprise data store—as well as sharing the same multi-party process execution. Below is a comparison between these two supply chain application network approaches.



4.5 PLANNING DEMAND IN SUPPLY CHAIN NETWORK

Demand planning is a supply chain management process that enables a company to project future demand and successfully customize company output — be it products or services — according to those projections.

4.5.1 Why is demand planning important?

Demand planning is the linchpin of an effective supply chain, serving two essential functions — which makes it doubly important to business.

First, there always exists the fundamental drive to protect the sale and ensure that expected revenues are generated. But retailers can't sell what they don't have in stock. And it doesn't take long for today's consumers to develop a lasting impression of a company, and whether it can meet supply demand. Demand planning works to see that retailers have exactly the right amount of inventory at the right place to avoid stock-outs and remain prepared for that next sale.

4.5.2 Elements of demand planning



Product portfolio management



Statistical forecasting



Demand sensing



Trade promotion management

4.5.3 Key steps of demand planning

Organizations vary widely in how they approach the demand planning process, but there is a general set of steps that businesses typically follow. Those steps include:

- Organizing and preparing data
- Making a preliminary forecast
- Integrating market data
- Reconciling bottom-up and top-down forecasts
- Developing a final forecast
- Using analytics to monitor project performance
-

4.5.4 Best practices for demand planning

In addition to establishing a precise set of implementation steps, successful companies usually engage in the following best practices for demand planning:

Carefully assembling data:

In order to process complex projections, effective demand planning requires ample amounts of data. Smart companies rely on metrics reports that help them prepare their data through increasingly sophisticated data mining and aggregation techniques.

Choosing the right tool:

There are numerous options when choosing demand planning software, but companies should try to be selective, based on their unique needs. Goal: Find a solution refined enough to reflect the subtleties of demand forecasting methods yet robust enough to handle reporting tasks.

Coordinating strategy:

Experienced demand planners typically begin their process by using descriptive analytics data to develop a testing baseline. Next, they shape the actual plan, devoting personnel and resources to cultivate and refine that plan, and then work on the exact implementation steps.

4.5.5 Demand planning best practices

Both supply and demand planning have known best practices that help companies better manage their inventories. Three of the most common demand planning best practices include:

Achieving a balance between modeling based on statistical data and forecasting

- Analyzing metrics gathered from the product, customer and location levels
- Understanding how forecasting should be carried out (by country, product, month, year, etc.)

By paying attention to historical data, as well as current and upcoming events, business leaders can better plan for periods of increased or decreased demand and accommodate their inventories based on that data.

4.6 INVENTORY IN SUPPLY CHAIN

4.6.1 What Is Inventory?

Inventory is the accounting of items, component parts and raw materials that a company either uses in production or sells. As a business leader, you practice inventory management in order to ensure that you have enough stock on hand and to identify when there's a shortage.

- ✓ Inventory, which describes any goods that are ready for purchase, directly affects an organization's financial health and prosperity.
- ✓ While there are many types of inventories, the four major ones are raw materials and components, work in progress, finished goods and maintenance, repair and operating supplies.

- ✓ While there are many ways to count and value your inventory, the importance lies in accurately tracking, analyzing and managing it. Insights gained from inventory evaluations are necessary for success as they help companies make smarter and more cost-efficient business decisions.

4.6.2 13 Types of Inventory:

There are four different top-level inventory types: raw materials, work-in-progress (WIP), merchandise and supplies, and finished goods. These four main categories help businesses classify and track items that are in stock or that they might need in the future. However, the main categories can be broken down even further to help companies manage their inventory more accurately and efficiently.



Figure 4.6.1 Types of Inventory

1. **Raw Materials:** Raw materials are the materials a company uses to create and finish products. When the product is completed, the raw materials are typically unrecognizable from their original form, such as oil used to create shampoo.
2. **Components:** Components are like raw materials in that they are the materials a company uses to create and finish products, except that they remain recognizable when the product is completed, such as a screw.
3. **Work In Progress (WIP):** WIP inventory refers to items in production and includes raw materials or components, labor, overhead and even packing materials.
4. **Finished Goods:** Finished goods are items that are ready to sell.

5. **Maintenance, Repair and Operations (MRO) Goods:** MRO is inventory — often in the form of supplies — that supports making a product or the maintenance of a business.
6. **Packing and Packaging Materials:** There are three types of packing materials. Primary packing protects the product and makes it usable. Secondary packing is the packaging of the finished good and can include labels or SKU information. Tertiary packing is bulk packaging for transport.
7. **Safety Stock and Anticipation Stock:** Safety stock is the extra inventory a company buys and stores to cover unexpected events.
8. **Decoupling Inventory:** Decoupling inventory is the term used for extra items or WIP kept at each production line station to prevent work stoppages.
9. **Cycle Inventory:** Companies order cycle inventory in lots to get the right amount of stock for the lowest storage cost.
10. **Service Inventory:** Service inventory is a management accounting concept that refers to how much service a business can provide in a given period. A hotel with 10 rooms, for example, has a service inventory of 70 one-night stays in each week.
11. **Transit Inventory:** Also known as pipeline inventory, transit inventory is stock that's moving between the manufacturer, warehouses and distribution centers. Transit inventory may take weeks to move between facilities.
12. **Theoretical Inventory:** Also called book inventory, theoretical inventory is the least amount of stock a company needs to complete a process without waiting.
13. **Excess Inventory:** Also known as obsolete inventory, excess inventory is unsold or unused goods or raw materials that a company doesn't expect to use or sell but must still pay to store.

4.6.3 Inventory Examples

Real-world examples can make inventory models easier to understand. The following examples demonstrate how the different types of inventory work in retail and manufacturing businesses.

- **Raw Materials/Components:** A company that makes T-shirts has components that include fabric, thread, dyes and print designs.

- **Finished Goods:** A jewelry manufacturer makes charm necklaces. Staff attaches a necklace to a preprinted card and slips it into cellophane envelopes to create a finished good ready for sale. The cost of goods sold (COGS) of the finished good includes both its packaging and the labor exerted to make the item.
- **Work In Progress:** A cell phone consists of a case, a printed circuit board, and components. The process of assembling the pieces at a dedicated workstation is WIP.
- **MRO Goods:** Maintenance, repair and operating supplies for a condominium community include copy paper, folders, printer toner, gloves, glass cleaner and brooms for sweeping up the grounds.
- **Packing Materials:** At a seed company, the primary packing material is the sealed bag that contains, for example, flax seeds. Placing the flax seed bags into a box for transportation and storage is the secondary packing. Tertiary packing is the shrink wrap required to ship pallets of product cases.
- **Safety Stock:** A veterinarian in an isolated community stocks up on disinfectant and dog and cat treats to meet customer demand in case the highway floods during spring thaw and delays delivery trucks.
- **Anticipated/Smoothing Inventory:** An event planner buys discounted spools of ribbon and floral tablecloths in anticipation of the June wedding season.
- **Decoupled Inventory:** In a bakery, the decorators keep a store of sugar roses with which to adorn wedding cakes – so even when the ornament team’s supply of frosting mix is late, the decorators can keep working. Because the flowers are part of the cake’s design, if the baker ran out of them, they couldn’t deliver a finished cake.
- **Cycle Inventory:** As a restaurant uses its last 500 paper napkins, the new refill order arrives. The napkins fit easily in the dedicated storage space.
- **Service Inventory:** A café is open for 12 hours per day, with 10 tables at which diners spend an average of one hour eating a meal. Its service inventory, therefore, is 120 meals per day.
- **Theoretical Inventory Cost:** A restaurant aims to spend 30% of its budget on food but discovers the actual spend is 34%. The “theoretical inventory” is the 4% of food that was lost or wasted.

- **Book Inventory:** The theoretical inventory of stock in the inventory record or system, which may differ from the actual inventory when you perform a count.
- **Transit Inventory:** An art store orders and pays for 40 tins of a popular pencil set. The tins are en route from the supplier and, therefore, in transit.
- **Excess Inventory:** A shampoo company produces 50,000 special shampoo bottles that are branded for the summer Olympics, but it only sells 45,000 and the Olympics are over — no one wants to buy them, so they're forced to discount or discard them.

4.6.4 Inventory Best Practices

The business saying “if you can’t measure it, you can’t manage it” applies to inventory management and best practices. While the first best practice is keeping track of your inventory, others include:

- **Carry Safety Stock:**
Also known as buffer stock, these products help keep companies from running out of materials or high-demand items. Once companies deplete their calculated supply, safety stock serves as a backup should the level of demand increase unexpectedly.
- **Invest in a Cloud-based Inventory Management Program:**
Cloud-based inventory management systems let companies know in real-time where every product and SKU are located globally. This data helps an organization be more responsive, up-to-date, and flexible.
- **Start a Cycle Count Program:**
Cycle counting benefits extend well past the warehouse by keeping stock reconciled and customers happy while also saving businesses time and money.
- **Use Batch/Lot Tracking:**
Record information associated with each batch or lot of a product. While some businesses log precise details, such as expiration dates that provide information about their products’ sellable dates, companies that do not have perishable goods use batch/lot tracking to understand their products’ landing costs or shelf lives.



Figure 4.6.2 Strengthen SCM

Inventory management is critical in strengthening companies supply chain because it helps to stabilize the dynamics between customer demand, storage space, and cash restraints.

4.6.5 What Is Inventory Turnover?

Inventory turnover is the number of times a company sells or uses an item in a specific timeframe, which can reveal whether a company has too much inventory on hand. To determine inventory turnover, use the following equations:

$$\text{Average inventory} = (\text{Beginning Inventory} + \text{Ending Inventory}) / 2$$

$$\text{Inventory turnover} = \text{Sales} / \text{Average Inventory}$$

4.5.6 What Is Inventory Analysis?

Inventory analysis is the study of how product demand changes over time and it helps businesses stock the right amount of goods and project how much customers will want in the future.

A well-known method for performing inventory analysis is ABC analysis. To perform an ABC analysis, group goods into three categories:

- **A inventory:** A inventory includes the best-selling products that require the least space and cost to store. Many experts say this represents about 20% of your inventory.
- **B inventory:** B items move at a similar rate to A items but cost more to store. Generally, this represents about 40% of your inventory.
- **C inventory:** The remainder of your stock costs the most to store and returns the lowest profits. C inventory represents the other 40% of your inventory.

ABC Analysis



Figure 4.6.3 ABC Analysis

4.5.7 Benefits of Inventory Analysis

Inventory analysis raises profits by lowering costs and supporting turnover. It also:

1. **Improves Cash Flow:** Inventory analysis helps you identify and reorder items you sell often, so you don't spend money on inventory that moves slowly.
2. **Reduces Stockouts:** When you understand which inventory customers want most, you can better anticipate demand and prevent stockouts.
3. **Increases Customer Satisfaction:** Analyzing inventory offers insight into what and how customers purchase goods.

4. **Reduces Wasted Inventory:** Understanding what, when and how much people buy minimizes the need to store obsolete products, as well as when products expire so you can have a strategy behind using them.
5. **Reduces Project Delays:** Learning about supplier lead times helps you understand when to reorder and how to avoid late shipments.
6. **Improves Pricing From Suppliers and Vendors:** Inventory analysis can lead you to order high volumes of products regularly rather than small volumes on a less reliable schedule. This regularity can put you in a stronger position to negotiate discounts with suppliers.
7. **Expands Your Understanding of the Business:** Reviewing inventory provides insights into your stock, customers and business.



Figure 4.6.4 Increase Sales & Revenue

4.7 LOGISTICS

"Logistics typically refers to activities that occur within the boundaries of a single organization and Supply Chain refers to networks of companies that work together and coordinate their actions to deliver a product to market. Also, traditional logistics focuses its attention on activities such as procurement, distribution, maintenance, and inventory management. Supply Chain Management (SCM) acknowledges all of traditional logistics and also includes activities such as marketing, new product development, finance, and customer service"



4.7.1 What is Logistics?

"Logistics is about getting the right product, to the right customer, in the right quantity, in the right condition, at the right place, at the right time, and at the right cost (the 7 Rs)"

In the past, various tasks were under different departments, but now they are under the same department and report to the same head as below,

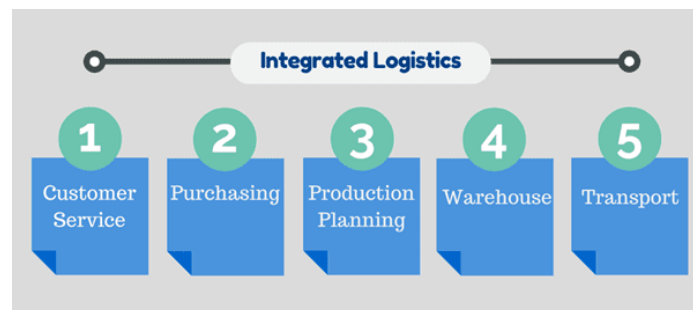
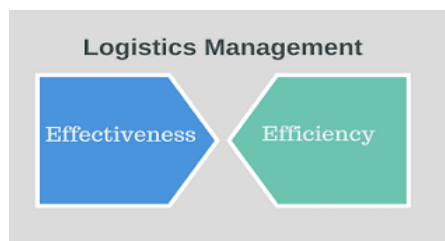


Figure 4.7.1 Integrated Logistics

4.7.2 What is Logistics Management?

"Logistics Management deals with the efficient and effective management of day-to-day activity in producing the company's finished goods and services" - Paul Schönsleben



4.7.3 What is Supply Chain?

"Supply Chain is the network of organizations that are involved, through upstream and downstream linkages, in the different processes and activities that produce value in the form of products and services in the hands of the ultimate consumer" - Martin Christopher

What is Supply Chain Management?

Each researcher defines supply chain management differently. However, we would like to provide the simple definition as below,

"Supply Chain Management (SCM) refers to the coordination of production, inventory, location, and transportation among the participants in a supply chain to achieve the best mix of responsiveness and efficiency for the market being served" -Michael Hugos

4.7.4 What is the Difference Between Inbound and Outbound Logistics?

"Inbound Logistics refers to movement of goods and raw materials from suppliers to your company. In contrast, Outbound Logistics refers to movement of finished goods from your company to customers"

To illustrate this term, we make a small graphic as below,

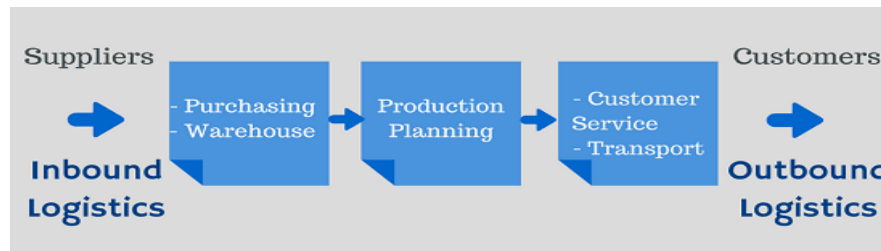


Figure 4.7.2 Inbound and Outbound Logistics

As you can see, purchasing and warehouse (distribution center) communicate with suppliers and they are sometimes called "supplier facing function". Production planning and inventory control function is the center point of this chart. Customer service and transport functions communicate with customers and they are sometimes called "customer-facing functions".

4.7.5 What is Transport and Logistics?

"Transport and Logistics refers to 2 types of activities, namely, traditional services such as air/sea/land transportation, warehousing, customs clearance and value-added services which including information technology and consulting"

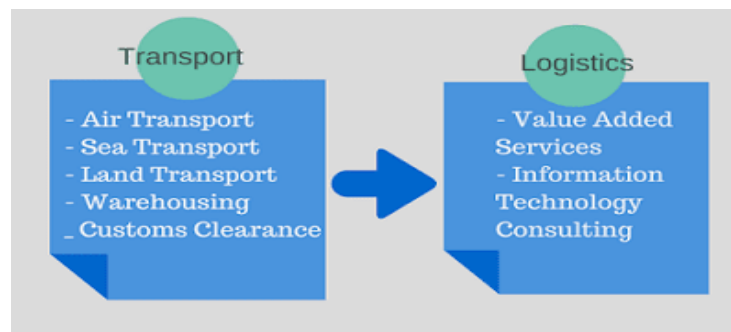


Figure 4.7.3 Transport and Logistics

4.7.6 What is International Logistics?

These are one of the most ambiguous groups of terms in international business out there. They are used interchangeably with international supply chains or international production and transportation activities. However, the most concise definition is as below,

"International Logistics focuses on how to manage and control overseas activities effectively as a single business unit. Therefore, companies should try to harness the value of overseas product, services, marketing, R&D and turn them into competitive advantage"

4.7.7 What is Third Party Logistics or 3PL?

The concept of 3PL appeared on the scene in the 1980s as a way to reduce costs and improve services which can be defined as below,

"Third Party Logistics or 3PL refers to the outsourcing of activities, ranging from a specific task, such as trucking or marine cargo transport to broader activities serving the whole supply chain such as inventory management, order processing and consulting."

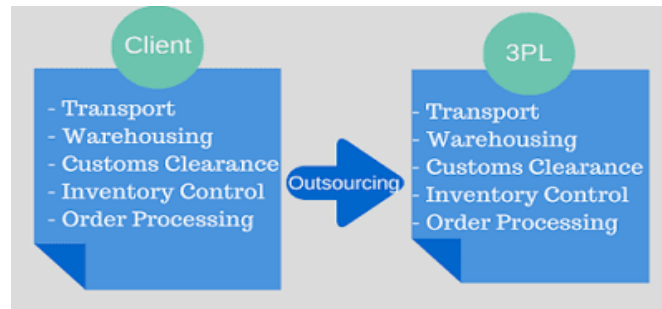


Figure 4.7.4 Third Party Logistics

In the past, many 3PL providers didn't have adequate expertise to operate in complex supply chain structures and processes. The result was the inception of another concept.

4.7.8 What is Fourth Party Logistics or 4PL?

The 4PL is the concept proposed by Accenture Ltd in 1996 and it was defined as below,

"Fourth Party Logistics or 4PL refers to a party who works on behalf of the client to do contract negotiations and management of performance of 3PL providers, including the design of the whole supply chain network and control of day-to-day operations"



Figure 4.7.5 Fourth Party Logistics

4.7.9 Top 7 Functions of Logistics Management

Logistics is as simple as it is complicated. It's never that easy to get a service or product from point A to B. Logistics and supply chain management are interwoven. The two work together to develop network topologies of logistical activities that service clients.

Logistics encompasses a lot more than just physical distribution. It has a lot of other functions as well. In this article, we will go through the seven most prominent logistical functions:



Figure 4.7.6 Logistics Functions

1. Processing Orders
2. Transportation
3. Managing Inventory
4. Warehousing
5. Packaging
6. Handling Materials and Storage
7. Monitoring

4.8 ANALYTICS APPLICATIONS IN HR & SUPPLY CHAIN

4.8.1 What is data analytics in HR?

Data analytics is the process of examining raw data with the goal of identifying trends and drawing conclusions. In the human resources field, data analytics is often deployed to help HR professionals make more informed decisions about the future of their workforce. There are a few notable subtypes of HR analytics, and we'll give a brief overview of how those are differentiated from one another below:

- **People analytics:** People analytics, also called talent analytics, involves collecting information about your organization's talent with the goal of improving the employee experience.
- **Workforce analytics:** Workforce analytics is a tactic that's closely tied to strategic HR planning. There are several goals of workforce analysis including managing staffing levels so that they are aligned with business needs

and priorities, optimizing your organization's structure, and preparing for the future of your workforce through informed succession planning.

- **Predictive HR analytics:** More future oriented than the two analytics tactics above, predictive analytics is an advanced strategy that uses techniques such as regression analysis, multivariate statistics, pattern matching, predictive modeling, and forecasting. Adoption rate of predictive analytics in HR is still low, but it can be useful for things such as predicting when employees are at risk of leaving the company.

4.8.2 What are the applications of data analytics in HR?

HR data analytics can be used to make improvements in nearly every area of human capital management. Ahead, we've included examples of five applications for HR data analytics to illustrate how this practice can be used to finetune different HR processes.

1. Recruitment

If you haven't heard, it's a tough time for recruiters. We're in the midst of an ongoing talent shortage, which means that sourcing and recruiting quality talent is more competitive than ever. Data analytics can help by revealing where your recruiting process can be improved.

Track the recruiting data that's a priority for your organization. Many **recruiting software platforms** offer features such as customizable dashboards as a part of their analytics functionality to help you do this.



Figure 4.8.1 A dashboard in Lever turns valuable recruiting data into easy-to-understand visualizations

2. Training and development

Are your training and development programs effective? That's an important question for HR professionals to answer, but without a method for measuring results, there's no way of knowing if your teams' resources and efforts were wasted on a training initiative.

Measure your employees' engagement with and comprehension of training programs. **Learning management systems (LMS)**, which are used to deliver training programs to employees, are often built to track learners' completion rates, time spent training, individual assessment scores, and more.



Figure 4.8.2 A dashboard in Absorb LMS shows learner logins, enrollments, and course completions

3. Attrition and retention

In Capterra's **Recruiting Strategy Survey**** from July 2021, 51% of employees with recruiting responsibilities revealed that the number of job openings that they're actively recruiting for is higher than average. The talent shortage is partially to blame, but **ongoing, record-high levels of turnover** is another major factor contributing to the surplus of open positions.

In order to get turnover in check, you need to understand what's contributing to it. **HR analytics tools** help with this by collecting data related to employee attrition, such as voluntary and involuntary turnover rates, average employee tenure, and employee retention rate.

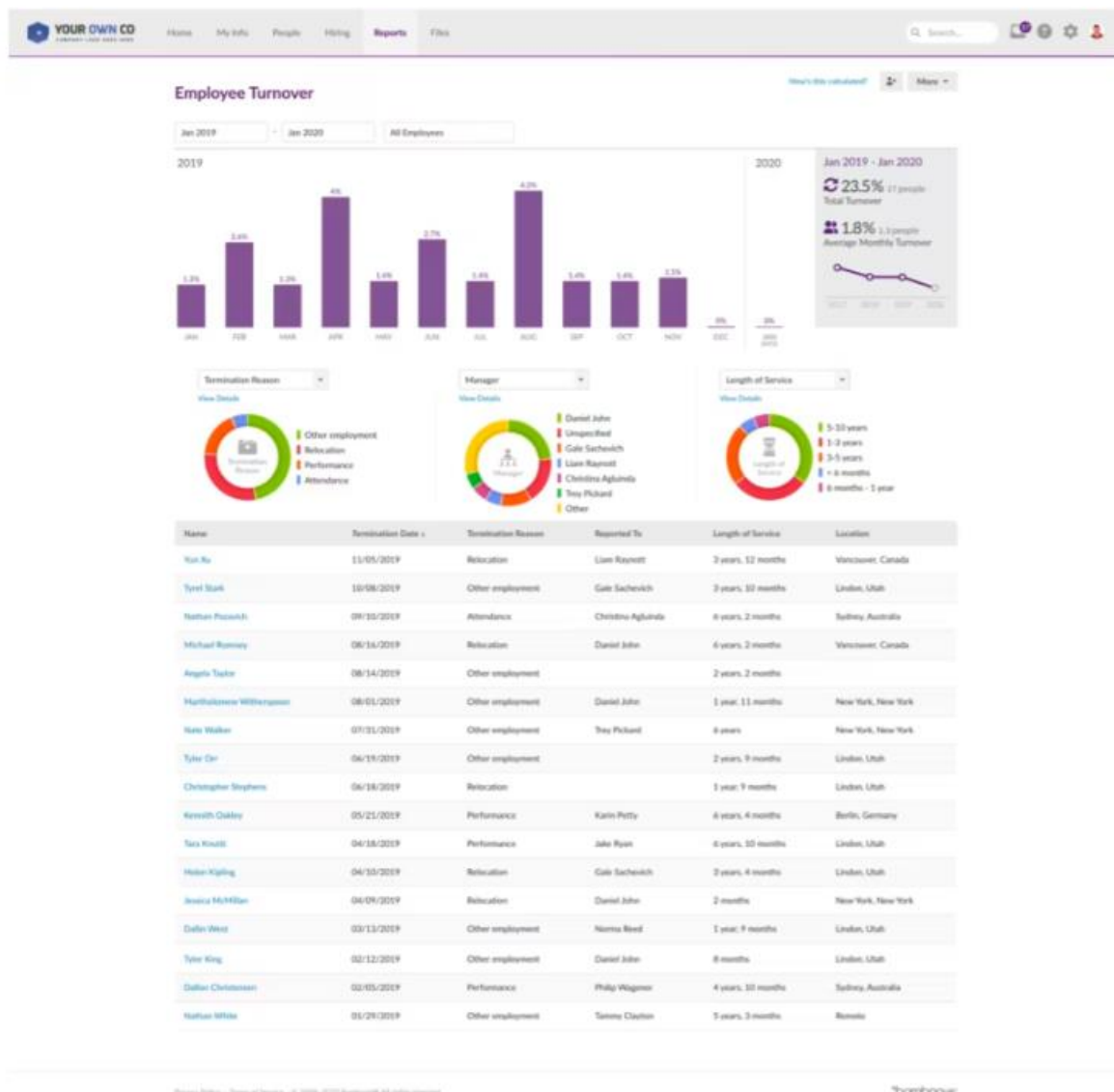


Figure 4.8.3 A dashboard shows tenure and termination reason for employees in BambooHR

4. Benefits and compensation

One of the biggest expenses for any business is employee compensation. This, coupled with the fact that we're currently in an **inflationary environment**, has made compensation analytics a more essential practice than ever for HR managers.

For instance, our **Recruiting Strategy Survey*** revealed that only 13% of employers are *not* considering increasing compensation as a means of attracting job seekers to their organization. So, if you're having a hard time recruiting quality talent and you're a part of that minority, it's time to take a more intentional approach to compensation planning. You can do this by comparing your organization's salary ranges to those from similar businesses, reviewing adoption rates of different benefits to determine where changes should be made, and assessing pay equity gaps.

Ask your **payroll software** vendor about the capabilities of their reporting and insights functionalities. There may be analytics features that can help guide your compensation strategy that you aren't currently taking advantage of.

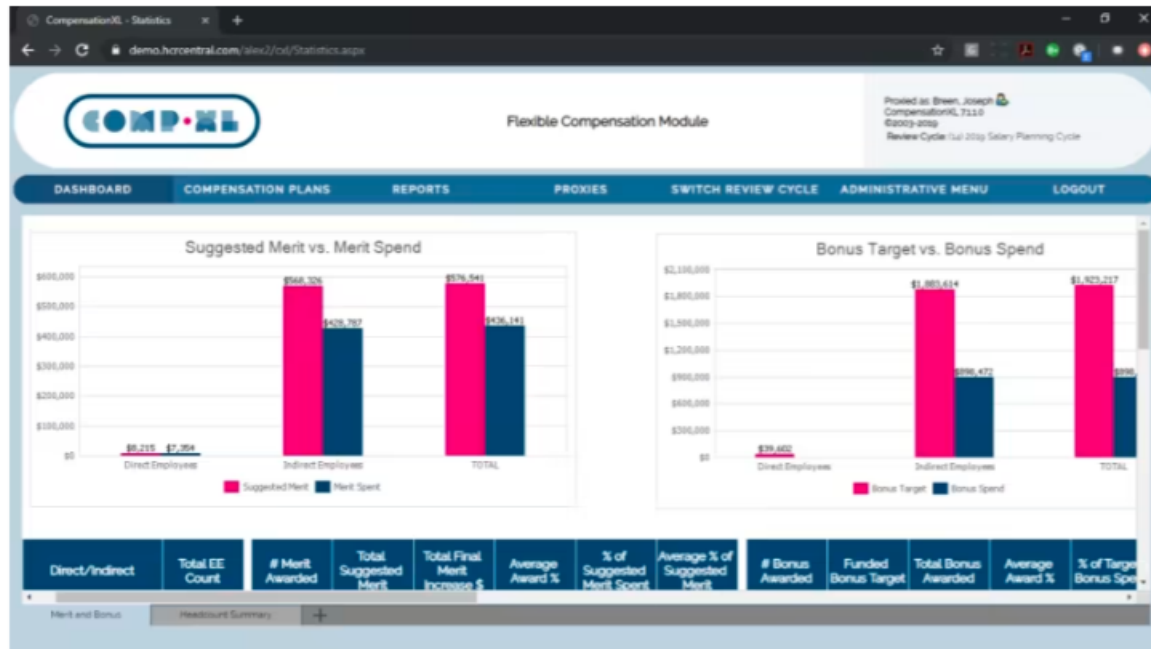


Figure 4.8.3 Compare your planned merit spending to suggestions based on industry trends in CompensationXL

5. Employee productivity

From the perspective of the HR department, productivity refers to how efficiently resources such as talent, money, and time are being used. Data analytics is particularly helpful when it comes to measuring productivity, but the *kind* of productivity data you should analyze depends on your goal.

For instance, some metrics (such as **revenue per full-time employee**) are helpful for judging the productivity of your workforce as a whole, while others (such as project completion rate) are more useful for measuring the performance of a team or individual. Further, you can measure how efficiently resources are being allocated by tracking the adoption rates of different tools or benefits.

Encourage department leaders to report on their team's productivity. Most businesses today use a **collaboration tool** or **project management platform**, and these tools often have reporting features that track everything from communication between employees to missed deadlines.

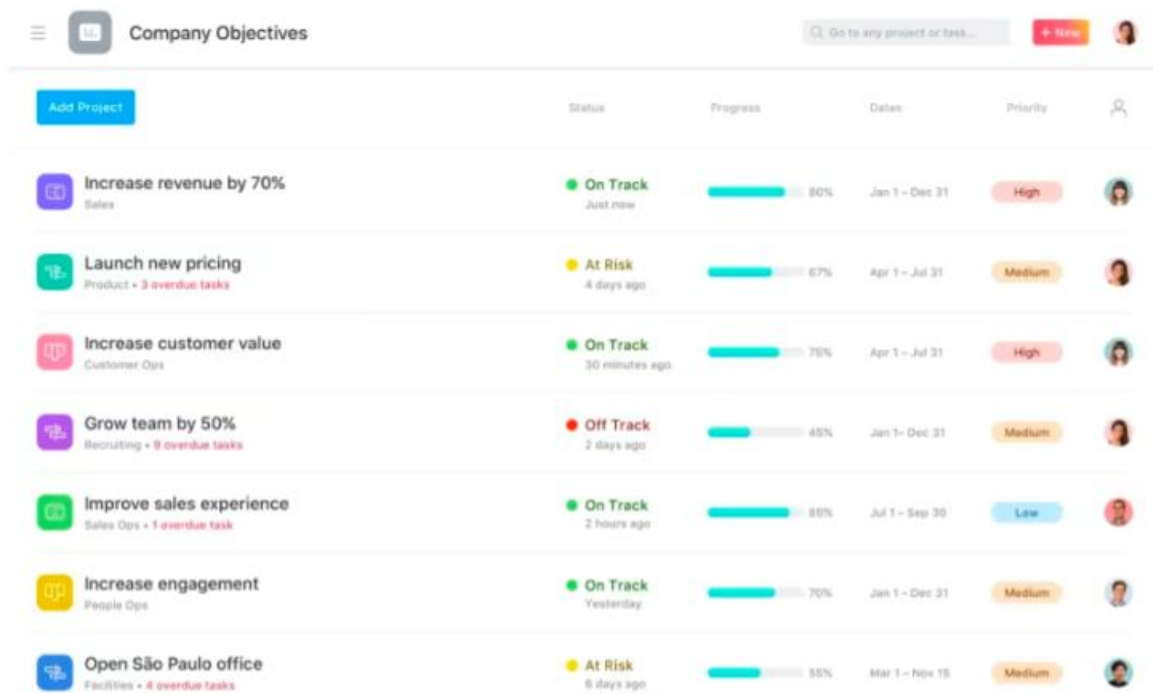


Figure 4.8.4 The status, progress, and priority of different company objectives organizes in Asana

4.8.3 Applications Of Analytics In Supply Chain Management

The top five sales applications of big data analytics in supply chain management are listed below.

So, let's understand the big data analytics use cases in supply chain management.

Supplier Relationship Management

Using big data and machine learning to apply supplier relationship management methods, effective supply chains can be created. Even though big data analytics isn't extensively used in supply chain firms, future supplier relationship management strategies will rely heavily on big data analytics and machine learning. For supply chain companies to develop effective relationship management strategies, they require reliable supplier data. A lot of the information they want is qualitative. Evaluations, audits, and assessments are all part of it.

Product Design & Development

In smart manufacturing, Big Data Analytics is beneficial for solving the company's problems at the speed required by the business. Big data analytics is a helpful tool for manufacturers, assisting them in developing plans, sharing data, designing predictive models, and connecting factories to control operations. Routing order pickup and delivery, as well as assigning orders to respective agents, are further benefits of using big data analytics in supply chain management.

Demand Planning

Incomplete or incomplete orders can damage the image of any company. Offering the appropriate product to the right person at the right time and place is critical to maintaining customer happiness and loyalty in the era of the customer. Smart businesses can use Big Data to gain a complete 360-degree perspective of their customers. This allows them to better foresee their requirements, understand their preferences, and provide a unique brand experience.

Logistics Management

Another excellent use of big data analytics in supply chain management is **logistics management**. For logistics systems, optimizing service experiences such as delivery speed, resource application, and geographic coverage is a constant issue. For logistics companies, both late and early delivery would be expensive. One of the major risk concerns for logistics organizations is the time gap between the anticipated and actual delivery. To reduce the danger of erroneous delivery timeframes, the use of big data analytics in supply chain management is beneficial. This will help you improve traceability, which guarantees that items are traceable from the point of manufacture to the point of sale.

This improved tracking system can aid in the integration of multiple supply chain entities and the smooth flow of goods. Big warehouses, for example, can reorganize pallets mechanically

at night to optimize scheduling for the next day with the help of big data analytics. Companies can also monitor the success of pickers in various picking zones to optimize future staffing.

Machine Maintenance

The use of big data in maintenance is expanding, and this trend is enhancing the way the maintenance team works. The shift toward data analytics is enhancing the efficacy of the maintenance department and, as a result, improving operational uptime. By merging machine data with other area data, the operation can effectively perceive the health and performance of its equipment. This is mainly done by installing sensors and using the data to replicate the operation of the device. Following the establishment of a model, the operation can use real-time data to predict when the machine would fail. Data analytics results in more than just the replacement of objects before they break down. Advanced data can be useful for having a better idea of how to extend the life of an existing asset.

UNIT – 5

MARKETING & SALES ANALYTICS

Marketing Strategy, Marketing Mix, Customer Behaviour - Selling Process - Sales Planning - Analytics applications in Marketing and Sales.

5. INTRODUCTION

Data has become the key ingredient in a business success model. With the right data, marketers, strategists, and managers can identify critical improvement areas and allocate resources for optimized results. For marketing and sales, data and business analytics has a special importance. Functions in both of these domains can achieve effective results with the help of accurate data and business analytics.

Here are 5 Benefits of Business analytics for marketing.

1. Ask the right questions

Marketing managers need to formulate the right question to achieve key performance metrics. Business analytics gives a company the capability to parse through massive troves of data and produce actionable insights from it.

2. Better define and target your essential demographics

BI tools can narrow down demographics to find the right audiences. Online business analytics courses offer valuable learning in creating brand messaging strategies.

3. Enhance the quality and speed of reports

Bulk data collection helps companies to get better insights which improve strategy creation and also keeps the marketing department in tune with the changing preferences and trends.

4. Look at the right data

Marketing professionals should prioritize performance metrics to achieve efficiency. XLRI PG certificate in business analytics imparts learning in analysis of key marketing statistics- the number of site visitors and how much time they spend on the website.

5. Analyze customers and their behavior

Business analytics marketing data can offer a clear picture of your customers, making it easy to tailor your online services. This helps in driving your content to the popular part of your website. By analyzing the popularity of articles and blog posts, companies can better understand which topics are more engaging, so that you can modify your content strategies to enhance site visits and conversions.

5.1 MARKETING ANALYTICS

Marketing analytics is the study of data to evaluate the performance of a marketing activity. By applying technology and analytical processes to marketing-related data, businesses can understand what drives consumer actions, refine their marketing campaigns and optimize their return on investment.

5.1.1 Why is Marketing Analytics Important?

- (i) Understanding the customer and market trends is important in today's time. Marketing analytics easily allows an understanding of big-picture trends by focusing on every detail.
- (ii) With the aid of providing you with a clear picture of the efforts and the returns, it allows you to easily depict which programs worked and also depicts the reasons why they failed or even succeeded.
- (iii) The market study is also essential to the business. Marketing analytics allows the monitoring of trends over time.
- (iv) Marketing analytics allows understanding of the return on investment by providing a clear picture of each program's working and reports.
- (v) By easily helping to study the market trends, marketing analytics facilitates proficiently forecasting future results.

Turning to search marketing, keywords also allow for understanding what even the mindset of the customers is.

Analytics help you find the keywords that can easily be used to optimize business processes by

(i) Customer surveys — One can easily infer the relative priorities of competing interests by the examination of keyword frequency data

(ii) Industry trends — It becomes really easy to understand and predict trends in customer behavior

(iii) Product design — It is vital to understand the solutions a customer is looking for.

Keywords can easily facilitate the understanding of what the customer wants

5.1.2 Features of Marketing Analytics

Marketing analytics tells you about the most profitable marketing activities of your business. The more effectively you target the right people with the right approach, the greater value you generate for your business.



Figure 5.1.1 Features of Marketing Analytics

1. Impressions

In digital marketing, impressions are the number of times any piece of your content has been shown on a person's screen. It can be an ad, a social media post, video etc. However, it is important to remember that impressions do not mean views, a view is an engagement, anytime somebody sees your video that is a view, but an impression would also include anytime they see your video in the recommended videos on YouTube or in their newsfeed on Facebook. The impression will be counted regardless of whether they watch your video or not.

2. Engagement rate

In social media marketing, engagement rate is an important metric. Engagement is when a user comments, likes, clicks, or otherwise interacts with any of your content. Engagement rate is a metric that measures the amount of engagement of your marketing campaign relative to each of the following:

- Reach
- Post
- Impressions
- Days
- Views

Engagement rate by reach is the percentage of people who chose to interact with the content after seeing it. It is calculated by the following formula. Reach is a more accurate measurement than follower count, because not all of your brand's followers may see the content while those who do not follow your brand may still be exposed to your content.

Engagement rate by post is the rate at which followers engage with the content. This metric shows how engaged your followers are with your content. However, this metric does not account for organic reach and as your follower count goes up your engagement by post goes down.

Engagement rate by Impressions is the rate of engagement relative to the number of impressions. If you are running paid ads for your brand, engagement rate by impressions can be used to gauge your ads effectiveness.

Average Daily engagement rate tells you how much your followers are engaging with your content daily. This is suitable for specific use cases for instance, when you want to know how much your followers are commenting on your posts or other content.

Engagement rate by views gives the percentage of people who chose to engage with your video after watching them. This metric however does not use unique views so it may double or triple count views from a single user.

3. Sessions

Sessions are another especially important metric in marketing campaigns that help you analyze engagement on your website. A session is a set of activities by a user within a certain period. For example, a user spent 10 minutes on your website, loading pages, interacting with your

content and completed an interaction. All these activities will be recorded in the same 10-minute session.

4. Conversion rate

Once you have engaged your audience the next step in the customers' journey is conversion. A conversion is when you make the customer or user complete a specific action. This desired action can be anything from a form submission, purchasing a product or subscribing to a service. The conversion rate is the percentage of visitors who completed the desired action.

5. Attribution

Attribution is a sophisticated model that helps you measure which channels are generating the most sales opportunities or conversions. It helps you assign credit to specific touchpoints on the customers journey and understand which touchpoints are driving conversions the most. But how do you know which touchpoint to attribute to a specific conversion? Well, that depends on which attribution models you are using. There are four common attribution models.

First touch attribution models assign all the credit to the first touchpoint that drove the prospect to your website. It focuses on the top of the marketing efforts funnel and tells you what is attracting people to your brand

Last touch attribution models assign credit to the last touchpoint. It focuses on the last touchpoint the visitor interacted with before they converted.

Linear attribution model assigns an equal weight to all the touchpoints in the buyer's journey.

Time decay attributions is based on how close the touchpoint is to the conversion, where a weighted percentage is assigned to the most recent touchpoints. This can be used when the buying cycle is relatively short.

What model you use is based on what product or subscription you are selling and what is the length of your buyer cycle. While attribution is very important in identifying the effectiveness of your channels, to get the complete picture you need to look at how each touchpoint drives conversion.

6. Customer lifetime value

Businesses prefer retaining customers over acquiring new ones, and one of the main reasons is that attracting new customers has a cost. The customer acquisition cost is the total cost that you incur as a business acquiring a customer. The customer acquisition cost is calculated by dividing the marketing and sales cost by the number of new customers.

5.1.3 *Marketing analytics methods*



Figure 5.1.1 Marketing Analytics Methods

Marketing Analytics methods can be broken down into the following main categories.

- Marketing mix modeling
- Demand forecasting
- Competitor analytics
- Trend analytics

5.2 MARKETING STRATEGY

A strategy is like a map that helps you reach your goals using a predefined route. Marketers use such strategies to fulfil their marketing objectives. These marketing strategies connect the dots of a marketing plan to help the business reach the target customers using apt channels and aid business growth in the long run.

5.2.1 What Is Marketing Strategy?

A marketing strategy is the combination of all the business's decisions and actions to increase sales and achieve a sustainable competitive advantage by understanding and fulfilling the needs and wants of the customers.

This definition of marketing strategy stands on three key pillars. These are:

- **What:** Marketing strategy is the overall game plan or the roadmap marketers use to achieve a business's marketing objectives and goals.
- **Why:** It aims to achieve a unique market position backed by a sustainable competitive advantage that results in more sales and profits for the business.
- **How:** Every marketing strategy works directly or indirectly by fulfilling the needs and wants of the customers.

In simple terms, a marketing strategy is an achievable and actionable focused set of steps devised by marketers to achieve a marketing objective.

5.2.2 Components Of A Marketing Strategy

Even though there are numerous types of marketing strategies, all of them consists of these six components.

- **Target Market:** It is the customer segment to whom all the marketing activities are directed.
- **Business Offering:** The product or service offered by the business.
- **Competitive Advantage:** The value proposition that separates the company from the competition.
- **Goal:** A milestone that decides whether the marketing strategy was successful or not.
- **Communication strategy:** How the company plans to communicate the marketing message to the target market. It includes communication channels and tactics to get more traction and conversion.
- **KPI:** Key performance indicators measure the business's performance and progress in the strategic marketing areas associated with its success.

5.2.3 Focus on the 7 P's

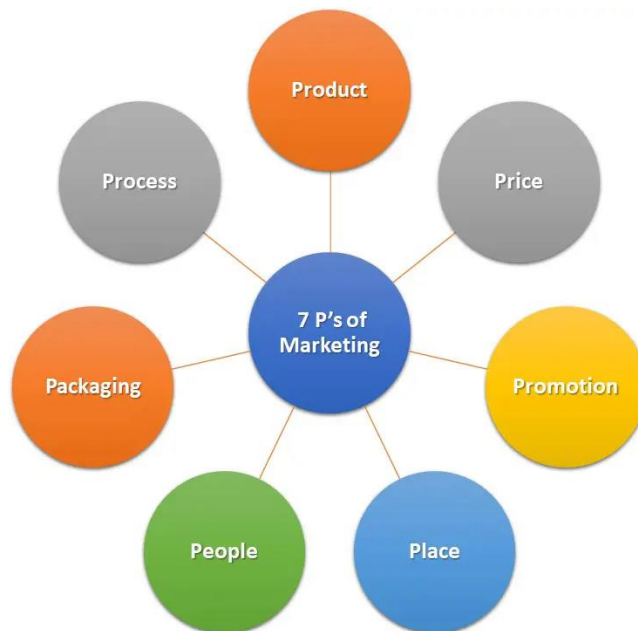


Figure 5.2.1 7 P's

As you develop your marketing strategy, you should focus on the traditional 7 P's of marketing:

- Product – how you satisfy customer needs
- Price – how much customers are willing to pay for your product
- Promotion – which channels you use to tell customers about your product
- Place – where you sell your product
- People – individuals who help sell your product to customers
- Packaging – how you present your product to the customer
- Process – how you deliver your product to customers

5.2.4 Types of marketing strategies

Various kinds of marketing strategies are generally available. Choosing a marketing strategy involves looking at your target market's needs, your target market's demographics, and the details of your products.

There are two primary categories of marketing strategy:

1. Business-to-business (B2B) marketing
2. Business-to-consumer (B2C) marketing

The various types of market strategies are listed below.

- Cause Marketing
- Relationship Marketing
- Worth of Word-of-Mouth Promotion
- Paid Marketing
- Diversity Marketing
- Transactional Marketing
- E-Marketing
- Undercover Marketing
- Offline Marketing

5.2.5 7 Steps for making a Great Marketing Strategy

Follow these seven essential steps to create an effective marketing plan. Let's delve deeper into each step's specifics in the following part:

1. Create a marketing strategy.
2. Establish buyer personas.
3. Establish your objectives.
4. Choose the right tools.
5. Analyze your media.
6. Audit and plan media campaigns.
7. Put it into action.

5.3 MARKETING MIX

5.3.1 What Is a Marketing Mix?

A marketing mix includes multiple areas of focus as part of a comprehensive marketing plan. The term often refers to a common classification that began as the four Ps: product, price, placement, and promotion.

- A marketing mix refers to a framework that uses the four Ps of product, price, placement, and promotion.
- This concept traces back to 1960, when marketing professor E. Jerome McCarthy first published it in a book entitled *Basic Marketing: A Managerial Approach*.
- The different elements of a marketing mix work in conjunction with one another with the ultimate purpose of generating higher sales.
- In addition to the 4 Ps, three approaches can also be integrated that include people, process, and physical evidence to reinforce a consumer-centric type of marketing strategy.
- This type of strategy extends beyond a product-focused marketing approach.

5.3.2 Elements of Marketing Mix

The four Ps classification for developing an effective marketing strategy was first introduced in 1960 by marketing professor and author E. Jerome McCarthy.¹ It was published in the book entitled *Basic Marketing: A Managerial Approach*. Depending on the industry and the target of the marketing plan, marketing managers may take various approaches to each of the four Ps. Each element can be examined independently, but in practice, they often are dependent on one another.

Marketing mix- 4P's



Figure 5.3.1 4P's

1. Product

The first element of the marketing mix is Product, and it depicts the tangible or intangible goods offered by organisations to customers to satisfy their needs and wants. In simple terms, it is a bundle of utilities. Product Mix means the important decisions related to a product like the design of the product, quality of the product, the quantity of the product, packaging of the product, etc. Besides these decisions, product assortment is also an essential part of the product mix. In simple terms, it means the number or volume of products and items a specific producer offers to the market. For example, ITC is an FMCG company and is dealing with different food products.

The essential components of the product mix are Branding, Packaging, and Labelling.

i) Branding: Under product mix, branding is one of the most important decisions taken by a marketer. Branding involves deciding whether the product of the firm will be marketed under a generic name or a brand name. The generic name is the name of the whole class of products. For example, soft drink is a generic name and Coca-Cola is a brand name.

ii) Packaging: It is the basic activity of every organisation and can be defined as a set of activities or tasks related to the designing, and production of an appropriate container, wrapper, bag, or box of the product. Packaging involves three levels; viz., Primary Packaging, Secondary Packaging, and Transportation Packaging.

iii) Labelling: It means putting identification marks on the package of the product. A label is basically the carrier of information regarding the product. The information provided on the label of a product includes the name of the product, its price, date of manufacturing, date of expiry, ingredients, warnings(if any), weight, etc.

2. Price

Price is the value of a product or service passed on by the buyer to the seller. As a customer is very sensitive about the price of a product, it is a crucial element of the marketing mix. A slight shift by the organisation in the price of a product can shift the customers towards the competitor's product. Therefore, Price Mix is an important decision and is related to the fixing of the price of a product or service. The decisions under price mix are related to demand of the commodity, price of competitors, etc.

While fixing the price of a commodity, the marketer should keep some factors in mind, such as pricing objectives, product cost, the extent of competition in the market, customer's demand and utility, government and legal regulations, and marketing methods used.

3. Place/Physical Distribution

It is essential to make the product or service available to the customer at the right place and at the right time, then only the customer would be able to purchase the product or service. Place is an element of marketing and is a process of transferring goods from the place of production to the place of consumption. Therefore, Place Mix is an important decision and is related to the physical distribution of the goods and services to the customers. The decisions under place mix include deciding the market for distribution, the channel of distribution, etc.

Hence, the place mix consists of Channels of Distribution and Physical Movement of Goods. The two different channels of distribution are direct channel and indirect channel. And the components of physical distribution include order processing, transportation, warehousing, and inventory.

4. Promotion

The last element of the marketing mix is promotion, which includes activities undertaken by the marketer to communicate with the customers and distribution channels so they can enhance the sales of the firm. Through promotional communication, an organisation's aim is to inform and persuade the customer to purchase the goods and inform him/her regarding the benefits of the product. Hence, Promotion Mix is an important decision and includes all decisions of an organisation related to the promotion of a sale of goods and services. Some of the important decisions under promotion mix are selecting a media to advertise the product, selecting promotional techniques, public relations, etc.

The four major elements of the promotion mix are Advertising, Sales Promotion, Personal Selling, and Public Relations.

i) Advertising: Advertising means promoting the idea of a product or service through a non-personal presentation by an identified sponsor. It is a paid form of promotional activity, and helps an organisation in connecting with the customers indirectly. The basic features of advertising are reach, choice, legitimacy, expressiveness, economy, and enhancing customer satisfaction.

ii) Sales Promotion: Sales Promotion is a short-term use of the incentives or other promotional activities of an organisation that helps in stimulating the customer in buying the product. Some of the sales promotion techniques used by organisations are refunds, discounts, rebates, gifts or premiums, samples, contests, usable benefits, etc.

iii) Personal Selling: As the name suggests, personal selling means selling products personally. It involves face-to-face communication and interaction between the buyer and the seller with the motive of sale. Personal selling does not mean influencing the buyer in purchasing what the seller is selling, but to provide satisfaction to the customer. Some of the basic features of personal selling are better response, personal interaction, relationships, etc.

iv) Public Relations: Public Relations means maintaining public relations with the public and by doing so, the organisations can create goodwill. According to the Chartered Institute of Public Relations, Public Relations is a strategic management function that adds value to an organisation by helping it to manage its reputation.

Marketing Mix Product

All products can be broadly classified into 3 main categories. These are :

1. **Tangible products:** These are items with an actual physical presence such as a car, an electronic device, and an item of clothing or a consumer good.
2. **Intangible products:** These are items that have no physical presence but can be felt indirectly. An insurance policy is an example of this. Online items such as software, applications or even music and video files are also intangible products.
3. **Services:** Services are also intangible products but they are the result of an economic activity that does not result in ownership. It is a process that creates benefits for customers. Services depend highly on who is performing them and remain difficult to reproduce exactly.

5.4 CUSTOMER BEHAVIOR

5.4.1 Consumer Behavior

Customer behavior refers to an individual's buying habits, including social trends, frequency patterns, and background factors influencing their decision to buy something. Businesses study customer behavior to understand their target audience and create more-enticing products and service offers.

There are three factors that influence customer behavior: personal, psychological, and social. Let's dive into each type.

4 Factors Influencing Consumer Behavior?

There are several factors that influence consumer behavior, but four of the most important are cultural, social, personal, and psychological factors. Let's take a closer look at each.

- **Cultural Factors** - Culture consists of the practices of a particular nation, people, or group. It might be determined by nationality, religion, location, or associations. What does this look like? An example of a consumer making decisions based on their culture could be a Midwesterner who buys ranch for their pizza—or their french fries, onion rings, or chicken fingers.
- **Social Factors** - These are the parts of a person's environment or background that affect their purchasing behaviors. These factors include family and friends as well as other social groups whose acceptance the consumer desires. When purchasing an item, a consumer might choose a brand or a company based on what their parents bought for the family growing up.
- **Personal Factors** - These include someone's age, occupation, finances, marital status, lifestyle, and values. A good consumer behavior example would be a parent in their 20s choosing to purchase an SUV instead of a minivan. Why? Because personal perception might indicate that a minivan is a vehicle for a middle-aged family.
- **Psychological Factors** - The way someone sees a product or service, as well as their personal beliefs and motivations, affect the way they purchase. For example, a consumer who enjoys working out will probably spend more money on fitness classes or dumbbells for home than someone who dislikes working out.

To ensure stable growth, a business has to meet consumer needs. This means adapting to emerging trends in the market, understanding ever-changing consumer demands and needs,

experimenting with strategies that work, and continually looking for ways to engage with consumers and potential clients. To optimize consumer engagement, marketers need to understand buying behavior, purchase decisions, and other psychological factors. For this, they rely on reports, data, and surveys.

Marketers use this information to align their strategies and decisions.



Figure 5.4.1 strategies and decisions

5.4.2 Seven O's framework of consumer behavior

The Seven O's framework of consumer behavior is a set of seven principles that can help marketers understand how people decide what to buy and use.

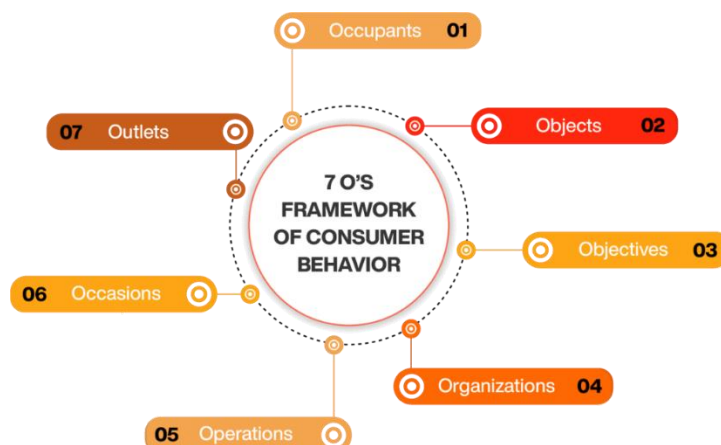


Figure 5.4.1 framework of consumer behavior

- **Occupants**

Occupants are the people who use a product or service.

- **Objects**

Objects are the things that occupy a person's time and attention. They can be tangible, like a car or computer, or intangible, like a song on the radio or a TV show.

- **Objectives**

Objectives are what you want to accomplish with your possessions or activities. You may have one objective for each object you use or goals that span multiple objects.

- **Organizations**

The organization has a significant impact on consumer behavior. An organization's mission statement, values, and vision all serve as a guide for consumer behavior.

- **Operations**

The operations of an organization influence consumers. For example, if a customer buys clothes from a store that does not serve the customers effectively, they will be less likely to buy from that store in the future. Additionally, if the prices are too high or the quality is low, it would be difficult for consumers to continue purchasing from this company.

- **Occasions**

Consumers buy things for different occasions, such as birthdays, holidays, etc., which can affect their buying behaviors.

- **Outlets**

Consumers shop at many grocery stores, drug stores, department stores, and online retailers; therefore, they must decide where they want to shop based on price and selection, among other factors.

What factors influence behavioral changes?

In this section, we shall discuss what influences consumer behavior. Here are a few factors that affect a consumer's behavioral pattern –

- **Cultural factors**

Culture is a set of beliefs, values, and behaviors that a group shares. It determines how we interact with each other and the environment around us. Cultural factors give rise to different styles of consumption.

- **Social factors**

Social factors are related to people's relationships with others in their group. These factors can indicate how consumers respond to one another's opinions regarding products or services on offer in the marketplace. For example, if your friends are buying a particular brand of car, then you may also decide to buy it even though you may not be interested in cars at all.

- **Personal factors**

Personal factors refer to individual tastes and preferences, which vary from one consumer to another. For instance, some people like spicy food while others do not like any spices in their

food items; hence, companies should try and understand their target audience to design their products accordingly.

- **Psychological factors**

Marketing strategies are based on consumer behavior and psychology. It is imperative to know what motivates consumers to buy your product or service. For example, if you sell luxury products, you will want to advertise them in ways that appeal to people who have an affinity for luxury.

- **Economic factors**

Economical factors are also important because they include price, supply and demand, competition, and many other things. The company should consider all these factors when developing a marketing strategy.

5.4.3 Benefits of Customer Behavior Analysis

In today's marketing industry, businesses are using data gathered from analyzing users' behavior to plan their marketing strategies. This data helps you make more informed decisions and limit the chance of failure. Therefore, if you haven't paid enough attention to behavior analytics, it's time to make a change.

- User behavior analytics helps you **Identify new trends and sales opportunities**.
- It enables you to **segment customers and design targeted campaigns** to increase the chance of conversion.
- In addition to customer segmentation, **customer behavior analysis helps you to understand the customer's personality** and design your customer journey accordingly.
- Behavioral analytics approaches provide you with a **clear vision about which marketing strategy attracted more consumers** to your brand.

One of the best use cases for customer behavior analytics is identifying trends before important social events. For example, if you are an e-commerce owner selling dresses online. By analyzing customers buying habits, you can easily identify what is trending this year and shift your focus on that. This way, you can attract more customers and seize the opportunity.

5.5 SALES ANALYTICS

5.5.1 What is a Sales Analysis?

Sales analysis is the process of integrating, analyzing, and understanding various data related to sales activities such as sales, customers, and transaction data.

It allows managers to look at sales from many aspects and decide what is working and what is not working.

It also provides information and reporting so sales managers and executives can ascertain what region is achieving its goals and which areas are not achieving goals, which salespeople are doing well and which are not, etc.

Benefits of Sales Analysis

A useful sales analysis has the following three benefits :

- ✓ Understanding customer needs
- ✓ Sharing sales skills and know-how
- ✓ Being aware of the market trend

What are the different types of sales analytics?

Most analysts break sales analytics applications into 3 or 4 categories in terms of their output. The categorization that makes most sense to us is:

- **Descriptive:** Summarized, prioritized observations from sales related data. Example: spike in sales of a certain product in professional services industry in a certain region.
- **Diagnostic:** Possible explanations for observations from sales related data. Example: Customers in automotive industry are spending less time to call a sales rep. Increased competition in the industry could be pushing companies to invest in our product.
- **Predictive:** Based on diagnostic results, predictions are made. Example: Leads from professional services industry in a certain region get higher scores. As a result, sales reps spend more time on those leads
- **Prescriptive:** Predictive analytics results are explained and sales rep is provided with a clear game plan. Example: Sales reps get tactics before calls to underline certain aspects of the product that are important to the customer.

5.5.1 Selling Process

What is Selling Process?

The selling process is the series of steps followed by a salesperson while selling a product. Selling Process is a complete cycle which starts from identifying the customers to closing the deal with them. It is more relevant in B2B business sales where the sales cycle is not short and might take a longer duration to close. In B2C the selling process may be transient and shorter.

7 Selling Process Steps

The process of selling a product covers various steps like prospecting, pre-approach, approach, presentation, handling objections, closing & follow-up with customers. The 7 steps of selling process are explained below in detail:

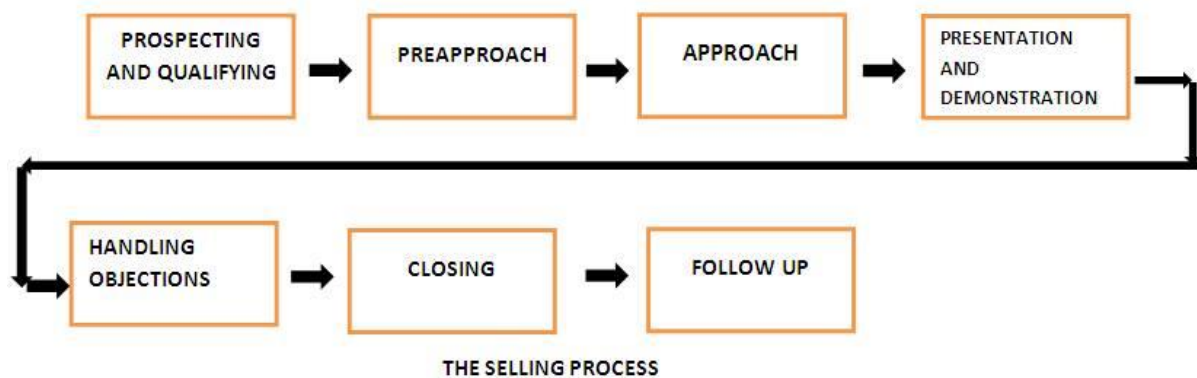


Figure 5.5.1 Selling Process

1. Prospecting

The first step in selling process in which potential customers are identified by the salesperson is called prospecting.

2. Pre-approach

The stage where the salesperson collects information about the potential customers and understands them before making the sales call is called pre-approach.

3. Approach

Approach is the step where the salesperson actually meets the customer for the first time.

4. Presentation

The step wherein the salesperson talks about how the product will satisfy the customer's needs and add value to his/her life is called presentation.

5. Handling Objections

In this step, the salesperson clarifies all the doubts and questions that the customer has and eliminates all his objections to buying the product.

6. Closing

The step in which the customer is asked to place and order for the product is called closing.

7. Follow-Up

This is the final step in the selling process where the salesperson follows up with the customers to ensure satisfaction and builds the relationship in order to repeat business with them. All the steps above help in increasing business sales and thereby increasing sales revenue & profits.

5.5.2 Sales Planning

What is sales planning?

Sales planning is the process of defining sales targets and creating a strategy that meets goals and achieves sales and marketing results. The sales plan works in collaboration with the marketing plan and the business plan.

all the resources, deciding on the amount to use, and describing the specific activities.

What are the benefits of sales planning software?

A sales planning solution can overcome these issues and transform sales performance by bringing together all the information required to produce an accurate plan. A comprehensive platform which also incorporates sales analytics can enable businesses to:

- **Gain greater insight into customers** through detailed customer segmentation, considering factors such as needs, preferences, and cross-selling opportunities
- **Produce accurate, detailed sales plans in much shorter timeframes**, planning quota for sales reps by allocating them customer portfolios and planning territory sales objectives by aggregating the sales targets for each rep
- **Simulate the impact of different sales strategies** on the rest of the business, and on the predicted outcome, before taking a decision
- **Integrate ML predictive forecasting and traditional methods**, using (at your discretion) figures generated by human estimates, sales pipeline analysis & inspection, and sales predictive analytics algorithms within a machine learning environment
- **Monitor and control sales performance in real time**, automatically conducting variation analyses between current data and targets within a multi-dimensional environment and dissecting your results in terms of volume, price, and mix effects
- **Launch re-planning cycles at any time** based on actual acquisition figures, updating and amending estimates generated by the previous planning cycle
- **Drive commercial efforts towards strategic targets** by integrating sales force incentives & compensation management with quota and territory planning
- **Exploit visibility of the lead journey**, customer behavior, and customer sales history, fostering sales planning efficacy thanks to integration with CRM systems

Objectives of Sales Plan:



Figure 5.5.2 Objectives of Sales Plan

Types of Sales Plans:

There is no unified sample sales plan template. They differ according to the company's purpose. Although most of the sales plans are rather similar, the following types can be distinguished:

- **Annual/quarterly/weekly sales plan** – It's a traditional business sales plan that features revenue goals, tactics, and specific time period by when it should be accomplished.
- **30 60 90 sales plan** – This milestone based sales plan specifies a goal to achieve within set milestones – 30, 60 or 90 days – and is ideal for new sales managers. Helps draw up strategic and tactical activities based on this plan.
- **Sales budget plan** – A sales plan that provides a forecast of factors that could influence revenue within a specific timeframe
- **Sales tactics plan**– This is a tactical sales plan that includes execution strategy, detailed daily or weekly plans such as email follow-up frequency, meeting appointments, and prescribed call sequences for different sales teams.
- **Territory-based sales plan** – It features tactics for sales teams across different territories, considering the working environment and market dynamics of a specific area.
- **Sales focus area plan** – This plan dives deep into different sales domains like sales compensation or sales training plan.

Sales Planning Process:



Figure 5.5.3 Sales Planning Process

❖ 1st Step of Building a Sales Planning Process is to Analyze where you are at?

To determine where you are going you first have to see where you have been.

Some questions to ask yourself are:

- What business are we in?
- Who are our customers?

- Where did most of my sales come from?
- Where do we want my sales to come from?
- What are some external/Internal factors that can impact my sales? Eg. Industry trends, Technology, Competition, Business Environment etc..

❖ 2nd Set your Goals

Your goals should be SMART (Specific, Measurable, Attainable, Relevant, Time bound)

Some samples:

- Achieve 1 Million in New Sales by December 31st, 2022
- Sell 2 Corporate wide Licenses by December 31st, 2022
- Grow Distributor Network by 30% by December 31st, 2022

❖ 3rd Step Develop the Sales Plan

This is the meat of your sales plan and is the most critical in helping you achieve your goals. It should outline all the strategies and tactics that you are going to use to overcome the obstacles that you may face in meeting your goals.

Analyze- Internal & External Factors



Strategy – Your Plan of Attack to Reduce Risk

A Strategy is a plan or action designed to achieve a particular goal. One way to develop sales strategies is to write down the risks/obstacles that you see in achieving your goals then write down some ideas on how you can overcome them. These include internal and external factors.

Tactics- Short-term actions to help you execute on your strategy

❖ Finally it's time to Execute on your strategy.

Action speaks louder than words. Now that you have taken the time to plan out your attack now it's time to act on it. Your The sales profession is one of the few professions where you can control how much you make or don't make by the actions we do or do not take. The plan is to help take away some of that mystery to help you use your creativity to hit those goals.

❖ Evaluate and Review Planning Processes

The world/market is constantly evolving if you write a plan and only look at it at the end of each year that's like looking in a rear view mirror instead of looking where you are going. Just as important as writing the plan it's important to:

- Review the plan quarterly in review meetings
- Review what went well, what did not, and make adjustments
- Score Card: Measure and Benchmark.

How to Write a Sales Plan:

Create a mission statement. Define your team's roles and responsibilities. Identify your target market. Outline your tools, software, and resources. Analyze your position in your industry. Plan your marketing strategy. Develop your prospecting strategy. Create an action plan. List your goals. Set your budget.

1. Create a mission statement.
2. Define your team's roles and responsibilities.
3. Identify your target market.
4. Outline your tools, software, and resources.
6. Plan your marketing strategy.
7. Develop your prospecting strategy.
8. Create an action plan.
9. List your goals.
10. Set your budget.

Describe the costs associated with hitting your sales goals. That usually includes:

- Pay (salary and commission)
- Sales training
- Sales tools and resources
- Contest prizes
- Team bonding activities
- Travel costs
- Food

Compare the sales plan budget to your sales forecast for accurate budgeting.

If you want to take your plan to the next level, read on to learn some tips for creating a highly effective sales plan.

Tips for Creating an Effective Sales Plan

We've gone over what you should include in a sales plan, including some examples and mockups.

Learn some tips and tricks for creating a sales plan that helps you hit target numbers and exceed your higher-ups' expectations.

- Use industry trends to strengthen your plan. When presenting your sales plan to a stakeholder, use industry trends to highlight why your plan will be effective.
- Specify the technology you'll use to track success. You can do this for internal reference or let stakeholders know how you'll measure success. Some tools you can consider include CRM and dashboard software.
- Support your budget proposal with hard facts and data. If you're creating a budget as a part of your plan, support it with previous performance data and sales forecasts.
- Create different plans for each team. If you create a sales plan for business development, inbound sales, outbound sales, field sales, and so forth, you can get even more granular and specific in your goals and KPIs.
- Get marketing's input. Marketing and sales alignment is critical for the success of your sales plan. The more input you have from marketing, the more you can align your lead generation, prospecting, and nurturing efforts.
- Talk with your sales reps to understand their challenges. It might be easy to get lost in numbers and forecasts. But it's important to know your sales representative's day-to-day to understand what will and will not prove effective or feasible.
- Complete an in-depth competitive analysis. You must know what the competition is doing well to create a plan that nudges your company in that direction.

5.6 ANALYTICS APPLICATIONS IN MARKETING AND SALES

In business analytics, different tools and techniques are available for analyzing, exploring, or investigating business data. These analytics tools and techniques are engaged in progressive business growth.

Nowadays, business analytics is used in various industries for more growth in a short duration of time to gain more profit than usual.

Starting with the basic definition of Business Analytics, “it is the study and exploration of statistical data, the formation of predictive models, deployment of the optimized technique, and communication of obtained output to business partners, customers, and other executives for the different business issues”.

Big data is used up to the maximum extent through qualitative and quantitative techniques to get desired business modeling and decision-making.

The role of a business analyst varies with the variation of business problems.

The first step is to recognize the business problem accurately and restructure it in flexible ways for analysis. Through this process, the massive size of data is used in analytical modeling and logical analysis for approaching business decisions.

With an interpretation of the business issue and becoming familiar with the accessible dataset, the business analyst gives insights into the growth of the business.

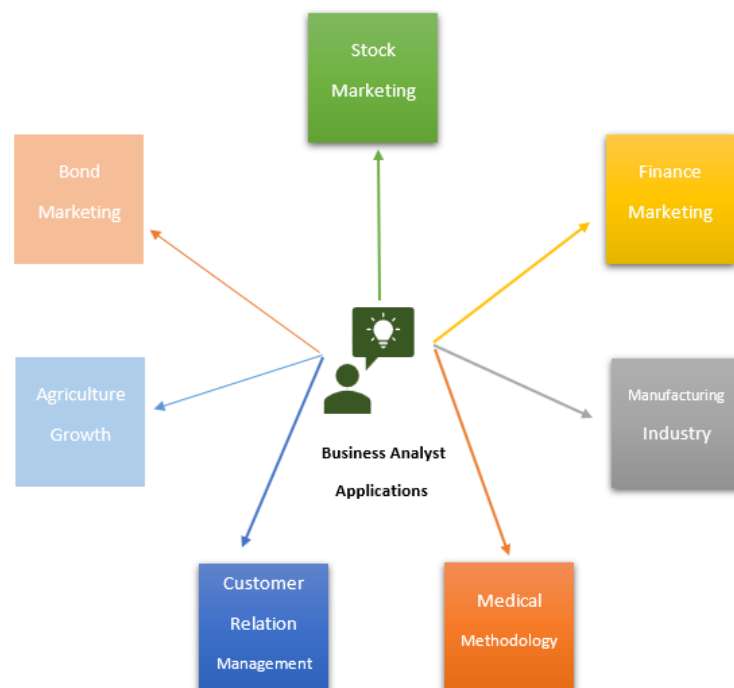


Figure 5.6.1 Business Analytics Applications

Applications of Analytics in Marketing and Sales:

- ✓ To identify the target audience
- ✓ To develop advertising technique to attract the customers
- ✓ Forecast the supply requirements
- ✓ To identify the potential markets
- ✓ To supply the potential markets
- ✓ To supply goods or services as per customer specifications

Importance of BA (Business Analytics) in Marketing and Sales:

➤ **Segmentation**

BA tools provide segment wise reports which can be used by marketing manager to ascertain where to focus his resources

➤ **Marketing mix optimization**

BA tools helps in identifying correct marketing mix(product, price, place and promotions)

➤ **Competitor Analysis**

Sales manager needs BA to compare the information with its competitors and to discover the areas of improvements.

➤ **Sales Performance Analysis**

BA can gauge the sales performance and helps to increase the performance.

Benefits of BA for Marketing and Sales

- ✓ Develop critical skills
- ✓ Track behaviour patterns of the customer
- ✓ Forecast trends
- ✓ Developing the quickness for numbers and statistics

Analytics use cases in marketing & sales

Marketing and sales are the most popular areas where companies have implemented data analytics use cases. Both sectors benefit from the use of data in separate ways:

With the help of big data, marketing managers can now make more accurate decisions. They collect data from various sources, such as social media, email communication, and other

platforms, to communicate with their customers to understand their needs and preferences better. Marketing specialists then analyze the collected data to improve the marketing strategy and drive more sales.

Data analytics can help salespeople sell more effectively. For example, with the help of data analytics, sales representatives can easily understand which products are most demanded by particular target groups and focus on selling them to increase their performance level.

In marketing and sales, data analytics can help businesses to:



Figure 5.6.2 Data Analytics Use case in marketing and sales

- Identify which customers are most likely to respond to a specific offer
- Find the best product mix for a given customer
- Find the best price for a given product or product mix
- Determine the effectiveness of different advertising or marketing campaigns
- Identify new markets for existing products or services